Hochschild Homology of the b-Calculus

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Abstract

Let X be a compact n-manifold with boundary, let ∂X have N components, and let x be a boundary defining function. We compute the continuous Hochschild homology of the algebra of b-pseudodifferential operators of integral order with Laurent coefficients at the boundary, of its quotient $\mathcal{A}^b = x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X)/x^{\infty}\Psi_b^{-\infty}(X)$, and of the ideals and quotients cut out by the two natural filtrations, by operator order and by powers of x. This is the b-calculus analogue of the computation of Melrose and Nistor for the cusp calculus. The final groups have the same form as theirs: they are expressed through the de Rham cohomology of ${}^bS^*X$, of its boundary, and of two auxiliary circles recording symbol order and boundary order. The proofs differ at the boundary. Conjugation by x acts on indicial families by the complex shift $\lambda \mapsto \lambda - i$, so the boundary algebras are Laurent series twisted by an automorphism rather than by a derivation, and their coefficients are entire functions of λ rather than Schwartz functions. We compute the homology of the boundary ideal by a weight decomposition and a crossed-product argument. We identify the connecting map to the residual ideal with the b-trace defect formula and evaluate the index functional on residual perturbations of the identity, where it reduces to a Wiener–Hopf winding number and to $-\frac{1}{2}$ times the b-eta invariant. For fully elliptic operators of positive order we prove the general index formula, with b-zeta and b-eta invariants, following Loya's thesis in codimension one, and we correct the sign of its boundary term. The computation for the Laurent symbol algebra $\mathcal{A}^b_\sigma$, which is the b-groupoid case of the Benameur–Nistor theorem, is retained as Part I. Each step is labelled as proved, quoted, or sketched; see Section 1.4.

1 Introduction

1.1 The algebras

Let X be a compact manifold with boundary of dimension n. Its boundary $\partial X = \bigsqcup_H H$ has N components. Fix a boundary defining function x, a collar $[0,\varepsilon)_x \times \partial X \hookrightarrow X$, and a smooth positive density ν_∂ on ∂X. Let $\Psi_b^m(X)$, $m \in \mathbb{Z}$, be the small calculus of classical b-pseudodifferential operators (Section 14). Conjugation by powers of x preserves each $\Psi_b^m(X)$ (Lemma 14.1), so

$$x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X) = \bigcup_{\ell\in\mathbb{Z},\, m\in\mathbb{Z}} x^\ell \Psi_b^m(X)$$

is an algebra. Following Melrose–Nistor [21, (8)–(10)], we consider its ideal of operators that are residual for both filtrations and the quotients by it:

$$\begin{aligned}
&\mathcal{I}^b = x^\infty\Psi_b^{-\infty}(X) = \bigcap_\ell x^\ell\Psi_b^{-\infty}(X), && \mathcal{A}^b = x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X)/\mathcal{I}^b,\\
&\mathcal{I}^b_\partial = x^{-\mathbb{Z}}\Psi_b^{-\infty}(X)/\mathcal{I}^b, && \mathcal{I}^b_\sigma = x^\infty\Psi_b^{\mathbb{Z}}(X)/\mathcal{I}^b,\\
&\mathcal{A}^b_\sigma = \mathcal{A}^b/\mathcal{I}^b_\partial, \qquad \mathcal{A}^b_\partial = \mathcal{A}^b/\mathcal{I}^b_\sigma, && \mathcal{A}^b_{\partial,\sigma} = \mathcal{A}^b/(\mathcal{I}^b_\sigma + \mathcal{I}^b_\partial).
\end{aligned}$$

The kernels of elements of $\mathcal{I}^b$ are exactly the smooth functions on X^2 that vanish to infinite order at $\partial(X^2)$. The algebra $\mathcal{A}^b_\sigma$ is the algebra of Laurent b-complete symbols treated in Part I, $\mathcal{A}^b_\partial$ is the algebra of Laurent series of indicial operators, and $\mathcal{A}^b_{\partial,\sigma}$ is the algebra of Laurent series of complete symbols at the boundary. These algebras form the diagram of extensions in Figure 1, which is the b-version of the diagram in [21, p. 4].

1.2 The main theorem

Set $Y = {}^bS^*X$ and $Z = Y|_{\partial X} = {}^bS^*_{\partial X}X$, and let Z_H be the part of Z over the component H. Let S_σ and S_∂ be circles, and write $d\theta_\sigma$ and $d\theta_\partial$ for their generators; they correspond to $d\log r$ (r a

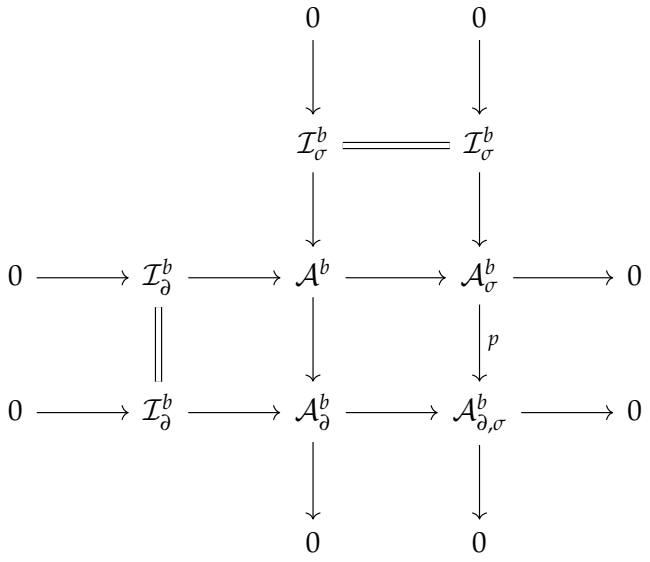


Figure 1: The extensions relating the b-algebras. Rows are the indicial (boundary) extensions; columns are the symbol (interior) extensions. Since $\mathcal{I}^b_\sigma \cap \mathcal{I}^b_\partial = 0$ in $\mathcal{A}^b$, the central square is a pull-back: $\mathcal{A}^b = \mathcal{A}^b_\partial \times_{\mathcal{A}^b_{\partial,\sigma}} \mathcal{A}^b_\sigma$ (Lemma 22.1). In addition, $0 \to \mathcal{I}^b \to x^{-\mathbb{Z}}\Psi^{\mathbb{Z}}_b(X) \to \mathcal{A}^b \to 0$ and $0 \to \mathcal{I}^b \to x^{-\mathbb{Z}}\Psi^{-\infty}_b(X) \to \mathcal{I}^b_\partial \to 0$.

fibre norm) and to $d\log x$. Homology is continuous Hochschild homology in the chain model of Section 16.1. Cohomology has complex coefficients. For β in $H^{2n-2}(Z)$, or in $H^{2n-2}(Y)$ after restriction to Z, write $\int_{Z_H}\beta$ for its integral over Z_H, oriented as part of ∂Y.

Standing hypotheses. X is connected and $\partial X \neq \varnothing$. For $n \geq 2$ every Z_H is connected and oriented. The case $n = 1$ is treated in Remark 20.9.

Theorem 1.1 (Hochschild homology of the b-calculus)**.** *Assume the standing hypotheses and the inputs listed in Section 1.4. Parts (i)–(vii) and (ix) hold for every $n \geq 1$. Part (viii) is stated for $n \geq 2$, with the $n = 1$ formula in Remark 20.9.*

(i) $\mathrm{HH}_0(\mathcal{I}^b) \cong \mathbb{C}$, *with dual basis the trace, and* $\mathrm{HH}_k(\mathcal{I}^b) = 0$ *for* $k > 0$.

(ii) $\mathrm{HH}_k(\mathcal{I}^b_\partial) \cong \mathbb{C}^N$ *for* $k = 0, 1$, *and* $\mathrm{HH}_k(\mathcal{I}^b_\partial) = 0$ *for* $k \geq 2$. *The boundary traces* $\mathrm{Tr}_{\partial,H}$ *of* (25) *form a dual basis in degree* 0, *and the cyclic cocycles* φ_H *of* (26) *form a dual basis in degree* 1.

(iii) $\mathrm{HH}_0(x^{-\mathbb{Z}}\Psi^{-\infty}_b(X)) \cong \mathbb{C}^N$, $\mathrm{HH}_1(x^{-\mathbb{Z}}\Psi^{-\infty}_b(X)) \cong \mathbb{C}^{N-1}$, *and the higher groups vanish. The connecting map* $\mathrm{HH}_1(\mathcal{I}^b_\partial) \to \mathrm{HH}_0(\mathcal{I}^b)$ *is the b-trace defect.*

(iv) $\mathrm{HH}_k(\mathcal{I}^b_\sigma) \cong H^{2n-k}(Y \times S_\sigma, Z \times S_\sigma)$.

(v) $\mathrm{HH}_k(\mathcal{A}^b_\sigma) \cong H^{2n-k}_L(Y \times S_\sigma) \cong H^{2n-k}(Y \times S_\sigma) \oplus H^{2n-k-1}(Z \times S_\sigma)$.

(vi) $\mathrm{HH}_k(\mathcal{A}^b_{\partial,\sigma}) \cong H^{2n-k}(Z \times S_\sigma \times S_\partial)$.

(vii) $\mathrm{HH}_k(\mathcal{A}^b_\partial) \cong \mathrm{HH}_k(\mathcal{A}^b_{\partial,\sigma})$ *for* $k \neq 1, 2$. *For* $k = 1, 2$, $\mathrm{HH}_k(\mathcal{A}^b_\partial)$ *is the kernel of a surjection* $\mathrm{HH}_k(\mathcal{A}^b_{\partial,\sigma}) \to \mathbb{C}^N$. *In the coordinates dual to the traces and cocycles of (ii), the surjections are* $-i(2\pi)^{-n}(\int_{Z_H}\beta_\partial)_H$ *for* $k = 1$ *and* $(2\pi)^{-n}(\int_{Z_H}\beta_0)_H$ *for* $k = 2$. *Here* β_∂ *is the coefficient of* $d\theta_\partial$ *written on the left, and* β_0 *is free of both circle factors.*

(viii) $\mathrm{HH}_k(\mathcal{A}^b) \cong \mathrm{HH}_k(\mathcal{A}^b_\sigma)$ *for* $k \neq 1, 2$. *Moreover*

$$\mathrm{HH}_1(\mathcal{A}^b) \cong \mathbb{C} \oplus H^{2n-1}(Y \times S_\sigma) \oplus H^{2n-3}(Z),$$
$$\mathrm{HH}_2(\mathcal{A}^b) \cong \ker\Big(H^{2n-2}(Y) \xrightarrow{(\int_{Z_H})_H} \mathbb{C}^N\Big) \oplus H^{2n-3}(Y) \oplus H^{2n-3}(Z \times S_\sigma).$$

The connecting map $\partial_2^{\mathcal{A}}$ *is* $(2\pi)^{-n}$ *times the displayed period map on* $H^{2n-2}(Y)$ *and vanishes on the other summands of* $\mathrm{HH}_2(\mathcal{A}^b_\sigma)$. *Its rank is* $N-1$ *(Remark 22.3). In particular, if* $N = 1$ *then* $\mathrm{HH}_2(\mathcal{A}^b) \cong \mathrm{HH}_2(\mathcal{A}^b_\sigma)$.

(ix) The index functional, which is the image of the trace under $\partial : \mathrm{HH}_1(\mathcal{A}^b) \to \mathrm{HH}_0(\mathcal{I}^b)$, *is nonzero. Hence* $\mathrm{HH}_k(x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X)) \cong \mathrm{HH}_k(\mathcal{A}^b)$ *for* $k \neq 1$, *and* $\mathrm{HH}_1(x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X))$ *is the kernel of the index functional on* $\mathrm{HH}_1(\mathcal{A}^b)$.

The displayed decompositions use a collar and the face splitting of Proposition 8.1. The formula for $\mathrm{HH}_1(\mathcal{A}^b)$ also uses a linear splitting of its short exact sequence. Remark 19.4 discusses how the generators depend on x and on the collar. If X has c components, each with nonempty boundary, the rank in (viii) becomes $N - c$, and the summand $\mathbb{C}$ in $\mathrm{HH}_1(\mathcal{A}^b)$ becomes $\mathbb{C}^c$ (Remark 22.3). Components without boundary contribute as in the closed case.

For $N = 1$, parts (ii) and (iv)–(viii) have the same form as the cusp-calculus result of Melrose and Nistor [21, (16)], which is stated for Hochschild cohomology, with ${}^cS^*X$ replaced by ${}^bS^*X$. Both cosphere bundles are diffeomorphic to S^*X (Proposition 4.1). The bookkeeping for several components is new.

1.3 Relation to Melrose–Nistor and novelty

Remark 1.2 (Comparison with Melrose–Nistor). Melrose and Nistor [21] compute the Hochschild homology of the full *cusp* calculus $x^{-\mathbb{Z}}\Psi_c^{\mathbb{Z}}(X)$. They say that they use the cusp calculus rather than the b-calculus "to avoid certain problems with completeness". They also remark that the cusp commutation relation $[x^2\partial_x, x] = x^2$, in place of the b-relation $[x\partial_x, x] = x$, "makes quite a lot of difference", and that "methods similar to those used here can be applied directly to the b-calculus" [21, pp. 3–4]. The present notes carry this out. The differences are as follows.

1. *Twist.* In the cusp case the boundary algebras are Laurent series $\Psi_{\mathrm{sus}}(\partial X)[[x]$ in which x and the suspension variable satisfy $[x, \partial_t] = x^2$ [21, (14)]; to leading order the twist is a derivation. In the b-case, $x^{-1}Ax$ has indicial family $I(A, \lambda - i)$ (Lemma 14.1), so the twist is the *automorphism* α of complex translation by $-i$. The associated graded algebra for the x-filtration is therefore not commutative, and Melrose–Nistor's x-adic spectral sequence [21, proof of Prop. 6] is replaced by a weight decomposition and a crossed-product computation (Section 18).

2. *Entire indicial families.* Indicial families in the small b-calculus are entire in λ and suspended on every horizontal strip; residual ones are Schwartz on every strip. The cusp indicial families are only Schwartz, or symbols, on the real line. Analyticity is what makes complex shifts possible. It is also the source of the completeness problem mentioned by Melrose–Nistor: an operator is invertible modulo $\mathcal{I}^b$ only if its indicial family is invertible for *all* $\lambda \in \mathbb{C}$ (Lemma 23.2). Geometric Dirac operators never satisfy this.

3. *Traces.* The boundary trace on $\mathcal{I}^b_\partial$ is the residue of $\mathrm{Tr}(x^w \cdot)$ at $w = 0$. The b-trace is the finite part of the same function. The connecting map for $\mathcal{I}^b \subset x^{-\mathbb{Z}}\Psi_b^{-\infty}$ is Melrose's b-trace defect

formula (Theorem 19.2). The extension of the boundary trace to $\mathcal{A}_\partial^b$ is Melrose's regularized suspended trace. Its Hochschild boundary is given by a *shift defect* (Proposition 21.3), which replaces [21, Lemma 13].

Melrose–Nistor's Appendix C [21, Prop. 26] lifts b-operators to cusp operators by the transcendental blow-up $t = e^{-1/x}$. This lift is useful for index formulas, but it does not identify the Laurent algebras: Laurent series in t and Laurent series in $x = 1/\log(1/t)$ are different algebras.

Novelty. We have not located a published computation of the Hochschild homology of the b-calculus operator algebra. Theorem 1.1 should therefore be read as a careful execution of the programme suggested in [21], not as a known result. The symbol part (v) is the b-groupoid case of the Benameur–Nistor theorem [2, 1].

1.4 Status of the proofs

The proofs use the inputs listed below. The references to Melrose–Nistor [21] are comparisons with the cusp calculus.

Proved in these notes.

- The structure theorems for $\mathcal{I}_\partial^b$ and $\mathcal{A}_\partial^b$ as twisted Laurent series (Propositions 15.3 and 15.4). These rest on (Q0) and (C3), and give only an algebraic surjection; no continuous section is claimed.
- The explicit H-unitality contractions for $\mathcal{I}^b$, $\Psi^{-\infty}(H)$ and $\mathcal{I}_\partial^b$ (Lemma 16.5).
- The Morita lemma for smoothing kernels (Lemma 16.7), and part (i), which also uses the kernel theorem (C3).
- The vanishing of the nonzero weights for $\mathcal{I}_\partial^b$ (Proposition 18.3), given the excision theorem (Q1).
- The coinvariants and invariants of the shift (Lemma 18.2).
- The trace Tr_∂, the cocycle φ, their nonvanishing, and their independence of choices.
- The b-trace defect formula on $x^{-\mathbb{Z}}\Psi_b^{-\infty}$ (Theorem 19.2) and the connecting map in (iii).
- Continuous HKR for the strip-Schwartz algebra $\mathcal{E}$ (Proposition 18.1) and for smooth Laurent, flat, and formal Laurent symbol coefficients (Lemma 5.2 and Proposition 5.3).
- The de Rham lemmas, the E^2-pages, and the leading-symbol isomorphisms for the three symbol algebras (Sections 9 and 20). The convergence proof uses the mixed complexes of Lemmas 20.2 and 20.3. The flat case also uses (Q1) (Lemma 20.4).
- The shift defect, its connecting cocycle given (Q1), and the residue and top-order formulas (Proposition 21.3 and Lemmas 21.5 and 21.6).
- Surjectivity and the boundary-period descriptions of the connecting maps, with their constants in trace-dual coordinates (Propositions 21.7 and 21.8). These use the inputs of (ii) and (Q1). In particular, $\operatorname{rank} \partial_2^{\mathcal{A}} = N - 1$ for connected X and $n \geq 2$ (Remark 22.3).
- For $n = 1$, the surjectivity of both boundary connecting maps of $\mathcal{A}_\partial^b$ and of $\partial_1^{\mathcal{A}}$, by explicit test cycles. The identification of connecting maps uses (Q1).

- The fibre-product description of $\mathcal{A}^b$, and the bound $N-1$ on the rank of $\partial_2^{\mathcal{A}}$. The bound uses $\dim \mathrm{HH}_1(\mathcal{I}_\partial^b) = N$ from (ii), and so inherits (Q1) and (Q2).

- The invertibility criterion (Lemmas 23.1 and 23.2).

- The Wiener–Hopf index formula for residual perturbations of the identity (Theorem 23.5), and the nonvanishing of the index functional. Part (ix) uses in addition (i), (Q0), (Q1) and (C3).

- The general index formula for fully elliptic operators of positive order (Theorem 24.16) is pending independent referee review. The section includes the reduction lemma (Lemma 24.1), the b-trace defect formula on the class $\mathcal{K}_\delta$ (Theorem 24.5), the resolvent and complex powers (Propositions 24.10 and 24.11), the b-zeta function and its heat form (Theorem 24.13), and the b-eta function (Lemma 24.15). The section adapts the arguments of [6, Chs. 2–8] to codimension one. These results use (Q0) and (C3) only, including the (C3) items marked there for Section 24. The sign of the boundary term differs from the printed form of [6, Thm. 8.6]; see Remark 24.17 and the explicit check in Example 24.18.

Quoted inputs.

(Q0) The small b-composition theorem, including the ideal property $x^K\Psi_b^m \circ \Psi_b^{m'} \subset x^K\Psi_b^{m+m'}$, and the trace-class property of $x^\varepsilon\Psi_b^{-\infty}$ for $\varepsilon > 0$ [20, 10].

(Q1) Wodzicki's excision theorem in the chain model of Section 16.1 (Theorem 16.3). Its topological input, exactness of the chain spaces, is proved (Lemma 16.2).

(Q2) The weight-zero part of the Hochschild homology of a crossed product by $\mathbb{Z}$ (Theorem 16.8). The cited proofs are algebraic. Its continuous version for $\mathcal{E} \rtimes_\alpha \mathbb{Z}$, and the reduction of the H-unital case to the unital case, are assumed, not proved.

(C3) Standard analysis and topology. This includes: nuclearity of $\mathcal{E}$, B_H, $B'_{H,N}$ and of the pieces of the b-calculus; the facts that $\widehat{\otimes}_\pi$ commutes with countable products and is exact on nuclear Fréchet spaces [26]; the smooth Schwartz kernel theorem, including $\dot{C}^\infty(X^2) = \dot{C}^\infty(X)\widehat{\otimes}\dot{C}^\infty(X)$; Borel extension of smooth jets, classical symbol quantization and asymptotic summation; Taylor expansion and asymptotic summation of conormal kernels at a face meeting the conormal set transversally; the suspended calculus on strips; the identity $\mathrm{Tr}(PQ) = \mathrm{Tr}(QP)$ when $\operatorname{ord} P + \operatorname{ord} Q < -\dim H$; mapping properties on weighted b-Sobolev spaces and the Calderón–Fedosov formula; and Lefschetz duality. For Section 24 it also includes: the parameter-dependent symbol calculus of Seeley [12, 13] in b-coordinates, with parameter-uniform remainder estimates (used in Lemma 24.9); the parameter-dependent suspended parametrix (Lemma 24.7); interpolation and elliptic regularity for the powers of P_0; and the Hilbert–Schmidt property of the inclusions $x^bH_b^N \hookrightarrow x^{b'}H_b^{N'}$ for $b > b'$ and $N - N' > n/2$.

Auxiliary arguments. The optional Čech route (Section 10) assumes the localization statement in Lemma 10.1. The Rees–Fedosov discussion (Section 11) is heuristic.

Dependencies: (i) uses (C3). (ii) and (iii) use (Q0)–(Q2) and (C3). Parts (v) and (vi) use the complete-symbol calculus (Q0), (C3), and the fixed coefficient stage argument. Part (iv) also uses (Q1). Parts (vii) and (viii) use (ii), the symbol calculations for $\mathcal{A}_\sigma^b$ and $\mathcal{A}_{\partial,\sigma}^b$, and excision (Q1). Part (ix) uses (i), (Q0), (Q1) and (C3) together with Theorem 23.5. The index formula (Theorem 24.16) uses (Q0) and (C3).

1.5 Organization

Part I keeps the computation for the symbol algebra $\mathcal{A}_\sigma^b$ (Sections 2–12). Part II treats the b-calculus conventions, the structure of the boundary algebras, the Hochschild tools, and the ideals $\mathcal{I}^b$, $\mathcal{I}_\partial^b$ and $x^{-\mathbb{Z}}\Psi_b^{-\infty}$. Part III assembles the remaining algebras and treats the index functional. Its last section proves the general index formula for fully elliptic operators of positive order.

Part I
The Laurent symbol algebra

2 The symbol algebra $\mathcal{A}_\sigma^b$

The phrase *the algebra generated by b-pseudodifferential operators* is ambiguous. At least four choices affect Hochschild homology: the allowed operator orders, the allowed powers of a boundary defining function, the residual ideal, and the locally convex completion. We use the standard algebra for which the homology has a finite-dimensional geometric description. It is the b-groupoid specialization of the Laurent symbol algebras studied in [2, 1].

Status of the result. Theorem 2.1 is the b-groupoid specialization of the Benameur–Nistor calculation [2, 1]. We give the scalar calculation in the chain model of Section 16.1. The global HKR proof is in Lemma 5.2. The fixed coefficient stages in Section 20.1 give convergence.

For one boundary hypersurface, the standard b-groupoid has underlying set

$$\mathcal{G}_b = X^\circ \times X^\circ \sqcup \bigsqcup_{H\in\pi_0(\partial X)} H \times H \times \mathbb{R}.$$

On a boundary component, composition adds the last coordinate. Its smooth structure is fixed by the normal ratio $t = \log(x/x')$, and its Lie algebroid is $A(\mathcal{G}_b) = {}^bTX$. For corners, one uses the standard groupoid with an $\mathbb{R}^j$-factor over each open codimension-j face [1].

Let $\mathcal{O}(X)$ be the ring generated by $C^\infty(X)$ and x_H^{-1} for all boundary hypersurfaces H. Benameur and Nistor define

$$\mathcal{A}_{b,L}(X;E) := \mathcal{O}(X)\left(\Psi^\infty(\mathcal{G}_b;E)/\Psi^{-\infty}(\mathcal{G}_b;E)\right)$$

with their topologically filtered topology and completed Hochschild complex [2, 1]. Here $\Psi^\infty(\mathcal{G}_b;E)$ denotes the $\mathcal{G}_b$-pseudodifferential operators acting on sections of E (their kernels are distributional sections over $\mathcal{G}_b$ of the pull-backs of E and E^* by the range and source maps); we write E throughout. The vector representation identifies this with the usual properly supported Laurent b-complete symbols. In particular, one may write its underlying filtered union heuristically as

$$\frac{\bigcup_{m,\ell} x^{-\ell}\Psi_b^m(X;E)}{\bigcup_\ell x^{-\ell}\Psi_b^{-\infty}(X;E)},$$

but this quotient notation does not by itself specify the topology used in the theorem.

Changing the defining function from x to $x' = e^f x$ gives

$$(x')^{-\ell} = e^{-\ell f} x^{-\ell}.$$

Hence the change only multiplies a Laurent generator by a smooth invertible function, and the algebra, its filtration, and the answer below are canonically independent of this choice. For

corners one allows an arbitrary finite Laurent monomial $\prod_H x_H^{-\ell_H}$, one factor for each boundary hypersurface.

Set $Y = {}^bS^*X = ({}^bT^*X \setminus 0)/\mathbb{R}_+$ and $Z = Y|_{\partial X}$. Thus $\partial Y = Z$. The symbol-order circle is denoted S^1_σ.

Theorem 2.1 (Hochschild homology; b-groupoid case of Benameur–Nistor)**.** *For every $k \geq 0$ the induced filtration of $\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E))$ has exactly one nonzero graded piece, and there is a canonical isomorphism*

$$\boxed{\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}(Y \times S^1_\sigma;\mathbb{C}).} \tag{1}$$

The answer is independent of E. If X has one embedded boundary hypersurface, then

$$\begin{aligned}\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong\ & H^{2n-k}(Y) \oplus H^{2n-k-1}(Y) \\ & \oplus H^{2n-k-1}(Z) \oplus H^{2n-k-2}(Z),\end{aligned} \tag{2}$$

where a cohomology group in negative degree is zero. In particular the homology vanishes for $k > 2n$.

The isomorphism (1) is canonical, and the Künneth decomposition for S^1_σ is canonical as well. The four-summand formula (2) uses one choice: a splitting of the Laurent face sequence (a collar, defining function, and cutoff; Proposition 8.1). The subspace $H^{2n-k}(Y \times S^1_\sigma)$ and the boundary quotient $H^{2n-k-1}(Z \times S^1_\sigma)$ are canonical.

Remark 2.2 (The algebraic Laurent quotient). The underlying filtered algebra of $\mathcal{A}_{b,L}$ is $\mathcal{A}^b_\sigma = x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X)/x^{-\mathbb{Z}}\Psi_b^{-\infty}(X)$, the quotient of the Introduction. Throughout, $\mathcal{A}^b_\sigma$ carries the Benameur–Nistor topology, in which the projective completion in decreasing order is taken at fixed pole order and the inductive limit is taken afterwards (3). Accordingly $\mathrm{HH}_*(\mathcal{A}^b_\sigma)$ means $\mathrm{HH}_*^{\mathrm{cont}}(\mathcal{A}_{b,L})$. Theorem 2.1 has the same form as the cusp result [21, Prop. 2], $\mathrm{HH}_*(\mathcal{A}_\sigma) \cong {}^bH^{2n-*}({}^cS^*X \times S_\sigma)$; this is a comparison, and the present proof does not use it. Here S^1_σ is the circle S_σ of Theorem 1.1.

$$\begin{array}{ccccc}
\mathcal{A}_{b,L} & \xrightarrow{\text{order filtr.}} & \operatorname{gr}\mathcal{A}_{b,L} & \xrightarrow{\text{HKR } (E^1)} & \Omega^*_L({}^bT^*X \setminus 0)_{\mathrm{hom}} \\
 & & & & \Big\downarrow{\scriptstyle d^1 = \delta_b} \\
\mathrm{HH}_*^{\mathrm{cont}}(\mathcal{A}_{b,L}) & \xleftarrow[\text{(convergence)}]{E^2 = E^\infty} & H_L^{2n-*}(Y \times S^1_\sigma) & \xleftarrow[\text{(duality)}]{*_b,\ \text{weight } 0} & H_*^{\pi_b,L}({}^bT^*X \setminus 0)_{\mathrm{hom}}
\end{array}$$

Figure 2: The computation passes from the filtered noncommutative algebra to its commutative associated graded (E^1), then from Poisson homology to Laurent de Rham cohomology (E^2). Only the weight-zero part survives, which forces $E^2 = E^\infty$. The fixed coefficient stages give convergence.

3 Continuous chains and the two filtrations

The topology is part of the theorem. For a topologically filtered algebra A, Benameur–Nistor do *not* take the completed projective tensor power of the whole locally convex space: multiplication need not be jointly continuous for that topology. In their notation the degree-k chain space is

$$C_k^{\mathrm{cont}}(A) := \varinjlim_N \big(F^N_{N,N}A\big)^{\widehat{\otimes}_\pi(k+1)}. \tag{3}$$

(Benameur–Nistor denote this chain space by $H_k(A)$; we avoid that notation, which suggests homology.) Here $F^N_{N,N}A$ is the diagonal cofinal family of filtration pieces of [1]; equivalently, one first takes the projective completion in the decreasing symbol-order direction and only then enlarges the support and Laurent bounds by an inductive limit. We apply this with $A = \mathcal{A}_{b,L}$. Whenever a tensor product below carries no decoration it denotes an elementary tensor in the dense filtered subcomplex of (3), not an ordinary completed tensor power of the whole algebra.

For a chain $a_0 \otimes \cdots \otimes a_k$, the Hochschild boundary is

$$b(a_0 \otimes \cdots \otimes a_k) = \sum_{j=0}^{k-1} (-1)^j a_0 \otimes \cdots \otimes a_j a_{j+1} \otimes \cdots \otimes a_k + (-1)^k a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}.$$

Multiplication is continuous on fixed filtration pieces, so b extends to the completed complex and satisfies $b^2 = 0$.

Laurent order and symbol order play different roles. The first is built into $\mathcal{O}(X)$; symbol order produces the spectral sequence. On every fixed piece in (3), filter elementary tensors by

$$\operatorname{ord}(a_0) + \cdots + \operatorname{ord}(a_k) \le p,$$

take the projective closure there, and then pass through the same projective/inductive limits. Denote the resulting subspace by $\mathcal{F}_p C_k$. Then $b(\mathcal{F}_p C_k) \subseteq \mathcal{F}_p C_{k-1}$. The filtration is exhaustive. Section 20.1 proves convergence by summing the decreasing-order corrections at fixed pole and order bounds in a mixed coefficient complex.

Discussion. **Why divide by smoothing operators?** The principal-symbol filtration sees an operator only through its full asymptotic symbol. Smoothing kernels form the kernel of that passage. Keeping them gives an extension problem, treated separately in Section 13; it is not a harmless change of notation.

4 The b-cotangent geometry

The Lie algebra of b-vector fields is

$$\mathcal{V}_b(X) = \{V \in C^\infty(X; TX) : V \text{ is tangent to } \partial X\}.$$

It is the space of smooth sections of a vector bundle ${}^bTX \to X$. Near a boundary point, choose product coordinates

$$(x, y_1, \dots, y_{n-1}).$$

Then

$$\mathcal{V}_b(X) = C^\infty(X)\text{-span}\{x\partial_x, \partial_{y_1}, \dots, \partial_{y_{n-1}}\}.$$

The dual local frame of ${}^bT^*X$ is

$$\frac{dx}{x}, \quad dy_1, \dots, dy_{n-1}.$$

Thus a b-covector has the form

$$\xi \frac{dx}{x} + \eta \cdot dy.$$

On ${}^bT^*X$ the tautological b-one-form and its differential are

$$\alpha_b = \xi \frac{dx}{x} + \eta \cdot dy, \qquad \omega_b = d\alpha_b = d\xi \wedge \frac{dx}{x} + d\eta \wedge dy.$$

The form ω_b is nondegenerate as a b-two-form. Its inverse is the b-Poisson tensor

$$\pi_b = x\partial_x \wedge \partial_\xi + \sum_j \partial_{y_j} \wedge \partial_{\eta_j}.$$

This is the geometric bracket appearing in the first nonzero differential of the Hochschild spectral sequence. We set $\{a, c\}_b = \pi_b(da, dc)$. With this convention $\{\xi, x\}_b = -x$ and $\{\eta_j, y_j\}_b = -1$. The commutator rule below is $\sigma([A, C]) = i\{a, c\}_b$. It is consistent with $[D_{y_j}, y_j] = -i$ and with $[xD_x, x] = -ix$ of (22), because xD_x has b-symbol ξ.

Proposition 4.1 (Anchor bridge). *(i) The anchor $a : {}^bTX \to TX$ becomes an isomorphism of Lie algebroids after adjoining Laurent coefficients:*

$$\mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma({}^bTX) \xrightarrow{a} \mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma(TX),$$

an isomorphism of $\mathcal{O}(X)$-modules preserving brackets.

(ii) There is a (noncanonical) isomorphism of smooth vector bundles ${}^bTX \cong TX$. Consequently $Y = {}^bS^*X \cong S^*X$ as sphere bundles over X, by a diffeomorphism carrying Z onto $S^*X|_{\partial X}$, and

$$H^*_L({}^bS^*X \times S^1_\sigma) \cong H^*_L(S^*X \times S^1_\sigma).$$

The isomorphism in (ii) is not induced by the anchor: the dual anchor $T^*X \to {}^bT^*X$ does not identify the homogeneous symbol spaces.

Proof. (i) Locally the anchor sends $x\partial_x$ to the ordinary vector field $x\partial_x$ and fixes ∂_{y_j}. Its localized inverse uses the Laurent coefficient x^{-1}: $\partial_x \mapsto x^{-1}(x\partial_x)$. The anchor of a Lie algebroid preserves brackets, hence so does its Laurent localization.

(ii) In a collar $[0, \varepsilon)_x \times \partial X$ choose a smooth function $h(x) > 0$ with $h = 1$ near $x = 0$ and $h(x) = x$ for $x \geq \varepsilon/2$. Define $\Phi : {}^bTX \to TX$ by $\Phi(x\partial_x) = h(x)\partial_x$, $\Phi(\partial_{y_j}) = \partial_{y_j}$ in the collar, and $\Phi = a$ outside it. Then Φ is smooth and fiberwise invertible, since $h > 0$ and a is invertible over X°. Its inverse transpose induces a diffeomorphism of cosphere bundles ${}^bS^*X \to S^*X$ over the identity of X, hence carries the restriction to ∂X onto the restriction to ∂X. A boundary defining function of S^*X pulls back to one of Y, so the diffeomorphism identifies the Laurent de Rham complexes.

Why not more: in the coordinates above the dual anchor is $\xi\, dx + \eta \cdot dy \mapsto (x\xi)\, dx/x + \eta \cdot dy$, so it kills the conormal direction at ∂X and does not map $T^*X \setminus 0$ into ${}^bT^*X \setminus 0$. For instance, the homogeneous function $\xi_b/|\zeta_b|$ pulls back to $x\xi/\sqrt{x^2\xi^2 + |\eta|^2}$, which is not Laurent-smooth near $x = \eta = 0$. So no chain-level isomorphism of homogeneous Laurent symbol complexes is induced by the anchor. □

5 The associated graded algebra

Filter $\mathcal{A}_{b,L}$ by operator order:

$$F_m\mathcal{A}_{b,L} = \text{image}\left(\bigcup_{\ell \geq 0} x^{-\ell}\Psi^m_b(X; E)\right), \qquad m \in \mathbb{Z}.$$

The filtration is exhaustive, separated in the complete-symbol sense, and multiplicative. Principal symbols give

$$\text{gr}_m\, \mathcal{A}_{b,L} \cong \mathcal{O}_L({}^bT^*X \setminus 0; \text{End}\, E)_m,$$

the Laurent functions that are homogeneous of degree m in the fibers. In a b-Darboux chart, a choice of local Weyl quantization gives the expansion

$$a\#c \sim ac + \frac{i}{2}\{a, c\}_b + \text{terms two orders lower}, \tag{4}$$

where changing quantization changes the individual lower-order symmetric terms. Without a chosen global symplectic connection, the full Weyl formula is not coordinate invariant. What is invariant, and all that is used below, is the commutator symbol

$$\sigma_{m+m'-1}([A, C]) = i\{a, c\}_b.$$

This one-order drop is the origin of the first nonzero differential.

Lemma 5.1 (Morita reduction)**.** *The coefficient bundle E does not change the Hochschild homology.*

Proof. The global module of complete symbols with coefficients in E is a finitely generated projective equivalence bimodule between the scalar algebra and its End E-coefficient algebra. The associated matrix-trace and corner maps are continuous and filtration preserving, and the standard matrix-unit homotopy proves Morita invariance. This is a global Morita equivalence; no partition of unity is used to glue local chain homotopies. The topologically filtered version, including graded bundles, is treated in [2, 1]. □

We therefore suppress E. The Hochschild–Kostant–Rosenberg map on the associated graded algebra is

$$\chi(a_0 \otimes \cdots \otimes a_j) = \frac{1}{j!} a_0\, da_1 \wedge \cdots \wedge da_j.$$

For $w \in \mathbb{Z}$ let $\Omega^j_L({}^bT^*X \setminus 0)_{(w)}$ be the Laurent j-forms of fiber weight w, that is, with $\mathcal{L}_{\mathcal{E}}\alpha = w\alpha$ for the radial (Euler) vector field $\mathcal{E}$, and put $\Omega^j_L({}^bT^*X \setminus 0)_{\mathrm{hom}} = \bigoplus_w \Omega^j_L({}^bT^*X \setminus 0)_{(w)}$. If a_i is homogeneous of degree m_i, then $a_0\, da_1 \wedge \cdots \wedge da_j$ has weight $m_0 + \cdots + m_j$, since d commutes with dilation.

Lemma 5.2 (Continuous HKR for the coefficient kernels)**.** *Let V be a compact smooth manifold with boundary W, with a boundary defining function x and a collar. Consider any of*

$$C^\infty(V), \qquad \dot{C}^\infty(V), \qquad C^\infty(V)[x^{-1}], \qquad C^\infty(W)[[x]], \qquad C^\infty(W)[[x]][x^{-1}].$$

Adjoin a polynomial variable t of weight -1, or a Laurent polynomial variable r of weight 1. Use the chains with completed smooth coefficient tensors, products for the Taylor coefficients, and a finite bound on the poles in each slot in the localized cases. At each fixed total weight take the direct sum over slot weights. Then

$$\chi_k(a_0 \otimes \cdots \otimes a_k) = \frac{1}{k!} a_0\, da_1 \wedge \cdots \wedge da_k$$

identifies the chain homology with the forms with the indicated coefficient behaviour and total weight, where dt has weight -1 and dr has weight 1. In the formal power-series case the normal one-form is dx, with formal differentiation in x. The maps commute with Taylor restriction in the chosen collar. For a free rank-one coefficient bimodule with both actions given by multiplication, placing its generator in a prescribed weight shifts both the chain complex and this answer by that weight.

Proof. Write $M = V$ for smooth coefficients and $M = W$ for formal coefficients. By the smooth kernel theorem the coefficient tensors are kernels on products of M. For flat coefficients they are kernels flat at the boundary in each slot: apply nuclear tensor exactness successively to the kernel of the Taylor map $C^\infty(V) \to C^\infty(W)[[x]]$, using Borel's lemma for its surjectivity. For formal coefficients the Taylor variables in the slots are independent, since the completed tensor product commutes with the countable products of coefficients. For Laurent coefficients one multiplies such kernels by finite negative powers in the individual slots. These identifications apply with two endpoint variables and any finite number of middle variables.

Use the augmented bar complex with two endpoints. On an open set $U \subset M^2$ of endpoint pairs, write $P_q(U)$ for kernels with the two endpoints in U and the q middle variables in M^q. There is a polynomial or Laurent polynomial variable in each slot and, in the formal cases, a Taylor variable in each slot. The bar differential is the alternating sum of the $q+1$ restrictions that identify adjacent slots, including their polynomial and Taylor variables. The augmentation from P_0 identifies the endpoints. We use the sheaves of these kernels, allowing bounds on the finite sets of slot weights and on poles to hold locally in the endpoints. A finite cover of the compact base supplies global bounds for global sections. The augmented bar complex is exact on stalks. Indeed, multiply a germ cycle by a smooth cutoff of the endpoints supported where it is a cycle, extend by zero, and contract the resulting global bar cycle. Endpoint multiplication commutes with the bar faces since they retain the endpoints. In the unital cases the global contraction is insertion of 1 in the first slot, including the augmentation degree.

For flat kernels, take $\kappa \in C^\infty([0,\infty))$ with $\kappa = 0$ on $[0,1/2]$ and $\kappa = 1$ on $[3/4,\infty)$. Replace insertion of 1 by

$$(hF)(z_0, z_1, \ldots) = \kappa\big(x(z_0)/x(z_1)\big)F(z_1, \ldots),$$

with the new polynomial or Laurent polynomial slot of weight zero. The formula is extended to the boundary by smoothness. On its support $x(z_1) \le 2x(z_0)$, and derivatives of the cutoff cost at most powers of $x(z_1)^{-1}$. Flatness of F in z_1 absorbs these powers and any prescribed powers of $x(z_0)^{-1}$, while flatness in the other slots is preserved. Taylor's integral formula bounds the resulting weighted seminorms by higher smooth seminorms of F, so the map of flat kernels is continuous. The first face of hF is F, since $\kappa(1) = 1$, and every subsequent face is h applied to the preceding face of F. Thus $b'h + hb' = 1$ for the augmented bar differential b'.

Near an endpoint pair on the diagonal, choose a product of two copies of a coordinate box, using a half-box with normal coordinate x at the boundary of V. The endpoint ring R is the term in bar degree zero with its two independent slot variables, using smooth coefficients in the flat case. Let K be the Koszul complex over R for the differences of the endpoint coordinates. The coordinates include the independent Taylor variable x in the formal cases and the adjoined variable t or r in all cases. Assign a Koszul generator the weight of the coordinate defining it. For flat coefficients set $K' = R_{\mathrm{f}} \otimes_R K$, with R_{f} the ideal of kernels flat in both endpoints. Set $K' = K$ in the other cases.

For P_q, R, and R_{f}, the Koszul complex with these coefficients has zero positive homology and degree-zero homology given by identifying the endpoints. Here are the division details. Impose the coordinate equalities one at a time. In a smooth coordinate the difference is injective as a multiplier on the previous restriction, by density of the complement of its zero set. A kernel vanishing at that equality is divisible by the difference using integration of the coordinate derivative along the segment between the two coordinate values. Substitution of one coordinate for the other gives a right inverse to restriction. The other variables are parameters. For flat kernels in a half-box do the tangential equalities before the normal equality. In the normal coordinate the quotient of a vanishing kernel is smooth by the same integral formula. Along the open part of either individual boundary face the normal difference is nonzero, so the quotient is flat there by ordinary division.

Continuity of all its derivatives gives flatness up to the corner. A flat kernel G on the normal diagonal extends to a separately flat kernel by $\kappa(x_R/x_L)G(x_L,\dots)$, with L, R the endpoints, by the ratio estimate above. In a formal normal variable the replacement $x_R = x_L + u$ is an invertible change of variables of formal series. Multiplication by u is injective, with cokernel restriction to $u = 0$. For the polynomial difference, division by $t_R - t_L$ gives the same two conclusions, and for the Laurent polynomial difference one first clears negative powers. These polynomial operations use finite sums. Localizing the smooth or formal calculation at the boundary variables in the slots preserves exactness, by clearing a finite common denominator. After endpoint identification the two endpoint denominators use the same boundary variable. These divisions and restrictions respect the smooth seminorms, Taylor product topology and weight decomposition. The Koszul assertion follows by induction, viewing the addition of each generator as the mapping cone of multiplication by its difference on the preceding Koszul complex.

Extend each coordinate of the box to a global function f_i by a cutoff equal to one on a smaller box. The Taylor and polynomial coordinates are already global. There is a local endpoint-linear chain map $\Phi : K' \to P_\bullet$, given by

$$\Phi_q\big(a\, e_{i_1} \wedge \cdots \wedge e_{i_q}\big) = a \sum_{\sigma \in S_q} \operatorname{sgn}(\sigma) \prod_{j=1}^{q} f_{i_{\sigma(j)}}(z_j),$$

where $z_1, \dots, z_q$ are the middle variables. In the flat case multiply the right side by

$$W_q(z_0, \dots, z_{q+1}) = \prod_{j=0}^{q} \kappa\big(x(z_{j+1})/x(z_j)\big),$$

with endpoints z_0, z_{q+1}. This product restricts on each bar face to W_{q-1}. On its support $x(z_0) \le 2^{q+1} x(z_j)$ for every j; derivatives of active cutoff factors cost at most powers of $x(z_0)^{-1}$. The endpoint flatness of a in z_0 therefore makes the whole expression separately flat. The middle bar faces of the antisymmetrization cancel in pairs, and the end faces give the Koszul differential with differences $f_i(L) - f_i(R)$. Both maps augment to the identity on the diagonal (in the flat case $W_0 = 1$ there). Exactness of the two augmented complexes makes Φ a quasi-isomorphism locally.

To justify passing to the diagonal, compare the total complexes of $K' \otimes_R K$ and $P_\bullet \otimes_R K$. In the first direction $\Phi \otimes 1$ is a quasi-isomorphism since the terms of K are finite free. In the other direction the Koszul calculation with coefficients in each term shows that the map on homology is the restriction of Φ obtained by identifying endpoints. Both comparisons use the first-quadrant double complexes with finite Koszul direction. After endpoint identification the Koszul differential of K' is zero. The bar differential becomes the Hochschild differential, including the cyclic face. The image of $a\, e_{i_1} \wedge \cdots \wedge e_{i_q}$ under $\chi_q \Phi_q$ is $a|_\Delta\, df_{i_1} \wedge \cdots \wedge df_{i_q}$. In the flat case the cutoff equals one near the full interior diagonal, so it has no effect on this calculation, and the resulting flat forms extend to the boundary. The chain identity $\chi b = 0$ follows by the Leibniz rule, first on elementary tensors and then on kernels by continuity. Thus χ induces the stated local isomorphism.

After endpoint identification the base is M, the distinguished first variable of the Hochschild chains. Its smooth functions act on that variable and commute with b and χ. A global cycle with zero HKR form has local boundary lifts by the local calculation. A finite partition of unity in the distinguished variable, supported in the sets where the lifts are defined, extends the lifts by zero and sums them to a global boundary lift. The same construction glues local antisymmetric cycle lifts of a form. The finite gluing preserves a finite pole bound and finite slot-weight support. The local coordinate formula for a cycle lift, with the diagonal extensions used above, is continuous on each fixed pole bound, and so is the finite gluing. The explicit formula for χ commutes with Taylor restriction and with the identification of a shifted coefficient copy with its algebra. □

Proposition 5.3 (The E^1-page)**.** *The HKR map identifies the first page of the order spectral sequence with homogeneous Laurent differential forms on ${}^bT^*X \setminus 0$, the fiber weight being the total symbol order:*

$$E^1_{p,q} \cong \Omega_L^{p+q}({}^bT^*X \setminus 0)_{(p)}.$$

For $\mathcal{I}^b_\sigma$ and $\mathcal{A}^b_{\partial,\sigma}$ the coefficients are flat and formal Laurent, respectively.

Proof. A fibre norm identifies the homogeneous functions of order m with r^m times functions on Y. A graded chain at order p has finitely many radial multidegrees $m_0 + \cdots + m_k = p$, since in a fixed stage each m_j has a fixed upper bound. Every finite set of such multidegrees occurs at a sufficiently large stage. The three coefficient spaces are the smooth Laurent functions on Y, the flat functions on Y, and the formal Laurent functions over Z. In the formal case use a norm constant in the collar coordinate. The associated graded product is multiplication in all three cases, so Lemma 5.2 applies with r. Its differential forms agree with the stated Laurent, flat and formal Laurent forms, respectively. In particular the identification commutes with taking the boundary Taylor series. □

6 The first differential is Poisson homology

For classical b-symbols a and c, symbolic composition gives

$$\sigma_{m+m'-1}([A, C]) = i\{a, c\}_b.$$

Consequently the first differential after HKR is, up to the harmless factor i, the Poisson differential

$$\delta_b = [\iota_{\pi_b}, d] = \iota_{\pi_b} d - d\iota_{\pi_b}.$$

Explicitly,

$$\begin{aligned}\delta_b(a_0\, da_1 \wedge \cdots \wedge da_j) = &\sum_{r=1}^{j} (-1)^{r+1} \{a_0, a_r\}_b\, da_1 \wedge \cdots \widehat{da_r} \cdots \wedge da_j \\ &+ \sum_{1 \le r < s \le j} (-1)^{r+s} a_0\, d\{a_r, a_s\}_b \wedge da_1 \wedge \cdots \widehat{da_r} \cdots \widehat{da_s} \cdots \wedge da_j. \end{aligned} \tag{5}$$

To see this differential directly, insert the order-one-lower part of (4) into each multiplication in the Hochschild boundary. The symmetric terms cancel after HKR antisymmetrization, while the commutator terms combine to (5).

Let $N = 2n$ be the dimension of ${}^bT^*X$, put $\mu_b = \omega_b^n/n!$, and let $\pi_b^\sharp\alpha$ be defined on covectors by $\beta(\pi_b^\sharp\alpha) = \pi_b(\beta, \alpha)$, and extend by exterior powers. For multivectors we use $\iota_{U\wedge V} = \iota_V \iota_U$. Symplectic duality is the map

$$*_b : \Omega_L^j({}^bT^*X \setminus 0) \longrightarrow \Omega_L^{N-j}({}^bT^*X \setminus 0), \qquad *_b\alpha = \iota_{\pi_b^\sharp\alpha}\mu_b,$$

and $*_b^2 = 1$ up to the standard degree sign. Applying the two sides to a monomial in the Darboux b-frame

$$\frac{dx}{x}, dy_1, \ldots, dy_{n-1}, d\xi, d\eta_1, \ldots, d\eta_{n-1}$$

gives, with the displayed convention for δ_b,

$$*_b\delta_b = (-1)^{j+1} d *_b \quad \text{on } \Omega_L^j. \tag{6}$$

The calculation remains valid at $x = 0$: the displayed frame is smooth as a b-frame, and both operators preserve finite Laurent pole order. Therefore a δ_b-cycle of degree j is carried to a closed Laurent form of degree $2n - j$, while boundaries are carried to exact forms.

The operator $*_b$ also shifts fiber weight. Since α_b and hence ω_b have weight 1, μ_b has weight n and π_b has weight -1. Contracting a j-form of weight w with π_b in each slot gives a j-vector of weight $w - j$, and contracting this into μ_b gives

$$*_b : \Omega_L^j({}^bT^*X \setminus 0)_{(w)} \xrightarrow{\cong} \Omega_L^{2n-j}({}^bT^*X \setminus 0)_{(w+n-j)}. \tag{7}$$

In particular δ_b, which lowers both j and w by one, preserves $w - j$, consistent with d preserving weight.

Corollary 6.1. *On the homogeneous complex of the E^1-page, Poisson homology is Laurent de Rham cohomology with complementary degree and shifted weight:*

$$H_j^{\pi_b,L}({}^bT^*X \setminus 0)_{(w)} \cong H_L^{2n-j}({}^bT^*X \setminus 0)_{(w+n-j)}, \qquad H_j^{\pi_b,L}({}^bT^*X \setminus 0)_{\mathrm{hom}} \cong H_L^{2n-j}({}^bT^*X \setminus 0)_{\mathrm{hom}}.$$

7 Homogeneity and the symbol-order circle

Choose a positive fiber norm r on ${}^bT^*X \setminus 0$. Then

$${}^bT^*X \setminus 0 \cong Y \times \mathbb{R}_+, \qquad (q, r) \longmapsto rq.$$

By Proposition 5.3 and (7), the column $E^1_{p,\bullet}$ in Hochschild degree k corresponds under $*_b$ to $(2n-k)$-forms of weight $p + n - k$. This bookkeeping matters: δ_b lowers fiber homogeneity by one, whereas $*_b$ changes both form degree and homogeneity.

The zero-weight reduction is elementary. The Euler vector field $\mathcal{E} = r\partial_r$ is dilation invariant, so d and $\iota_{\mathcal{E}}$ preserve weight. If a form α has nonzero weight w, Cartan's formula gives

$$(d\iota_{\mathcal{E}} + \iota_{\mathcal{E}} d)\alpha = \mathcal{L}_{\mathcal{E}}\alpha = w\alpha.$$

Thus $w^{-1}\iota_{\mathcal{E}}$ is a contracting homotopy of the weight-w subcomplex, and only weight 0 contributes to $H_L^*({}^bT^*X \setminus 0)_{\mathrm{hom}}$. By (7), in Hochschild degree k the only surviving column is $p = k - n$, that is, $q = n$ (Section 9).

At weight zero, restriction to $r = 1$ gives a unique splitting

$$\alpha(r, q) = \alpha_0(q) + \frac{dr}{r} \wedge \alpha_1(q).$$

Since $d(dr/r) = 0$, its differential is

$$d\alpha = d_Y\alpha_0 - \frac{dr}{r} \wedge d_Y\alpha_1.$$

Thus the weight-zero complex is the direct sum of the de Rham complex of Y and a copy shifted by one degree. The complete-symbol order is an integer grading with no preferred logarithm; cohomologically this two-term complex is canonically written as the de Rham complex of an auxiliary circle:

$$\alpha_0 + \frac{dr}{r} \wedge \alpha_1 \longleftrightarrow \alpha_0 + d\theta_\sigma \wedge \alpha_1.$$

The formal order variable dr/r is represented cohomologically by the generator $d\theta_\sigma$ of a circle. Hence

$$H_L^j({}^bT^*X \setminus 0)_{\mathrm{hom}=0} \cong H_L^j(Y \times S^1_\sigma). \tag{8}$$

This is the source of the extra S^1 in the answer; it is not a geometric circle in the fibers of ${}^bT^*X$, and no choice of fiber norm affects its cohomology class.

8 Laurent cohomology and the boundary face

Let V be a compact manifold with one embedded boundary hypersurface and defining function x. The Laurent de Rham complex is

$$\Omega_L^j(V) = \bigcup_{\ell \geq 0} x^{-\ell}\Omega^j(V), \qquad d : \Omega_L^j(V) \to \Omega_L^{j+1}(V).$$

Proposition 8.1 (Face decomposition)**.** *There is a split short exact sequence*

$$0 \to H^j(V) \to H_L^j(V) \xrightarrow{\text{Res}} H^{j-1}(\partial V) \to 0,$$

hence an isomorphism $H_L^j(V) \cong H^j(V) \oplus H^{j-1}(\partial V)$. A splitting sends $[\beta]$ to the class of $(-1)^{j-1}\widetilde{\beta} \wedge d(\chi \log x)$, where β is a closed $(j-1)$-form on ∂V, $\widetilde{\beta}$ is a collar extension, and χ is a cutoff supported in the collar and equal to one near the boundary.

Proof. Work in a collar and write a Laurent form uniquely as

$$\alpha(x) + \frac{dx}{x} \wedge \beta(x),$$

where the tangential forms $\alpha(x), \beta(x)$ are Laurent families with a finite principal part and a smooth Taylor tail. Filter the collar complex by pole order and then by normal jet. On its associated graded, a term has a definite normal weight q. For $q \neq 0$, Cartan's identity gives the contracting homotopy

$$h_q = q^{-1}\iota_{x\partial_x}, \qquad dh_q + h_q d = 1.$$

Induction over the finite negative weights removes the principal part; the remaining positive Taylor tail is smooth and belongs to the ordinary de Rham complex. This jet-filtration argument, rather than freezing all coefficients at $x = 0$, is the Laurent Poincaré lemma. To globalize, multiply the collar homotopies by a cutoff equal to one near ∂V; the error terms contain a derivative of the cutoff, are supported away from ∂V, and are therefore smooth. The only new normal class is the weight-zero generator dx/x.

In particular, after writing

$$\alpha = \alpha_0 + \frac{dx}{x} \wedge \beta_0 + \text{terms of nonzero normal weight},$$

the residue is

$$\text{Res}_{\partial V} : \Omega_L^j(V) \longrightarrow \Omega^{j-1}(\partial V), \qquad \text{Res}_{\partial V}(\alpha) = \beta_0|_{\partial V}.$$

A direct calculation in the displayed order gives

$$\text{Res}(d\alpha) = -d_{\partial V}\,\text{Res}(\alpha).$$

Thus residue is a chain map to the degree-shifted boundary complex. The Laurent Poincaré lemma identifies its homotopy kernel with $\Omega^*(V)$, yielding the quasi-isomorphism

$$\Omega_L^*(V) \simeq \Omega^*(V) \oplus \Omega^{*-1}(\partial V).$$

If β is closed of degree $j-1$, let $p : [0,\varepsilon)_x \times \partial V \to \partial V$ be the collar projection. Choose χ with support in this collar and choose $\widetilde{\beta} = p^*\beta$ on an open set containing $\operatorname{supp} d(\chi \log x)$. Extend it

arbitrarily beyond that set. Then $d\widetilde{\beta} = 0$ everywhere that $d(\chi \log x)$ is nonzero, so the representative below is closed. This collar construction gives the cohomological splitting

$$\beta \longmapsto (-1)^{j-1}\widetilde{\beta} \wedge d(\chi \log x).$$

Its residue is β, because $\widetilde{\beta} \wedge dx/x = (-1)^{j-1}(dx/x) \wedge \widetilde{\beta}$ near ∂V. If $x' = e^f x$, then $d \log x' = d \log x + df$; changing the defining function, collar, or cutoff changes this splitting by an ordinary/exact correction. Hence the direct sum is natural as a filtered extension but its explicit vector-space splitting is not canonical. □

Discussion. **Where do higher poles go?** A pole such as $x^{-q-1}dx$, $q > 0$, looks more singular than dx/x, but it is $-q^{-1}d(x^{-q})$. Higher Laurent coefficients occur in the complex and are essential for closure under composition; they add no new normal cohomology. Only the logarithmic coefficient survives.

Apply Proposition 8.1 to $V = Y \times S^1_\sigma$, whose boundary is $Z \times S^1_\sigma$. Then use

$$H^r(W \times S^1) \cong H^r(W) \oplus H^{r-1}(W).$$

For $r = 2n - k$ this gives exactly the four summands in (2).

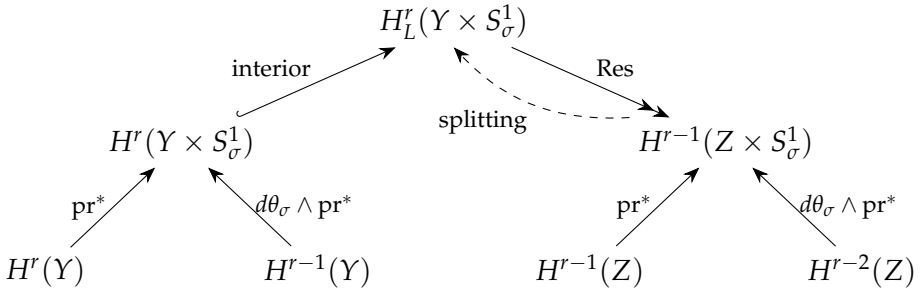


Figure 3: The four summands of (2), with $r = 2n - k$. The top row is the split exact sequence of Proposition 8.1 for $V = Y \times S^1_\sigma$; only the dashed splitting depends on choices. The bottom row is the Künneth decomposition for S^1_σ. The four summands come from the ordinary class, the symbol-order class $d\theta_\sigma \leftrightarrow d \log r$, the boundary logarithm $d \log x$, and their product.

For a manifold with corners, the same local argument has one generator $d \log x_H$ for every boundary hypersurface H through a point. If $\mathcal{F}_j(V)$ denotes the codimension-j faces, then

$$H^r_L(V) \cong \bigoplus_{j \geq 0} \bigoplus_{F \in \mathcal{F}_j(V)} H^{r-j}(F; o(N_F)). \tag{9}$$

Here $o(N_F)$ is the orientation line of the conormal directions to F. If the hypersurfaces have been globally cooriented and ordered, these lines are trivialized and the untwisted formula results. Indeed, apply the one-face residue successively. Residues for distinct hypersurfaces anticommute with the graded sign, so the summand attached to $F = H_1 \cap \cdots \cap H_j$ is represented by

$$\widetilde{\gamma} \wedge d \log x_{H_1} \wedge \cdots \wedge d \log x_{H_j}, \qquad [\gamma] \in H^{r-j}(F).$$

The product of the nonzero-weight contractions gives an acyclic complement. Changing the order of the hypersurfaces changes the orientation sign of the exterior logarithmic factors; the local

system $o(N_F)$ records exactly this ambiguity. Iterated collar extensions provide a noncanonical splitting. This is only a sketch of (9): the Laurent Poincaré lemma above, applied one hypersurface at a time, reduces $\Omega_L^*(V)$ to the logarithmic (b-) de Rham complex, and (9) is then the standard b-de Rham theorem for manifolds with corners (for one boundary hypersurface see [20]; the corner case follows by iterating it over the boundary faces). The corner formula is not used elsewhere in these notes.

9 Collapse and assembly of the computation

Combining HKR, the Poisson differential, symplectic duality, and homogeneity gives the second page. Its support in a single row gives degeneration. The convergence argument is in Section 20.1.

Proposition 9.1 (The E^2-page and degeneration)**.** *For the order spectral sequence of* $C^{\mathrm{cont}}_\bullet(\mathcal{A}_{b,L})$*,*

$$E^2_{p,q} \cong \begin{cases} H_L^{2n-p-q}(Y \times S^1_\sigma), & q = n, \\ 0, & q \neq n. \end{cases}$$

Consequently $d^r = 0$ *for all* $r \geq 2$ *and* $E^2 = E^r$ *for every* $r \geq 2$*.*

Proof. By Proposition 5.3, $E^1_{p,q}$ is the space of Laurent k-forms of weight p, $k = p + q$, and $d^1 = i\delta_b$ by Section 6. By (6) and (7), $*_b$ identifies $E^2_{p,q}$ with the cohomology of the Laurent de Rham complex in degree $2n - k$ and weight $p + n - k = n - q$. By Section 7, nonzero weights are acyclic, so $E^2_{p,q} = 0$ unless $q = n$; for $q = n$ the weight is zero and (8) gives $H_L^{2n-k}(Y \times S^1_\sigma)$. The differential d^r maps $E^r_{p,q}$ to $E^r_{p-r,q+r-1}$. For $r \geq 2$ the rows q and $q + r - 1$ are distinct, so at least one of source and target vanishes, and $d^r = 0$. □

The bidegree argument gives degeneration. Convergence to homology is proved in Section 20.1.

Proposition 9.2 (Source dictionary)**.** *The algebra* $\mathcal{A}_{b,L}(X; E)$ *is the one-face b-groupoid specialization of the algebras treated by Benameur–Nistor.*

Proof. The dictionary has six entries.

Groupoid. The differential groupoid is $\mathcal{G}_b$, with unit space X and Lie algebroid ${}^bTX$.

Laurent multipliers. The coefficient ring is precisely $\mathcal{O}(X) = C^\infty(X)[x_H^{-1}]$.

Algebra and topology. The algebra is $\mathcal{O}(X)(\Psi^\infty(\mathcal{G}_b; E)/\Psi^{-\infty}(\mathcal{G}_b; E))$, equipped with the topologically filtered completion of [1]; its continuous Hochschild complex is (3).

Anchor condition. The relevant property of the Lie algebroid is that its anchor becomes an isomorphism of Lie algebroids once the boundary defining functions are inverted. For ${}^bTX$ this is Proposition 4.1(i). We do not use, and do not claim, any chain-level identification of b-symbols with ordinary symbols through the anchor; see the end of the proof of Proposition 4.1.

Local models and coefficients. Locally near ∂X the groupoid $\mathcal{G}_b$ is the product of the pair groupoid of the boundary chart with the b-groupoid of $[0, \varepsilon)$. This product structure is the kind of local model their hypotheses require; we have not compared it with their precise formulation beyond this. The global Morita lemma removes the coefficient bundle E.

Spectral sequence. The Laurent HKR identification and the convergent order spectral sequence of [2, 1] give the same cohomology groups as the calculation here. □

Proof of Theorem 2.1. By Proposition 9.1, $E^2_{p,q}$ vanishes unless $q = n$. Section 20.1 proves that every class in degree k has a representative of order at most $k - n$ and identifies the class by its leading HKR form at order $k - n$. Hence

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) = \mathcal{F}_{k-n}\,\mathrm{HH}_k \cong E^2_{k-n,n} \cong H_L^{2n-k}(Y \times S^1_\sigma).$$

The filtration below $k - n$ is zero. The leading-symbol map and the identification (8) are independent of the full-symbol splitting and the fibre norm. Proposition 8.1 and the Künneth formula give (2). □

10 A Čech–de Rham route (conditional)

This optional route uses local HKR and Čech descent to compute the E^2-page. It assumes the chain localization statement of Lemma 10.1. Section 20.1 supplies the abutment.

Choose a finite good cover $\mathcal{U} = \{U_i\}$ of $Y = {}^bS^*X$. Near $Z = \partial Y$, take product sets $[0, \varepsilon)_x \times V_i$, and require every nonempty intersection to be contractible and collar adapted. Write $\widetilde{U}_I = U_I \times \mathbb{R}_+$ for the corresponding conic open set in ${}^bT^*X \setminus 0$.

Complete-symbol composition is local in ${}^bT^*X \setminus 0$, so for a conic open set $\widetilde{U}$ there is an algebra $\mathcal{A}_{b,L}(\widetilde{U})$ of Laurent complete symbols on $\widetilde{U}$, with restriction homomorphisms for $\widetilde{U}' \subset \widetilde{U}$. Let $\mathcal{C}_q(U)$ be its Benameur–Nistor completed degree-q chain space: one forms the diagonal pieces $F^N_{N,N}$ over $\widetilde{U}$ first, takes their completed projective tensor powers, and only then takes the strict inductive limit in N, as in (3). Restriction of every tensor factor makes $U \mapsto \mathcal{C}_q(U)$ a presheaf of filtered complexes.

Form the double complex

$$K^{p,q} = \check{C}^p(\mathcal{U}, \mathcal{C}_q), \qquad D = \delta_{\check{C}} + (-1)^p b. \tag{10}$$

The total degree is $q - p$, since b lowers Hochschild degree and the Čech differential raises p.

Lemma 10.1 (Localization for the optional Čech argument; hypothesis)**.** *The augmentation*

$$C^{\mathrm{cont}}_\bullet(\mathcal{A}_{b,L}) \longrightarrow \mathrm{Tot}\, K^{\bullet,\bullet}$$

is a filtration-preserving quasi-isomorphism, and it remains a quasi-isomorphism after passing to any fixed order quotient.

Remark 10.2 (Status of Lemma 10.1)*.* The augmentation is *not* a quasi-isomorphism row by row: a chain $a_0 \otimes a_1$ whose factors are supported in disjoint open sets restricts to zero on every $\widetilde{U}_i$, although it is a nonzero chain. Such chains are, however, far from the diagonal, and the needed input is that continuous Hochschild chains localize to an arbitrarily small neighbourhood of the diagonal: the subcomplex of $\mathcal{U}$-small chains, all of whose factors are supported in a common $\widetilde{U}_i$, is a filtered deformation retract of the full complex, and on $\mathcal{U}$-small chains the Čech resolution is exact by a partition of unity. For the symbol algebra of a closed manifold, localization of this kind underlies the computation of Brylinski–Getzler [15]; for fine algebras it is Teleman's localization of the Hochschild complex [25]. We assume the corresponding statement for the topologically filtered Laurent algebra, compatibly with the order filtration; we do not prove it here.

Filter (10) by total symbol order. The associated graded of the total complex is the total complex of $\check{C}^p(\mathcal{U}, \mathrm{gr}\,\mathcal{C}_q)$, on which multiplication is commutative. Compute its homology by first taking

Hochschild homology in each Čech degree. The local HKR map and the local Koszul homotopy, which commute with restriction, give

$$\check{C}^p\big(\mathcal{U}, \Omega_L^q({}^bT^*X \setminus 0)_{\mathrm{hom}}\big). \tag{11}$$

This is an intermediate page, not yet E^1 of the total complex: the Čech differential remains. Each weight-w piece of the sheaf of Laurent forms with a fixed pole bound is a module over $C^\infty(Y)$, hence fine, and the union over pole bounds is still Čech acyclic because multiplication by a partition of unity on Y preserves pole bounds and weight. Since the cover is finite, the Čech direction is bounded and (11) has Čech cohomology only in degree $p = 0$, equal to the global forms. Hence

$$E^1(\mathrm{Tot}\, K) \cong \Omega_L^*({}^bT^*X \setminus 0)_{\mathrm{hom}},$$

the same page as in Proposition 5.3. The first order-lowering part of the symbol product gives $d^1 = i\delta_b$, exactly as in (5).

For a closed manifold these are exactly the local filtered steps in the Brylinski–Getzler calculation [15]: complete symbols pass to homogeneous functions, HKR gives forms, and the first quantum correction is the Poisson boundary. The present descent argument adds the Laurent normal-weight contraction, the residue complex at Z, and the strict topologically filtered completion required at the boundary.

Proposition 10.3 (Čech computation of the second page)**.** *Granting Lemma 10.1, the total filtered complex* (10) *has*

$$E^2_{p,q} \cong H_L^{2n-p-q}(Y \times S^1_\sigma; \mathbb{C}) \text{ for } q = n, \qquad E^2_{p,q} = 0 \text{ otherwise.}$$

This identification is independent of the good cover and the partition of unity.

Proof. Symplectic duality (6) converts the Poisson differential to the Laurent de Rham differential with complementary degree, with the weight shift (7). The Euler homotopy removes nonzero fiber weights. At weight zero a form is $\alpha_0 + (dr/r) \wedge \alpha_1$, so (8) identifies the remaining complex with Laurent forms on $Y \times S^1_\sigma$; the row $q = n$ is singled out as in Proposition 9.1. Refining the cover gives a common refinement, and the two refinement maps are homotopic on the augmented Čech resolution. This proves independence of all auxiliary choices. □

The boundary term can also be read directly in the descent complex. One must not write the generally false short exact sequence $0 \to \Omega^*(Y) \to \Omega_L^*(Y) \to \Omega^{*-1}(Z) \to 0$: a Laurent form with zero residue can still have higher poles. Instead let $\mathcal{K}^* = \ker(\mathrm{Res}_Z)$. The nonzero normal-weight contraction in the proof of Proposition 8.1 gives a quasi-isomorphism

$$\Omega^*(Y) \longrightarrow \mathcal{K}^*.$$

The exact sequence

$$0 \longrightarrow \mathcal{K}^* \longrightarrow \Omega_L^*(Y) \xrightarrow{\mathrm{Res}_Z} \big(\Omega^{*-1}(Z), -d_Z\big) \longrightarrow 0$$

therefore yields the derived decomposition

$$R\Gamma(Y; \Omega_L^*) \simeq R\Gamma(Y; \Omega^*) \oplus R\Gamma(Z; \Omega^{*-1}), \tag{12}$$

after choosing a collar splitting. This is precisely the interior/boundary decomposition used in (2); it also shows why the splitting, unlike the residue filtration, is not canonical.

Finally, Lemma 10.1 identifies the abutment of the order filtration on (10) with $\mathrm{HH}_*^{\mathrm{cont}}(\mathcal{A}_{b,L})$, compatibly with filtrations. The proposition identifies its second page, the bidegree argument of Proposition 9.1 gives $E^2 = E^\infty$, and Section 20.1 proves convergence.

11 A Rees–Fedosov reorganization

This section makes the deformation behind symbol composition explicit. It is a heuristic reorganization of the computation, not an independent proof: nothing in it is used in the proof of Theorem 2.1, the E^2-page already follows from Sections 5–8, and the formal Hochschild homology computed below lacks the order circle. It has two logically separate parts: a formal Lie-algebroid formality statement and a finite-order Rees comparison with classical symbols, which is elementary.

This is also the deformation-theoretic refinement of the Brylinski–Getzler filtration: their principal-symbol and Poisson differentials are the first two coefficients of the Rees star product below [15].

11.1 The Rees deformation

The algebraic Rees algebra of the order filtration is

$$\mathcal{R}_F(\mathcal{A}_{b,L}) = \bigoplus_{m\in\mathbb{Z}} t^m F_m \mathcal{A}_{b,L} \subset \mathcal{A}_{b,L}[t,t^{-1}].$$

It records both specialization to the associated graded and the return to complete symbols at $t = 1$. Since $[F_m, F_{m'}] \subset F_{m+m'-1}$, its commutator is divisible by the Rees parameter in an associated-graded chart. Equivalently, for a symbol of fixed upper order $a \sim a_m + a_{m-1} + a_{m-2} + \cdots$, factor off the bookkeeping degree t^m and encode the descending expansion by

$$a_m + \hbar a_{m-1} + \hbar^2 a_{m-2} + \cdots . \tag{13}$$

The power t^m records the absolute order. After it is factored off in a fixed sector, we write the same Rees parameter as $\hbar$; its successive powers record loss of order. Thus (13) is an $\hbar$-adic chart on one upper-order sector, not a global $\mathbb{C}[[\hbar]]$-module structure on the union of all orders.

On Laurent functions on $M = {}^bT^*X \setminus 0$, full symbol composition in this chart becomes

$$a \star c = \sum_{j\geq 0} \hbar^j C_j(a,c), \qquad C_0(a,c) = ac, \tag{14}$$

where C_j is b-bidifferential and lowers the unscaled symbol order by j. Its antisymmetric first term is

$$C_1(a,c) - C_1(c,a) = i\{a,c\}_b,$$

so $[a,c]_\star = i\hbar\{a,c\}_b + O(\hbar^2)$.

The space M has the symplectic Lie algebroid ${}^bTM$ with symplectic form ω_b. Choose a torsion-free b-symplectic connection and use the homogeneous cotangent Fedosov construction, so that positive fiber dilation acts equivariantly. The Fedosov Weyl bundle has a flat differential of the form

$$D_F = -\delta + \nabla + \frac{i}{\hbar}\operatorname{ad}(r_F), \qquad D_F^2 = 0.$$

Flat sections, transported to Laurent functions by the Fedosov–Taylor map, should give a star product equivalent to (14). Fedosov quantization of symplectic Lie algebroids is developed by Nest–Tsygan [22]; ${}^bTM$ with ω_b is a symplectic Lie algebroid, but we have not checked the homogeneous version or the comparison with the b-symbol product in detail. Equivalent star products induce isomorphic Hochschild complexes, so the choice of connection does not affect the answer.

11.2 Hochschild-chain formality

Calaque–Dolgushev–Halbout prove formality theorems for Hochschild chains in the Lie-algebroid setting [16]; Dolgushev's smooth-manifold theorem is the corresponding prototype [17]. Their chain complexes are built from Lie-algebroid jets and polydifferential data; we have not shown that they agree with the Laurent chains used here, so the following is a heuristic. Twisting their module quasi-isomorphism by the Maurer–Cartan element defining (14) would give

$$(C_\bullet(\mathcal{O}_L(M)((\hbar)),\star), b_\star) \simeq (\Omega^\bullet_L(M)((\hbar)), \mathcal{L}_{\pi_\hbar}), \tag{15}$$

where $\pi_\hbar = \hbar\pi_b + O(\hbar^2)$ is the formal Poisson tensor and $\mathcal{L}_{\pi_\hbar} = [\iota_{\pi_\hbar}, d]$.

The formal tensor $\pi_\hbar$ is symplectic over $\mathbb{C}((\hbar))$; let $\omega_\hbar$ be its inverse. The formal symplectic star operator $*_{\omega_\hbar}$ satisfies the same Darboux calculation as (6) and converts $\mathcal{L}_{\pi_\hbar}$ to d, up to an invertible scalar and the degree sign. Therefore

$$\mathrm{HH}_k^{\mathbb{C}((\hbar))}\big(\mathcal{O}_L(M)((\hbar)),\star\big) \cong H_L^{2n-k}(M;\mathbb{C})((\hbar)). \tag{16}$$

The formality maps are b-polydifferential. Hence they preserve finite Laurent pole order: locally $x\partial_x(x^{-\ell}) = -\ell x^{-\ell}$, and tangential b-derivatives do not create a larger pole.

Equation (16) concerns the extended formal star algebra, before restriction to classical homogeneous expansions and before Borel realization. It must not be read as the complete-symbol answer. In particular, $H_L^*(M)$ itself has no order-circle factor: that factor appears only after the homogeneous order complex is formed.

Remark 11.1 (Where the order circle does, and does not, come from). The complex in (15) is relative to the base field $\mathbb{C}((\hbar))$. Thus $d\hbar = 0$, and it is incorrect to insert an additional class $d\hbar/\hbar$ into (16). The single class giving S^1_σ is the radial zero-weight class dr/r in Section 7.

One can instead compute absolute Hochschild homology over $\mathbb{C}$ for the Laurent extension of a Rees algebra. Then a class $d\log t$ can appear. The Rees grading ties the exponent of t to fiber homogeneity, so on the diagonal order complex $d\log t$ and $-d\log r$ describe the same order direction. After specialization at $t = 1$, only the radial class remains. Either convention yields one order circle, never two.

11.3 From formal series to classical complete symbols

Formal formality does not by itself compute the continuous homology of $\mathcal{A}_{b,L}$. What follows directly from the classical symbol calculus is a comparison at every finite depth of the order filtration.

Proposition 11.2 (Finite-order Rees comparison)**.** *After fixing a homogeneous classical symbol map, for every $m \in \mathbb{Z}$ and $N \geq 0$, the first $N+1$ symbol components give an isomorphism, continuous at every fixed pole bound,*

$$F_m\mathcal{A}_{b,L}/F_{m-N-1}\mathcal{A}_{b,L} \cong \bigoplus_{j=0}^{N} \mathcal{O}_L(M)_{m-j}. \tag{17}$$

Under (17)*, operator composition is the star product* (14) *modulo $\hbar^{N+1}$. These isomorphisms are compatible with truncation, completed projective tensor powers on each diagonal piece, and the Hochschild boundary. Consequently the Rees and classical order spectral sequences have the same E^1-page and the same Poisson differential d^1.*

Proof. There are three points to check.

Algebraic realization. Taylor expansion in symbol order gives (17); no infinite summation is needed at finite N. The b-symbol composition formula shows coefficient by coefficient that the two products agree modulo $\hbar^{N+1}$. Its first antisymmetric correction is the b-Poisson bracket, proving the assertion about d^1.

Topology. For a fixed Laurent bound, both sides of (17) are finite direct sums of nuclear Fréchet spaces. Truncation and the coefficient maps are continuous. Taking completed projective tensor powers first and the strict inductive limit over diagonal pieces afterward is exactly the order of operations in (3); hence the comparison extends to the finite quotients of the continuous Hochschild complex.

Inverse limit. Fix a pole bound ℓ. If $x^{-\ell}A \in F_{m-N-1}$ with $A \in \Psi_b^m$, then the full symbol of A has order $m-N-1$, so $x^{-\ell}\Psi_b^m \cap F_{m-N-1} = x^{-\ell}\Psi_b^{m-N-1}$. The b-symbol Borel lemma realizes every compatible sequence of truncations at this pole bound by a classical symbol, uniquely modulo smoothing; thus

$$\varprojlim_N x^{-\ell}\Psi_b^m / x^{-\ell}\Psi_b^{m-N-1} \cong x^{-\ell}\Psi_b^m / x^{-\ell}\Psi_b^{-\infty}.$$

The statement is false without the pole bound: an element of $\varprojlim_N F_m/F_{m-N-1}$ may need pole orders $\ell_N \to \infty$ as $N \to \infty$, and so need not come from F_m. Section 20.1 uses this asymptotic completeness at fixed bounds after passing to the mixed coefficient complex. □

Remark 11.3 (Scope of the Rees–Fedosov picture). Formality would organize the local differential beyond its first Poisson term, and Proposition 11.2 identifies the finite classical jets to which such a calculation applies. On the homogeneous Poisson complex the Euler contraction makes every nonzero weight acyclic, and the zero-weight splitting $\Omega_L^j(Y) \oplus (dr/r) \wedge \Omega_L^{j-1}(Y)$ has cohomology $H_L^j(Y \times S^1_\sigma)$. This reproduces the E^2-page of Proposition 9.1, but only because those steps are the ones already carried out in Sections 6–7; neither (15) nor (16) is used. The formal answer (16) is a different group (over $\mathbb{C}((\hbar))$, for all of M, without the order circle), and nothing here licenses specialization of formal Hochschild homology to continuous Hochschild homology. The Rees picture is therefore a heuristic explanation of where the Poisson differential comes from, not a second proof of Theorem 2.1.

$$\begin{array}{ccccc}
C_\bullet^{\mathrm{cont}}(\mathcal{A}_{b,L}) & \xrightarrow[\text{(localization)}]{\check{\text{C}}\text{ech}} & \mathrm{Tot}\,\check{C}^\bullet(\mathcal{U}, \mathcal{C}_\bullet) & \xrightarrow[(E^2)]{\text{HKR, } \delta_b,\ *_b} & H_L^{2n-\bullet}(Y \times S^1_\sigma) \\
\Big\downarrow \text{Rees jets} & & & & \\
C_\bullet(\mathcal{O}_L(M)((\hbar)), \star) & \xrightarrow[\text{formality}]{} & \left(\Omega_L^\bullet(M)((\hbar)), \mathcal{L}_{\pi_\hbar}\right) & \xrightarrow[*_{\omega_\hbar}]{} & H_L^{2n-\bullet}(M)((\hbar))
\end{array}$$

Figure 4: Top row: the Čech route globalizes the filtered HKR–Poisson calculation; it depends on the localization hypothesis of Lemma 10.1. Bottom row (dashed): the Rees–Fedosov picture, a heuristic only (Remark 11.3). Finite Rees jets compare classical symbols with the formal star product at each finite order, but the formal Hochschild homology lives over $\mathbb{C}((\hbar))$, on all of $M = {}^bT^*X \setminus 0$, and has no order circle; it is not mapped to the answer.

12 Checks and consequences

12.1 The closed-manifold case

If $\partial X = \varnothing$, Laurent cohomology is ordinary cohomology and $Z = \varnothing$. Formula (2) reduces to

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}(X;E)) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X),$$

equivalently $H^{2n-k}(S^*X \times S^1_\sigma)$. This is the Brylinski–Getzler computation [15].

12.2 Degree zero and residue traces

For the completed chain complex (3), degree-zero Hochschild homology is

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}) = \mathcal{A}/\operatorname{im}(b : C_1^{\mathrm{cont}} \to \mathcal{A}).$$

We abbreviate the image by $[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}$; it need not be closed. Its Hausdorffization is the different quotient

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A})_{\mathrm{H}} = \mathcal{A}/\overline{[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}}.$$

Theorem 2.1 computes the former group; by itself it does not prove that the commutator image is closed (but see below for $k = 0$). Continuous traces factor through the Hausdorffization.

Now assume $n \geq 2$, X connected and $\partial X \neq \varnothing$. Then Y is connected, since its fibers S^{n-1} are connected, and Z has one component over each component of ∂X. The manifold Y is orientable, being diffeomorphic to S^*X (Proposition 4.1(ii)), a cooriented hypersurface in the symplectic manifold $T^*X \setminus 0$; hence its boundary Z is a closed orientable manifold of even dimension $2n-2$. (It is not a contact manifold, which would require odd dimension.) In (2) with $k = 0$, $H^{2n}(Y) = 0$ and $H^{2n-1}(Z) = 0$ for dimension reasons, and $H^{2n-1}(Y) = 0$ because Y is a connected compact manifold with nonempty boundary. Hence

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H^{2n-2}(Z) \cong \mathbb{C}^{\pi_0(\partial X)}.$$

Continuous traces are continuous linear functionals vanishing on $[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}$, so they form a space of dimension at most $|\pi_0(\partial X)|$. Conversely, the boundary-face residue functionals, one for each component of ∂X (see [1]; compare the residue traces of the cusp calculus in [21]), are continuous traces; granting their linear independence, the space of continuous traces has dimension exactly $|\pi_0(\partial X)|$. In that case $[\mathcal{A},\mathcal{A}]_{\mathrm{cont}}$ is the joint kernel of finitely many continuous functionals, so it is closed and $\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}) = \mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A})_{\mathrm{H}}$. If ∂X is connected, the trace is the b-residue trace up to scale.

12.3 A product cylinder

Let $X = [0,1] \times N$, where N is closed of dimension $n-1$, and put

$$W = S(\mathbb{R} \oplus T^*N).$$

Then $Y \simeq [0,1] \times W$ and $Z = W \sqcup W$. Therefore

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H^{2n-k}(W) \oplus H^{2n-k-1}(W)^{\oplus 3} \oplus H^{2n-k-2}(W)^{\oplus 2}.$$

This example separates the order-circle contribution from the two boundary faces.

Part II
The residual and boundary ideals

13 From symbols to operators

Part I treats the quotient of the operator algebra by *all* smoothing operators with Laurent coefficients. To recover the operator algebra one has to choose the residual ideal and its topology (Section 2). We make the choice of Melrose–Nistor [21, (8)–(10)]: the small calculus, Laurent powers $x^{-\mathbb{Z}}$, and the ideal $\mathcal{I}^b = x^\infty \Psi_b^{-\infty}$ of operators with kernels in $\dot{C}^\infty(X^2)$. With these choices the remaining pieces are $\mathcal{I}^b$, the boundary ideal $\mathcal{I}^b_\partial$, and the extensions of Figure 1. The H-unitality of the ideals (Lemma 16.5) puts the excision theorem at our disposal. At the level of homology, the extension

$$0 \to \mathcal{I}^b_\partial \to \mathcal{A}^b \to \mathcal{A}^b_\sigma \to 0$$

replaces the extension by $\Psi^{-\infty}_{b,L}$ that was discussed in earlier versions of these notes. Its connecting map is computed in Section 22.

14 Conventions for the b-calculus

We follow the companion notes [10], whose conventions are recalled here.

14.1 Kernels and coordinates

Extend $\mu_b = \frac{dx}{x}\nu_\partial$ from the collar to a positive b-density on X. Operators act on functions, and the kernel of A is defined by $Au(p) = \int_X K_A(p,p')u(p')\,\mu_b(p')$. The b-stretched product X^2_b is obtained from X^2 by blowing up $H \times H$ for every component H. It has front faces ff_H and the lifts lb, rb of $\partial X \times X$ and $X \times \partial X$. Near ff_H we use the projective coordinates

$$(x', s, y, y'), \qquad s = \frac{x}{x'} \in [0,\infty), \qquad z = \log s, \tag{18}$$

in which $\mathrm{ff}_H = \{x' = 0\}$, $\mathrm{lb} = \{s = 0\}$ and the lifted diagonal is $\Delta_b = \{s = 1,\ y = y'\}$. Near rb one uses $(x, s' = x'/x, y, y')$ instead. We write $K_A = \kappa_A(x', s, y, y')$ in these coordinates.

The space $\Psi_b^{-\infty}(X)$ consists of the operators whose kernels lift to smooth functions on X^2_b that vanish to infinite order at $\mathrm{lb} \cup \mathrm{rb}$. The space $\Psi_b^m(X)$ consists of kernels that are classical conormal of order m at Δ_b, smooth up to the front faces, and vanish to infinite order at $\mathrm{lb} \cup \mathrm{rb}$. These spaces are closed under composition (Q0) [10, Thm. "Small b-composition"].

14.2 Indicial families

For $A \in \Psi_b^m(X)$, the restriction $\kappa_A(0, s, y, y')$ of its kernel to ff_H is the convolution kernel of the normal operator. The indicial family is its Mellin transform,

$$I_H(A,\lambda) = \int_0^\infty s^{-i\lambda}\kappa_A(0,s,\cdot,\cdot)\frac{ds}{s} = \int_{\mathbb{R}} e^{-iz\lambda}\kappa_A(0,e^z,\cdot,\cdot)\,dz, \tag{19}$$

regarded as an operator on H with kernel relative to ν_∂. This is the convention of [10, Sect. "The normal operator"]. For differential operators it gives $I(P,\lambda) = x^{-i\lambda}Px^{i\lambda}|_{\partial X}$; in particular, with $D_x = -i\partial_x$,

$$I(xD_x,\lambda) = \lambda. \tag{20}$$

The inversion formula is $\kappa_A(0, e^z) = \frac{1}{2\pi}\int e^{iz\lambda} I_H(A,\lambda)\, d\lambda$. At $z = 0$ it reads

$$\kappa_A(0, 1, y, y') = \frac{1}{2\pi}\int_{\mathbb{R}} I_H(A,\lambda)(y, y')\, d\lambda. \tag{21}$$

Lemma 14.1 (Conjugation by powers of x)**.** *Let $A \in \Psi_b^m(X)$, $m \in \mathbb{Z} \cup \{-\infty\}$, and $w \in \mathbb{C}$. Then $x^{-w}Ax^w \in \Psi_b^m(X)$, its kernel is $s^{-w}K_A$ near each front face, and*

$$I_H(x^{-w}Ax^w, \lambda) = I_H(A, \lambda - iw).$$

In particular $\alpha(A) := x^{-1}Ax$ has $I_H(\alpha(A),\lambda) = I_H(A, \lambda - i)$. The operator

$$D(A) := \frac{d}{dw}\Big|_{w=0} x^w A x^{-w} = [\log x, A]$$

has kernel $z\,K_A$, lies in $\Psi_b^m(X)$, and satisfies $I_H(D(A),\lambda) = i\,\partial_\lambda I_H(A,\lambda)$. The map D is a derivation of $x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X)$ that commutes with α and satisfies $D(x) = 0$.

Proof. Since $(x^{-w}Ax^w u)(p) = \int x(p)^{-w}K_A(p,p')x(p')^w u(p')\,\mu_b$, the new kernel is $(x'/x)^w K_A$, which equals $s^{-w}K_A$ near the front faces. On X_b^2 the function x'/x is a smooth positive multiple of $\rho_{\mathrm{rb}}/\rho_{\mathrm{lb}}$. The factor $(x'/x)^w$ is therefore smooth on the interior and near Δ_b, where $s = 1$, and it grows at most like a power of $\rho_{\mathrm{lb}}\rho_{\mathrm{rb}}$; infinite-order vanishing at lb and rb absorbs this growth. Since $s^{-i\lambda}s^{-w} = s^{-i(\lambda - iw)}$, the Mellin transform is shifted as stated. Differentiating s^w at $w = 0$ gives $\log s = z$, and $ze^{-iz\lambda} = i\partial_\lambda e^{-iz\lambda}$. Differentiating $x^w(AB)x^{-w} = (x^wAx^{-w})(x^wBx^{-w})$ shows that D is a derivation. It commutes with α because conjugations by powers of x commute, and $D(x) = 0$ because x commutes with x^w. □

Fix a b-vector field equal to $x\partial_x$ in the collar and put $h = -i\,x\partial_x$ there; we write $h = xD_x$. Then $I(h,\lambda) = \lambda$, and in the collar

$$[h, x^k] = -ik\,x^k. \tag{22}$$

14.3 Entire functions that are Schwartz on strips

For a Fréchet space F let $\mathcal{S}_{\exp}(\mathbb{R}; F)$ be the space of smooth $g : \mathbb{R} \to F$ with $\sup_z e^{N|z|}p(g^{(j)}(z)) < \infty$ for all N, j and continuous seminorms p. Let $\mathcal{E}(F)$ be the space of entire $f : \mathbb{C} \to F$ with

$$\sup_{|\operatorname{Im}\lambda| \le N} \langle\lambda\rangle^j p(f(\lambda)) < \infty \quad \text{for all } N, j, p.$$

By Cauchy's estimates on slightly wider strips, every derivative of f satisfies the same bounds. Write $\mathcal{E} = \mathcal{E}(\mathbb{C})$.

Lemma 14.2 (Paley–Wiener)**.** *The Fourier transform $\widehat{g}(\lambda) = \int e^{-iz\lambda}g(z)\,dz$ is a topological isomorphism $\mathcal{S}_{\exp}(\mathbb{R}; F) \to \mathcal{E}(F)$. It carries convolution to pointwise product, and multiplication by e^{-z} to the shift $f \mapsto f(\cdot - i)$.*

Proof. If $g \in \mathcal{S}_{\exp}$, then $\widehat{g}$ is entire, and for $|\operatorname{Im}\lambda| \le N$ we have $|\lambda^j\widehat{g}(\lambda)| \le \int e^{N|z|}|g^{(j)}(z)|\,dz$. Conversely, let $f \in \mathcal{E}(F)$ and $g(z) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{iz\lambda}f(\lambda)\,d\lambda$. For $z > 0$, Cauchy's theorem moves the contour to $\operatorname{Im}\lambda = N$; the vertical sides vanish by the decay on the strip. This gives $|g(z)| \le e^{-Nz}\frac{1}{2\pi}\int |f(\xi + iN)|\,d\xi$. For $z < 0$ one uses $\operatorname{Im}\lambda = -N$, and derivatives are handled in the same way. Finally, $\int e^{-iz(\lambda - i)}g\,dz = \int e^{-iz\lambda}e^{-z}g\,dz$. □

Definition 14.3 (Model algebras)**.** For a component H, let $B_H = \mathcal{E}(\Psi^{-\infty}(H))$ with pointwise composition, and let $B'_H = \Psi^{\mathbb{Z}}_{\mathrm{sus,ent}}(H)$ be the algebra of entire families $\lambda \mapsto a(\lambda)$ of pseudodifferential operators on H of integral order m that are suspended of order m uniformly on every strip $|\operatorname{Im}\lambda| \le C$ [10, 8]. By Lemma 14.2, B_H is exactly the set of indicial families of dilation-invariant operators on the model $M_H = [0,\infty)_x \times H$ whose kernels $\beta(x/x', y, y')$ satisfy $\beta(e^z, \cdot) \in \mathcal{S}_{\exp}(\mathbb{R}; C^\infty(H^2))$. For $b \in B'_H$ we write $b(xD_x)$ for the dilation-invariant operator on M_H with indicial family b, and $\alpha(b)(\lambda) = b(\lambda - i)$.

15 Taylor series at the front face

Lemma 15.1 (Composition at the front face)**.** *Let $A, A' \in \Psi_b^{-\infty}(X)$ have kernels supported in $\{x, x' < \varepsilon\}$, and near ff_H let $\kappa_A \sim \sum_{j\ge 0} x'^j \kappa_j(s,y,y')$ and $\kappa_{A'} \sim \sum_{l\ge 0} x''^l \kappa'_l(s,y,y')$ be the Taylor expansions. Then $\kappa_{AA'} \sim \sum_k x'^k \kappa''_k$, with*

$$\kappa''_k(s) = \sum_{j+l=k} \int_0^\infty t^j\, \kappa_j(s/t) \circ \kappa'_l(t)\, \frac{dt}{t}, \tag{23}$$

where $\circ$ denotes composition of kernels on H.

Proof. In the coordinates (18),

$$K_{AA'}(x,y;x',y') = \int_0^\infty \int_H \kappa_A\Big(x'', \frac{x}{x''}, y, y''\Big)\kappa_{A'}\Big(x', \frac{x''}{x'}, y'', y'\Big)\, \nu_\partial(y'')\frac{dx''}{x''}.$$

Substituting $x'' = x't$ gives $\int \kappa_A(x't, s/t, \cdot) \circ \kappa_{A'}(x', t, \cdot)\, dt/t$. The second factor decreases faster than any power of t and of t^{-1}, and the first decreases faster than any power of s/t and of t/s. Both bounds hold uniformly with all x'-derivatives, and $x't < \varepsilon$ on the support. Hence one may differentiate under the integral sign in x' at $x' = 0$. Expanding $\kappa_A(x't, \cdot) = \sum_j (x't)^j \kappa_j$ gives (23). □

Definition 15.2 (Twisted Laurent series and the Taylor map)**.** For an algebra $\mathcal{B}$ of entire families, closed under α, let $\mathcal{B}[[x;\alpha]$ be the space of formal series $\sum_{k\ge -\ell} x^k b_k$, $b_k \in \mathcal{B}$, with the product

$$(x^j a)(x^k b) = x^{j+k}\alpha^k(a)\, b, \qquad \alpha(a)(\lambda) = a(\lambda - i). \tag{24}$$

This product is associative: both bracketings of $(x^i a)(x^j b)(x^k c)$ equal $x^{i+j+k}\alpha^{j+k}(a)\alpha^k(b)c$. Every coefficient of a product is a finite sum. For $A \in x^{-\ell}\Psi_b^{-\infty}(X)$ the kernel has, near ff_H, an expansion $\kappa_A \sim \sum_{k\ge -\ell} x'^k \kappa_k(s,y,y')$, because $x^{-\ell} = x'^{-\ell} s^{-\ell}$. Put $\beta_k = s^{-k}\kappa_k$, let b_k be the Mellin transform of β_k, and define

$$T_H(A) = \sum_{k\ge -\ell} x^k b_k \in B_H[[x;\alpha], \qquad T = (T_H)_H.$$

The term $x^k b_k$ is the operator $x^k \circ b_k(xD_x)$ on M_H: its kernel is $x^k\beta_k(s) = x'^k\kappa_k(s)$.

Proposition 15.3 (Structure of $\mathcal{I}^b_\partial$)**.** *The Taylor map*

$$T : x^{-\mathbb{Z}}\Psi_b^{-\infty}(X) \longrightarrow \bigoplus_H B_H[[x;\alpha]$$

is a surjective algebra homomorphism with kernel $\mathcal{I}^b$. It is continuous on each piece $x^{-\ell}\Psi_b^{-\infty}(X)$. Hence $\mathcal{I}^b_\partial \cong \bigoplus_H B_H[[x;\alpha]$. In addition, $T(D(A)) = \sum_k x^k\, i\partial_\lambda b_k$, and the constant term of $T_H(A)$ is $I_H(A,\cdot)$ when $A \in \Psi_b^{-\infty}(X)$.

Proof. Kernel. If $T(A) = 0$, the kernel vanishes to infinite order at every front face, and also at lb and rb. Hence it lies in $\dot{C}^\infty(X_b^2) = \dot{C}^\infty(X^2)$, which is $\mathcal{I}^b$. The converse is clear.

Cut-off. Let $\chi \in C_c^\infty([0,\varepsilon))$ equal 1 near 0. The kernel $(1-\chi(x)\chi(x'))K_A$ vanishes near the front faces and to infinite order at lb and rb, so it lies in $\mathcal{I}^b$. Modulo $\mathcal{I}^b$ we may therefore assume that kernels are supported in the collar.

Multiplicativity. For $A, A' \in \Psi_b^{-\infty}$, Lemma 15.1 gives the Taylor coefficients of AA'. Since $\kappa_j = s^j\beta_j$ and $\kappa'_l = s^l\beta'_l$,

$$s^{-k}t^j\kappa_j(s/t)\kappa'_l(t) = (s/t)^{-l}\beta_j(s/t)\,\beta'_l(t) \qquad (j+l=k),$$

and $(s/t)^{-l}\beta_j(s/t)$ is the kernel of $\alpha^l(b_j(xD_x))$ by Lemma 14.1. Hence the k-th coefficient of $T(AA')$ is $\sum_{j+l=k}\alpha^l(b_j)b'_l$, which is the k-th coefficient of $T(A)T(A')$ by (24). From the definitions, $T(x^{-\ell}A) = x^{-\ell}T(A)$ and $T(Ax^{-\ell}) = T(A)x^{-\ell}$: the kernel $x'^{k-\ell}\kappa_k$ gives $\beta = s^\ell\beta_k$, which is the kernel of $\alpha^{-\ell}(b_k)$, while $(x^k b_k)x^{-\ell} = x^{k-\ell}\alpha^{-\ell}(b_k)$. Since $x^{\ell'}\Psi_b^{-\infty}x^{-\ell'} = \Psi_b^{-\infty}$, multiplicativity on $x^{-\mathbb{Z}}\Psi_b^{-\infty}$ follows.

Surjectivity. Let F_H be the Fréchet space of smooth functions of $(s,y,y') \in (0,\infty)\times H^2$ that decrease rapidly, with all derivatives in $s\partial_s$ and y, y', as $s \to 0$ and as $s \to \infty$. Given $(b_k)_{k\geq -\ell}$, the functions $\kappa_k = s^k\beta_k$ lie in F_H by Lemma 14.2. Let $\{p_j\}$ be an increasing sequence of seminorms defining F_H, and let $\phi \in C_c^\infty([0,1))$ equal 1 near 0. By Borel's argument, $G = \sum_{k\geq 0} x'^k\phi(x'/\varepsilon_k)\kappa_{k-\ell}$ converges in $C^\infty([0,1);F_H)$ and has the prescribed Taylor series, provided ε_k is so small that the k-th term has C^{k-1}-norm, measured with p_k, at most 2^{-k}. Put $K = x'^{-\ell}\chi(x)\chi(x')G$ near ff_H, extended by zero. In the chart (x,s') near rb we have $x' = xs'$ and $s = 1/s'$, and the rapid decrease as $s \to \infty$, uniform in x', gives smoothness and infinite-order vanishing at rb. No continuous linear right inverse is claimed: the Borel map $\prod_k F_H \to C^\infty([0,1/2];F_H)$ has none. A Fréchet space with a continuous norm, such as $C^\infty([0,1/2];F_H)$ when F_H has one, cannot contain a copy of $\prod_k \mathbb{C}$, which has no continuous norm. A continuous linear right inverse would embed one. Continuity of T itself is clear. The last assertions follow from Lemma 14.1. □

Proposition 15.4 (Structure of $\mathcal{A}_\partial^b$)**.** *The Taylor map extends to a surjective homomorphism $x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X) \to \bigoplus_H B'_H[[x;\alpha]]$ with kernel $x^\infty\Psi_b^{\mathbb{Z}}(X)$, so that $\mathcal{A}_\partial^b \cong \bigoplus_H B'_H[[x;\alpha]]$.*

Proof. We use (Q0) in the following form: $\Psi_b^{\mathbb{Z}}$ is an algebra; for $K \geq 0$, $x^K\Psi_b^m \circ \Psi_b^{m'}$ and $\Psi_b^{m'} \circ x^K\Psi_b^m$ lie in $x^K\Psi_b^{m+m'}$ (the latter by Lemma 14.1); and conormal kernels admit Taylor expansion at ff_H, which Δ_b meets transversally, and asymptotic summation (C3).

Model identity. On the model M_H, the operators $x^j a(xD_x)$ with $a \in B'_H$ satisfy, exactly,

$$(x^j a(xD_x))(x^k b(xD_x)) = x^{j+k}(\alpha^k(a)b)(xD_x).$$

Indeed, $a(xD_x)x^k = x^k(x^{-k}a(xD_x)x^k)$, and by Lemma 14.1 the last factor has indicial family $a(\lambda - ik)$. Dilation-invariant operators compose by multiplying indicial families.

Cut-off model operators. For $a \in B'_H$, put $Q_k(a) = \chi(x)\,x^k a(xD_x)\,\chi(x')$, transplanted to X via the collar, where $\chi \in C_c^\infty([0,\varepsilon))$ equals 1 on $[0,\varepsilon/2]$. We claim

$$Q_j(a)Q_k(b) - Q_{j+k}(\alpha^k(a)b) \in x^\infty\Psi_b^{\mathbb{Z}}(X).$$

The difference is $\chi x^j a(xD_x)(\chi^2-1)x^k b(xD_x)\chi$. Its kernel is an integral over x'' of kernels evaluated at x/x'' and x''/x', where $x, x' < \varepsilon$ and $x'' \geq \varepsilon/2$ on the support of $\chi^2 - 1$. Both kernels are smooth there and decrease faster than any power of x/x'', resp. x'/x'', wherever x and x' stay away from the support of $\chi^2 - 1$. Near ff the difference therefore vanishes to infinite order. Away from ff it is an

operator of order $\le m+m'$ supported where $x, x' \ge \varepsilon/2$ or near the singular set $x \approx x''$. Its kernel need not be smoothing: for $a = b = 1$ the difference is the multiplication operator $\chi(\chi^2-1)x^{j+k}$. But it vanishes to infinite order at ff, so it lies in $x^\infty\Psi_b$, which is the kernel of the Taylor map T. This is all that is used below.

Taylor expansion. By transversality, the kernel of $A \in x^{-\ell}\Psi_b^m$ has a Taylor expansion at ff_H whose coefficients are dilation-invariant kernels conormal at $\{s=1, y=y'\}$, rapidly decreasing as $|z| \to \infty$. As in Proposition 15.3, this defines $T_H(A) = \sum x^k a_k$ with $a_k \in B'_H$, and

$$A - \sum_{k<K} Q_k(a_k) \in x^K \Psi_b^m(X) + \mathcal{I}^b \quad \text{for every } K.$$

Multiplicativity. Write $A = \sum_{j<K} Q_j(a_j) + R$ and $A' = \sum_{k<K} Q_k(a'_k) + R'$, with $R, R' \in x^K\Psi_b + \mathcal{I}^b$. By the claim and the ideal property, $AA' \equiv \sum_{j+k<K} Q_{j+k}(\alpha^k(a_j)a'_k)$ modulo $x^K\Psi_b$, since $\mathcal{I}^b$ and $x^\infty\Psi_b$ are contained in $x^K\Psi_b$. Since K is arbitrary, $T(AA') = T(A)T(A')$.

Kernel and surjectivity. $T(A) = 0$ means that $A \in x^K\Psi_b^m$ for every K. Conversely, given $\sum_{k\ge-\ell} x^k a_k$ with $a_k \in B'_{H,m}$, asymptotic summation of the kernels of $Q_k(a_k)$ (C3) produces $A \in x^{-\ell}\Psi_b^m$ with this Taylor series. □

Example 15.5 (The model $X = [0,1)$)**.** Let $\partial X = \{0\}$. Then $B = \mathcal{E}$, B' is the algebra of entire functions of λ with classical symbol expansions on strips, and

$$\mathcal{I}_\partial^b \cong \mathcal{E}[[x;\alpha]] = \Big\{ \sum_{k\ge-\ell} x^k\, b_k(xD_x) : b_k \in \mathcal{E} \Big\}.$$

The rule (24) is $b(xD_x)\, x^k = x^k\, b(xD_x - ik)$, which follows from (22). The operator $h = xD_x$ does not belong to $\mathcal{I}_\partial^b$, but it is a multiplier of $\mathcal{I}_\partial^b$ and an element of $\mathcal{A}_\partial^b$, and $[h, x^k b(h)] = -ik\, x^k b(h)$. The x-adic associated graded algebra is again $\mathcal{E}[x, x^{-1}; \alpha]$, which is not commutative. This is where the b-case departs from [21, proof of Prop. 6].

16 Hochschild tools

16.1 The chain model

All algebras in these notes are treated in one chain model.

Definition 16.1 (Filtered nuclear algebras and their chains)**.** A *filtered nuclear algebra* is an increasing union $\mathfrak{A} = \bigcup_{N\ge0} \mathfrak{A}_N$ of nuclear Fréchet spaces, each closed in the next, such that multiplication maps $\mathfrak{A}_N \times \mathfrak{A}_M \to \mathfrak{A}_{N+M}$ continuously. Its chain spaces are

$$C_k(\mathfrak{A}) = \varinjlim_N \mathfrak{A}_N^{\widehat{\otimes}(k+1)},$$

with completed projective tensor products. Write d_i, $0 \le i < k$, for multiplication of the factors in slots i and $i+1$, and d_k for $a_0 \otimes \cdots \otimes a_k \mapsto a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}$. Then $b = \sum_{i=0}^k (-1)^i d_i$ maps the N-th piece in degree k continuously into the $(k+1)N$-th piece, and $b' = \sum_{i=0}^{k-1}(-1)^i d_i$. If J is a closed ideal, then J and $\mathfrak{A}/J$ are filtered nuclear algebras with pieces $J_N = J \cap \mathfrak{A}_N$ and $\mathfrak{A}_N/J_N$.

The filtrations used are as follows.

- $x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X)$: $\mathfrak{A}_N = x^{-N}\Psi_b^N(X)$, with the Fréchet topology of classical conormal kernels on X_b^2 (symbol seminorms at Δ_b, and C^∞ seminorms weighted by $\rho_{\mathrm{lb}}^{-M}\rho_{\mathrm{rb}}^{-M}$ away from Δ_b).

- $x^{-\mathbb{Z}}\Psi_b^{-\infty}(X)$: $\mathfrak{A}_N = x^{-N}\Psi_b^{-\infty}(X)$, with the corresponding C^∞ topology.
- $\mathcal{I}^b$: the single Fréchet space $\dot{C}^\infty(X^2)$.
- $\mathcal{A}^b$, $\mathcal{A}^b_\sigma$, $\mathcal{I}^b_\sigma$, $\mathcal{I}^b_\partial$, $\mathcal{A}^b_\partial$, $\mathcal{A}^b_{\partial,\sigma}$: the induced quotient and subspace filtrations.

For $\mathcal{A}^b_\sigma$ these pieces are the diagonal pieces $F^N_{N,N}$ of (3), so Part I uses the same model. Via the Taylor map (Propositions 15.3 and 15.4), the pieces of $\mathcal{I}^b_\partial$ and $\mathcal{A}^b_\partial$ become $x^{-N}\prod_{k\geq 0} x^k B_H$ and $x^{-N}\prod_{k\geq 0} x^k B'_{H,N}$, where $B'_{H,N}$ consists of the elements of order $\leq N$, with suspended seminorms on strips. All ideals that occur ($\mathcal{I}^b$, the smoothing ideal, and the ideals of infinite order of vanishing at the boundary) are kernels of continuous maps (full symbol, Taylor series at the front faces), hence closed. On a closed subspace the induced Fréchet topology coincides with any finer Fréchet topology for which the inclusion is continuous, by the open mapping theorem. For example, $\mathcal{I}^b$ has the same topology whether it is regarded as $\dot{C}^\infty(X^2)$ or as a subspace of $x^{-N}\Psi_b^N$.

For $R = B[[x;\alpha]$ the completed tensor products of nuclear Fréchet spaces commute with countable products. A k-chain is therefore a family $(c_{\mathbf{j}})$ indexed by multi-indices $\mathbf{j} = (j_0, \ldots, j_k)$ with all $j_i \geq -N$ for some N, where $c_{\mathbf{j}} \in B^{\widehat{\otimes}(k+1)}$. For $B = \mathcal{E}(\Psi^{-\infty}(H))$, the space $B^{\widehat{\otimes}(k+1)}$ consists of entire functions of $(\lambda_0, \ldots, \lambda_k)$ with values in $\Psi^{-\infty}(H^{k+1})$, Schwartz on products of strips. In this picture, by (24), d_i evaluates slot i at $\lambda - ij_{i+1}$ and slot $i+1$ at λ, composes in the H-variables, and gives the new slot the exponent $j_i + j_{i+1}$. The *weight* $g = \sum_i j_i$ is preserved by every face map.

Lemma 16.2 (Exactness of chain spaces)**.** *Let $J \subset \mathfrak{A}$ be a closed ideal of a filtered nuclear algebra. Then for every k*

$$0 \to \mathcal{K}_k \to C_k(\mathfrak{A}) \to C_k(\mathfrak{A}/J) \to 0$$

is exact. Here $\mathcal{K}_k$ is the direct limit over N of the closed sums $\sum_i \mathfrak{A}_N^{\widehat{\otimes} i} \widehat{\otimes} J_N \widehat{\otimes} \mathfrak{A}_N^{\widehat{\otimes}(k-i)}$.

Proof. For Fréchet spaces, $\widehat{\otimes}_\pi$ carries quotient maps to quotient maps. When one factor is nuclear it also preserves injections, so it is exact on short exact sequences of nuclear Fréchet spaces [26, Chs. 43, 50]. Apply this factor by factor to $0 \to J_N \to \mathfrak{A}_N \to \mathfrak{A}_N/J_N \to 0$, and then pass to the direct limit. The limit is taken algebraically, as a filtered colimit of vector spaces, which is exact. The transition maps are injective by nuclearity, and homology is computed algebraically, so no strictness or regularity of the LF topology is needed. □

16.2 H-unitality and excision

A filtered nuclear algebra is *topologically H-unital* if its b'-complex admits a contraction E, with $b'E + Eb' = 1$ in all degrees ($b' = 0$ on C_0), that maps $\mathfrak{A}_N^{\widehat{\otimes}(k+1)}$ continuously into $\mathfrak{A}_N^{\widehat{\otimes}(k+2)}$.

Theorem 16.3 (Excision; quoted input (Q1))**.** *Let $J \subset \mathfrak{A}$ be a closed ideal of a filtered nuclear algebra, and suppose J is topologically H-unital. Then $C(J) \to \mathcal{K}$ is a quasi-isomorphism, where $\mathcal{K}$ is the kernel of Lemma 16.2. Consequently there is a long exact sequence*

$$\cdots \to \mathrm{HH}_k(J) \to \mathrm{HH}_k(\mathfrak{A}) \to \mathrm{HH}_k(\mathfrak{A}/J) \xrightarrow{\partial} \mathrm{HH}_{k-1}(J) \to \cdots .$$

To compute $\partial[c]$, lift c to a chain $\tilde{c}$ of $\mathfrak{A}$, which is possible by Lemma 16.2; then $\partial[c]$ is the class of $b\tilde{c} \in \mathcal{K}$. This is Wodzicki's theorem [27]. His proof is algebraic, and in the present model its inputs are the exactness of Lemma 16.2 and the contraction E. We have not rewritten the proof in the completed setting, so we list the theorem as an input. No continuous linear section of $\mathfrak{A} \to \mathfrak{A}/J$ is assumed. The Taylor maps have no such section, because of the Borel-map obstruction.

Lemma 16.4 (Connecting maps on cocycles)**.** *In the situation of Theorem 16.3, let τ be a $(k-1)$-cocycle on J, and let $\hat{\tau}$ be a continuous $(k-1)$-cochain on $\mathfrak{A}$ that extends τ and is such that $b\hat{\tau}$ vanishes whenever one of its arguments lies in J. Then $b\hat{\tau}$ descends to a cocycle $\overline{b\hat{\tau}}$ on $\mathfrak{A}/J$, and $\langle \tau, \partial[c]\rangle = \langle \overline{b\hat{\tau}}, c\rangle$ for every cycle c.*

Proof. The hypothesis says that $\hat{\tau}|_{\mathcal{K}}$ is a cocycle on $\mathcal{K}$. It restricts to τ on $C(J)$, so under the quasi-isomorphism it represents $[\tau]$. Hence $\langle \tau, \partial[c]\rangle = \langle \hat{\tau}, b\tilde{c}\rangle = \langle b\hat{\tau}, \tilde{c}\rangle = \langle \overline{b\hat{\tau}}, c\rangle$. □

Lemma 16.5 (Explicit contractions)**.** *(a) Let V, W be Fréchet spaces with a continuous pairing $\langle\cdot,\cdot\rangle : W \times V \to \mathbb{C}$, and let $\mathcal{K} = V\widehat{\otimes}W$ carry the product $(v\otimes w)(v'\otimes w') = \langle w, v'\rangle v \otimes w'$. Choose $\psi \in W$ and $\tilde{\psi} \in V$ with $\langle \psi, \tilde{\psi}\rangle = 1$. Then*

$$E\big((v\otimes w)\otimes c\big) = (v\otimes\psi)\otimes(\tilde{\psi}\otimes w)\otimes c$$

is a continuous contraction of the b'-complex. This applies to $\Psi^{-\infty}(H) = C^\infty(H)\widehat{\otimes}C^\infty(H)$ and to $\mathcal{I}^b = \dot{C}^\infty(X)\widehat{\otimes}\dot{C}^\infty(X)$, with the pairings given by ν_∂ and μ_b.

(b) Let $B = \mathcal{E}(\Psi^{-\infty}(H))$ and let R be $B[[x;\alpha]$, $B \rtimes_\alpha \mathbb{Z} = \bigoplus_k x^k B$, or the same algebras with $\mathcal{E}$ in place of B. Let $\phi(\zeta) = e^{-\zeta^2}$. On the component $\mathbf{j} = (j_0, \dots, j_k)$ define EF as the component $(0, j_0, \dots, j_k)$ given by

$$(EF)(\mu, \lambda_0, \dots; v, v', u, u', \dots) = \phi(\mu - \lambda_0 + ij_0)\,\psi(v')\tilde{\psi}(u)\,F(\lambda_0, \dots; v, u', \dots).$$

Here (v, v') are the kernel variables of the new slot, (u, u') those of the old slot 0, and $\int_H \psi\tilde{\psi}\,\nu_\partial = 1$. Then E is a continuous contraction of the b'-complex of R. It preserves weights and pole bounds.

Consequently $\mathcal{I}^b$, $\Psi^{-\infty}(H)$, $\mathcal{I}^b_\partial$, $B \rtimes_\alpha \mathbb{Z}$, $\mathcal{E}[[x;\alpha]$ and $\mathcal{E} \rtimes_\alpha \mathbb{Z}$ are topologically H-unital.

Proof. In both cases $d_0E = 1$ and $d_{i+1}E = Ed_i$ for $i \geq 0$. This gives $b'E = 1 - Eb'$. For (a): $(v\otimes\psi)(\tilde{\psi}\otimes w) = v\otimes w$, and $d_1E((v\otimes w)\otimes a_1\otimes\cdots) = (v\otimes\psi)\otimes(\tilde{\psi}\otimes w)a_1\otimes\cdots = Ed_0(\cdots)$, because $(v\otimes w)a_1 = v\otimes(w\cdot a_1)$.

For (b), merging the new slot (exponent 0, variable μ) with the old slot 0 (exponent j_0) sets $\mu = \lambda - ij_0$ and $\lambda_0 = \lambda$, by (24). Then $\phi(\lambda - ij_0 - \lambda + ij_0) = \phi(0) = 1$, and the kernel factors give $\int \psi\tilde{\psi} = 1$; hence $d_0E = 1$. Merging the old slots 0 and 1 sets $\lambda_0 = \lambda - ij_1$ and $\lambda_1 = \lambda$, and turns the prefactor into $\phi(\mu - \lambda + i(j_0 + j_1))$. This is the prefactor of E on the merged component, whose first exponent is $j_0 + j_1$; hence $d_1E = Ed_0$. For $i \geq 2$ the identity $d_iE = Ed_{i-1}$ is clear. Finally, $|\phi(\zeta)| = e^{-(\operatorname{Re}\zeta)^2 + (\operatorname{Im}\zeta)^2}$. Together with the decay of F in λ_0, this shows that EF is Schwartz on products of strips, with continuous dependence on F. □

Lemma 16.6 (Inner derivations [7, 1.3.3])**.** *Let $\mathfrak{A}$ be unital and $h \in \mathfrak{A}$. Put $s_h(a_0\otimes\cdots\otimes a_k) = \sum_{i=0}^k (-1)^{i+1} a_0\otimes\cdots\otimes a_i\otimes h\otimes a_{i+1}\otimes\cdots\otimes a_k$. Then $bs_h + s_hb = L_h$, where $L_h(a_0\otimes\cdots\otimes a_k) = \sum_i a_0\otimes\cdots\otimes[h, a_i]\otimes\cdots\otimes a_k$.*

Proof. This is a direct computation. For $k = 0$: $bs_h(a_0) = -b(a_0\otimes h) = [h, a_0]$. □

Lemma 16.7 (Morita invariance for smoothing kernels)**.** *Let $\mathcal{K} = V\widehat{\otimes}W$, and let ψ and $\tilde{\psi}$ be as in Lemma 16.5(a), and put $e = \tilde{\psi}\otimes\psi$. Let $\mathfrak{A}$ be unital or topologically H-unital. Then*

$$\iota(a_0\otimes\cdots\otimes a_k) = (e\otimes a_0)\otimes\cdots\otimes(e\otimes a_k)$$

and the generalized trace $\operatorname{Tr}_*\big((\kappa_0\otimes a_0)\otimes\cdots\otimes(\kappa_k\otimes a_k)\big) = \operatorname{Tr}(\kappa_0\cdots\kappa_k)\,a_0\otimes\cdots\otimes a_k$ are chain maps with $\operatorname{Tr}_*\,\iota = 1$. Moreover, $C(\mathcal{K}\widehat{\otimes}\mathfrak{A}) = \iota C(\mathfrak{A}) \oplus C'$ with $C' = \ker\operatorname{Tr}_*$ acyclic. If $\mathfrak{A}$ is graded by a weight and the contraction preserves weights, the statement holds weight by weight.

Proof. Write a chain as $(v_0 \otimes w_0 \otimes a_0) \otimes \cdots \otimes (v_k \otimes w_k \otimes a_k)$ and rearrange it as the *letters* $\ell_i = w_i \otimes v_{i+1} \in U := W \widehat{\otimes} V$ (indices mod $k+1$), together with $a_0 \otimes \cdots \otimes a_k$. Let $\varepsilon(w \otimes v) = \langle w, v \rangle$. The face d_i applies ε to the letter ℓ_i, multiplies a_i with a_{i+1} (for $i = k$ it puts $a_k a_0$ in slot 0), and leaves the other letters unchanged. Split $U = \mathbb{C}u_0 \oplus U'$, where $u_0 = \psi \otimes \tilde{\psi}$ and $U' = \ker \varepsilon$. Because $\varepsilon(U') = 0$, the number m of letters lying in U' is preserved by b. For $m = 0$ all letters are u_0, so every slot is $\tilde{\psi} \otimes \psi = e$; this summand is $\iota C(\mathfrak{A})$, and b acts on it as the Hochschild boundary of $\mathfrak{A}$. Projection onto it is Tr_*, since $\prod_i \varepsilon(\ell_i) = \mathrm{Tr}(\kappa_0 \cdots \kappa_k)$.

For $m \geq 1$, filter by the position w of the last U'-letter. A face d_i with $i < w$ lowers w, the face d_w vanishes, and the faces d_i with $w < i \leq k$ preserve w. In the associated graded complex only the faces $d_{w+1}, \ldots, d_k$ survive, with alternating signs $(-1)^i$. They multiply consecutive elements of the block $(a_{w+1}, \ldots, a_k, a_0)$, which is a b'-complex of $\mathfrak{A}$ of length $k - w + 1 \geq 1$, tensored with the spectators $a_1, \ldots, a_w$ and the earlier letters. That b'-complex is contractible: by $1\otimes$ in the unital case, or by E, inserted at the front of the block. In each degree the filtration is finite, so C' is acyclic. □

Theorem 16.8 (Crossed products by $\mathbb{Z}$; quoted (Q2))**.** *Let $\mathfrak{B}$ be unital or topologically H-unital (with a contraction compatible with α), let α be an automorphism, and let $\mathfrak{B} \rtimes_\alpha \mathbb{Z} = \bigoplus_k x^k \mathfrak{B}$ with the product* (24)*. Let $C^{(0)}$ be the weight-zero part of its Hochschild complex. Then there are natural short exact sequences*

$$0 \to \mathrm{HH}_q(\mathfrak{B})_\alpha \to H_q(C^{(0)}) \to \mathrm{HH}_{q-1}(\mathfrak{B})^\alpha \to 0,$$

where $V_\alpha = V/(1 - \alpha_*)V$ and $V^\alpha = \ker(1 - \alpha_*)$. In the unital case, and algebraically, $C^{(0)}$ is quasi-isomorphic to the mapping cone of $1 - \alpha_*$ on $C(\mathfrak{B})$ [3, 9, 4]. We assume the same statement in the chain model of Section 16.1 for $\mathfrak{B} = \mathcal{E}$. We also assume the reduction of the H-unital case to the unital one (apply the unital case to the unitization, then Theorem 16.3 weight by weight). Neither step is proved here.

17 The residual ideal

Proposition 17.1. *The ideal $\mathcal{I}^b$ is topologically H-unital, and $\mathrm{HH}_0(\mathcal{I}^b) \cong \mathbb{C}$ by the trace $\mathrm{Tr}(A) = \int_X K_A(p,p)\, \mu_b(p)$, while $\mathrm{HH}_k(\mathcal{I}^b) = 0$ for $k \geq 1$. The same holds for $\Psi^{-\infty}(H)$.*

Proof. By the kernel theorem, $\dot{C}^\infty(X^2) = \dot{C}^\infty(X) \widehat{\otimes} \dot{C}^\infty(X)$ (the Schwartz kernel theorem for flat functions), and composition is the product of Lemma 16.5(a) for the pairing $\int_X wv\, \mu_b$. Apply Lemma 16.5(a) and Lemma 16.7 with $\mathfrak{A} = \mathbb{C}$, for which $C(\mathbb{C})$ is $\mathbb{C}$ in degree 0 up to an acyclic complex. The generalized trace is Tr. □

18 The boundary ideal

We first treat the scalar model $\mathcal{E}[[x;\alpha]$ of Example 15.5. The general case then follows by Lemma 16.7.

18.1 The algebra $\mathcal{E}$

Chains of $\mathcal{E}$ are entire functions $F(\lambda_0, \ldots, \lambda_k)$ that are Schwartz on products of strips, and

$$(bF)(\lambda_0, \ldots, \lambda_{k-1}) = \sum_{i<k} (-1)^i F(\ldots, \lambda_i, \lambda_i, \ldots) + (-1)^k F(\lambda_0, \ldots, \lambda_{k-1}, \lambda_0).$$

Let $\chi_0 = 1$, $\chi_1(F) = (\partial_{\lambda_1} F)(\lambda, \lambda)\, d\lambda$, and $\chi_k = 0$ for $k \geq 2$. Let $\iota_0 = 1$ and $\iota_1(f\, d\lambda) = f(\lambda_0)\psi_1(\lambda_1 - \lambda_0)$, where $\psi_1(\zeta) = \zeta e^{-\zeta^2}$.

Proposition 18.1 (HKR for $\mathcal{E}$). *(a) The maps χ and ι are chain maps between $C(\mathcal{E})$ and the complex $\mathcal{E} \leftarrow \mathcal{E}\,d\lambda$ with zero differential. They satisfy $\chi\iota = 1$, and they commute with the diagonal action of α.*

(b) (Division lemma.) *If $G \in \mathcal{E}(\mathbb{C}^2)$ and $G(\lambda,\lambda) = 0$, then $G = (\lambda_0 - \lambda_1)G_1$ with $G_1 \in \mathcal{E}(\mathbb{C}^2)$ depending continuously on G.*

(c) The Koszul complex $0 \to \mathcal{E}(\mathbb{C}^2) \xrightarrow{\lambda_0-\lambda_1} \mathcal{E}(\mathbb{C}^2) \to \mathcal{E} \to 0$, whose last map is restriction to the diagonal, is exact. Restriction has the continuous right inverse $f \mapsto f(\lambda_0)e^{-(\lambda_0-\lambda_1)^2}$.

(d) χ is a quasi-isomorphism. Thus $\mathrm{HH}_0(\mathcal{E}) = \mathcal{E}$, $\mathrm{HH}_1(\mathcal{E}) = \mathcal{E}\,d\lambda$, and $\mathrm{HH}_q(\mathcal{E}) = 0$ for $q \geq 2$.

Proof. (a) On C_1, $bF(\lambda) = F(\lambda,\lambda) - F(\lambda,\lambda) = 0$, so $\chi_0 b = 0$. For $F \in C_2$,

$$bF(\mu_0,\mu_1) = F(\mu_0,\mu_0,\mu_1) - F(\mu_0,\mu_1,\mu_1) + F(\mu_0,\mu_1,\mu_0).$$

Applying ∂_{μ_1} at $\mu_0 = \mu_1$ gives $\partial_3 F - (\partial_2 + \partial_3)F + \partial_2 F = 0$, so $\chi_1 b = 0$. Also $\chi_1\iota_1(f) = f\,\psi_1'(0) = f$. Both maps are built from diagonal evaluations and are therefore translation-equivariant.

(b) Put $G_1(\lambda_0,\lambda_1) = -\int_0^1 (\partial_2 G)(\lambda_0, \lambda_0 + t(\lambda_1 - \lambda_0))\,dt$, which is entire. If $|\operatorname{Re}(\lambda_0 - \lambda_1)| \geq 1$, then $|G_1| \leq |G|$. Otherwise every point of the segment has real part within 1 of $\operatorname{Re}\lambda_1$. Cauchy estimates for $\partial_2 G$ on slightly wider strips then give decay in both variables. Derivatives are estimated in the same way.

(c) Multiplication by $\lambda_0 - \lambda_1$ is injective on entire functions; exactness in the middle is (b); and the stated map is a right inverse.

(d) Put $A = \mathcal{E}$ and $M = A\widehat{\otimes}A$, with the right action on the first variable u and the left action on the second variable v. Multiplication by $u - v$ and diagonal restriction are bimodule maps. Tensor the split short exact sequence in (c) with $A^{\widehat{\otimes}k}$. This gives a short exact sequence of coefficient complexes

$$0 \longrightarrow C_\bullet(A,M) \xrightarrow{u-v} C_\bullet(A,M) \xrightarrow{\mathrm{diag}} C_\bullet(A,A) \longrightarrow 0.$$

Here $C_k(A,M) = M\widehat{\otimes}A^{\widehat{\otimes}k}$. Order the variables of this complex as $(u,\lambda_1,\dots,\lambda_k,v)$. The first Hochschild face multiplies on the right of M, and the last face multiplies on its left. In the displayed order these are the first and last adjacent multiplications. Thus the differential agrees with b' on $C_{k+1}(A)$. The map $F \mapsto e^{-(\mu-u)^2}F$ inserts the new first variable μ and contracts the full b'-complex by the same face identities as in Lemma 16.5(b), with all exponents zero. Hence $H_k(C_\bullet(A,M)) = 0$ for $k > 0$. In degree zero the homology identifies with A by diagonal restriction, by exactness of the full bar complex at $C_1(A)$ and $C_0(A)$. Under this identification multiplication by $u - v$ induces zero.

The long exact sequence gives zero homology in degrees $k \geq 2$ and an isomorphism $\mathrm{HH}_1(A) \to A$ by its connecting map. For $F \in C_1(A,A)$, choose the lift $e^{-(u-v)^2}F(u,w)$, with middle variable w. Its boundary is

$$e^{-(u-v)^2}\big(F(u,u) - F(u,v)\big).$$

Dividing by $u - v$ and restricting to $u = v = \lambda$ gives $(\partial_2 F)(\lambda,\lambda)$. This identifies the connecting map with χ_1 after adjoining $d\lambda$. In degree zero χ_0 is the identity, since A is commutative. □

18.2 Coinvariants of the shift

Lemma 18.2. *Let $\alpha f(\lambda) = f(\lambda - i)$ on $\mathcal{E}$. Then $\ker(1-\alpha) = 0$, and the image of $1 - \alpha$ is the closed hyperplane $\{f : \int_{\mathbb{R}} f\,d\lambda = 0\}$, with a continuous inverse given by*

$$(1-\alpha)^{-1}f = \widehat{g_f}, \qquad g_f(z) = \frac{\check{f}(z)}{1 - e^{-z}}, \qquad \check{f}(z) = \frac{1}{2\pi}\int e^{iz\lambda} f(\lambda)\,d\lambda.$$

Consequently $\mathcal{E}_\alpha \cong \mathbb{C}$ by $f \mapsto \frac{1}{2\pi}\int f\,d\lambda$ and $\mathcal{E}^\alpha = 0$. The same holds for $\mathcal{E}\,d\lambda$.

Proof. By Lemma 14.2, $1-\alpha$ corresponds to multiplication of $\check f \in \mathcal{S}_{\exp}$ by $1-e^{-z}$. This function vanishes on $\mathbb{R}$ only at $z=0$, where its zero is simple, so the kernel is zero. If $\int f = 2\pi\check f(0) = 0$, then $\check f/(1-e^{-z})$ is smooth by the one-variable division lemma. For $z \geq 1$ the factor $(1-e^{-z})^{-1}$ is bounded with bounded derivatives. For $z \leq -1$ it is bounded by $2e^{z}$, again with derivatives. Hence the quotient lies in $\mathcal{S}_{\exp}$ and depends continuously on f. Conversely, $\int (1-\alpha)g\,d\lambda = 0$ by a contour shift, as in the proof of Proposition 18.4(a). The shift preserves $d\lambda$. □

18.3 The weight decomposition

Proposition 18.3 (Nonzero weights are acyclic)**.** *Let $R = B[[x;\alpha]$ with $B = \mathcal{E}$ or $B = \mathcal{E}(\Psi^{-\infty}(H))$. Then:*

(a) $C(R) = C^{(0)}(R) \oplus C^{\neq 0}(R)$ as complexes, where $C^{(0)}(R)$ consists of the components of weight zero;

(b) $C^{(0)}(R) = C^{(0)}(B \rtimes_\alpha \mathbb{Z})$;

(c) $C^{\neq 0}(R)$ is acyclic.

Hence $\mathrm{HH}_*(R) \cong H_*(C^{(0)}(B \rtimes_\alpha \mathbb{Z}))$.

Proof. (a) Projection onto the components of weight zero is defined componentwise and commutes with b, because every face preserves the weight.

(b) If all $j_i \geq -\ell$ and $\sum j_i = 0$, then $j_i \leq k\ell$, so only finitely many multi-indices occur. A chain of weight zero is therefore a finite sum, that is, a chain of the polynomial crossed product. The converse is clear.

(c) Let $\tilde R = R \oplus \mathbb{C}[h]$, where h acts as the multiplier $h \cdot x^k a = x^k(\lambda - ik)a$ and $x^k a \cdot h = x^k a\lambda$. This is a unital algebra containing R as an ideal, with $\tilde R/R = \mathbb{C}[h]$ and the obvious linear section. By Lemma 16.5(b) and Theorem 16.3, $C(R) \to \mathcal{K} = \ker(C(\tilde R) \to C(\mathbb{C}[h]))$ is a quasi-isomorphism. Both complexes split by weight, and the map respects the splitting, since h has weight 0. By (22), $[h, x^k a] = -ik\,x^k a$, so $L_h = -ig$ on chains of weight g. The homotopy s_h of Lemma 16.6 preserves $\mathcal{K}$, weights, and pole bounds. Therefore $\Sigma = \sum_{g\neq 0}(-ig)^{-1} s_h P_g$, where P_g is the projection to weight g, is a well-defined contracting homotopy of $\mathcal{K}^{\neq 0}$. Hence $H(\mathcal{K}^{\neq 0}) = 0$, and so $H(C^{\neq 0}(R)) = 0$. □

18.4 Generators and the theorem

For $A \in \mathcal{I}_\partial^b$ with $T_H(A) = \sum_k x^k b_k$, define

$$\mathrm{Tr}_{\partial,H}(A) = \frac{1}{2\pi}\int_{\mathbb{R}} \mathrm{Tr}_H\; b_0(\lambda)\,d\lambda, \tag{25}$$

$$\varphi_H(A,B) = \mathrm{Tr}_{\partial,H}\big(A\,D(B)\big), \qquad T(D(B)) = \sum_k x^k\, i\partial_\lambda b_k. \tag{26}$$

Proposition 18.4. *(a) $\mathrm{Tr}_{\partial,H}$ is a continuous trace on $\mathcal{I}_\partial^b$.*

(b) $\mathrm{Tr}_{\partial,H} \circ D = 0$, and φ_H is a continuous cyclic Hochschild 1-cocycle on $\mathcal{I}_\partial^b$.

(c) There are elements e'_H and 1-cycles z_H of weight zero with $\mathrm{Tr}_{\partial,H'}(e'_H) = \delta_{HH'}$ and $\varphi_{H'}(z_H) = \delta_{HH'}$.

Proof. (a) By (24), for $j+k=0$,

$$\operatorname{Tr}_\partial((x^j a)(x^k b)) = \frac{1}{2\pi}\int \operatorname{Tr}(a(\lambda - ki)b(\lambda))\,d\lambda, \quad \operatorname{Tr}_\partial((x^k b)(x^j a)) = \frac{1}{2\pi}\int \operatorname{Tr}(b(\lambda + ki)a(\lambda))\,d\lambda,$$

and both vanish if $j+k\neq 0$. Shift the contour by $\lambda \mapsto \lambda - ki$. This is justified by Cauchy's theorem on a rectangle, because the integrand is Schwartz on the strip and the vertical sides therefore tend to zero. The trace property of Tr_H on smoothing operators then gives equality.

(b) $\operatorname{Tr}_\partial(D(b)) = \frac{i}{2\pi}\int \operatorname{Tr}\,\partial_\lambda b_0\,d\lambda = 0$. Using (a) and the Leibniz rule, $(b\varphi)(a,b,c) = \operatorname{Tr}(abDc) - \operatorname{Tr}(aD(b)c) - \operatorname{Tr}(abDc) + \operatorname{Tr}(caDb) = 0$, and $\varphi(b,a) = -\operatorname{Tr}(D(b)a) = -\varphi(a,b)$.

(c) Let e_H be a rank-one projection in $\Psi^{-\infty}(H)$, and let $f, g \in \mathcal{E}$. Put $e'_H = fe_H$ with $\int f = 2\pi$, and $z_H = (fe_H)\otimes(ge_H)$ in weight zero. Then $bz_H = fge_H - gfe_H = 0$ and $\varphi_H(z_H) = \frac{i}{2\pi}\int fg'\,d\lambda$. For $g = e^{-\lambda^2}$ and $f = -c\lambda e^{-\lambda^2}$, this equals $\frac{ic}{\pi}\int \lambda^2 e^{-2\lambda^2}d\lambda \neq 0$; normalize c. The components $H' \neq H$ do not see these elements. □

Theorem 18.5. $\mathrm{HH}_k(\mathcal{I}^b_\partial) \cong \mathbb{C}^N$ *for* $k = 0, 1$, *and* $\mathrm{HH}_k(\mathcal{I}^b_\partial) = 0$ *for* $k \geq 2$. *The functionals* $\operatorname{Tr}_{\partial,H}$ *and* φ_H *are dual bases in degrees* 0 *and* 1. *This uses (Q1) and (Q2).*

Proof. By Proposition 15.3, $\mathcal{I}^b_\partial = \bigoplus_H R_H$. Excision for the split ideals R_H (Theorem 16.3) reduces the computation to a single $R_H = B_H[[x;\alpha]$. Proposition 18.3 reduces it further to the weight-zero part of $B_H \rtimes_\alpha \mathbb{Z} = \Psi^{-\infty}(H)\widehat{\otimes}(\mathcal{E}\rtimes_\alpha \mathbb{Z})$. Lemma 16.7, applied weight by weight, reduces it to $C^{(0)}(\mathcal{E}\rtimes_\alpha\mathbb{Z})$. Theorem 16.8 and Proposition 18.1(d) then give exact sequences with $\mathrm{HH}_q(\mathcal{E})_\alpha$ and $\mathrm{HH}_{q-1}(\mathcal{E})^\alpha$. By Lemma 18.2, $\mathrm{HH}_0(\mathcal{E})_\alpha \cong \mathbb{C}$, $(\mathcal{E}\,d\lambda)_\alpha \cong \mathbb{C}$, and all invariants vanish. Hence $H_0 \cong \mathbb{C}$, $H_1 \cong \mathbb{C}$, and $H_q = 0$ for $q \geq 2$. By Proposition 18.4, $\operatorname{Tr}_{\partial,H}$ and φ_H are nonzero on these one-dimensional groups. □

Remark 18.6. In the language of [21, (16), (68)], $\mathrm{HH}_k(\mathcal{I}^b_\partial) \cong \bigoplus_H H^{1-k}(S_\partial)$. The trace $\operatorname{Tr}_{\partial,H}$ is a residue in x (Lemma 19.1) and corresponds to $d\theta_\partial$. The class of φ_H corresponds to 1. In the cusp case, Melrose–Nistor reach the same answer by an x-adic spectral sequence whose E^1-term is commutative; here the weight argument replaces it.

19 The b-trace and the Laurent residual algebra

Lemma 19.1 (Diagonal residues)**.** *Let* $A \in x^{-\ell}\Psi_b^{-\infty}(X)$. *The function* $\operatorname{Tr}(x^w A) = \int_X x^w K_A(p,p)\,\mu_b(p)$ *is holomorphic for* $\operatorname{Re} w > \ell$, *extends meromorphically to* $\mathbb{C}$ *with at most simple poles at* $w \in \{-k : k \geq -\ell\}$, *and*

$$\operatorname{Res}_{w=0}\operatorname{Tr}(x^w A) = \sum_H \operatorname{Tr}_{\partial,H}(T(A)).$$

The same holds for holomorphic families $A(w)$, *with residue* $\sum_H \operatorname{Tr}_{\partial,H}(T(A(0)))$.

Proof. On Δ_b the coordinates are $s=1$, $x'=x$, and $y=y'$, so $K_A(p,p) \sim \sum_{k\geq -\ell} x^k \kappa_k(1,y,y)$. Moreover $\kappa_k(1) = \beta_k(1) = \frac{1}{2\pi}\int b_k\,d\lambda$ by (21). Away from the collar the integral is entire. In the collar, the term x^k contributes $\int_0^\varepsilon x^{w+k-1}dx = \varepsilon^{w+k}/(w+k)$, and the Taylor remainder of order M is holomorphic for $\operatorname{Re} w > -M$. The residue at $w=0$ is $\int_H \kappa_0(1,y,y)\nu_\partial$, which is $\frac{1}{2\pi}\int \operatorname{Tr}_H b_0\,d\lambda$. □

We define the b-trace by ${}^b\operatorname{Tr}(A) = \mathrm{FP}_{w=0}\operatorname{Tr}(x^w A)$ for $A \in x^{-\mathbb{Z}}\Psi_b^{-\infty}(X)$. On $\mathcal{I}^b$ it is the trace, and on $\Psi_b^{-\infty}$ it is the b-trace of [10, Sect. "The b-trace defect"].

Theorem 19.2 (b-trace defect). *For $A, B \in x^{-\mathbb{Z}}\Psi_b^{-\infty}(X)$ with $T_H(A) = \sum x^k a_k$ and $T_H(B) = \sum x^k b_k$,*

$$ {}^b\operatorname{Tr}([A,B]) = \sum_H \operatorname{Tr}_{\partial,H}\big(D(A)B\big) = \sum_H \frac{i}{2\pi} \sum_k \int_{\mathbb{R}} \operatorname{Tr}_H\big((\partial_\lambda a_k)(\lambda + ki)\, b_{-k}(\lambda)\big)\, d\lambda. $$

For $A, B \in \Psi_b^{-\infty}(X)$ this is $\frac{i}{2\pi}\sum_H \int \operatorname{Tr}_H(\partial_\lambda I_H(A,\lambda)\, I_H(B,\lambda))\, d\lambda$, which is the formula of [10, Thm. "b-trace defect formula"].

Proof. For $\operatorname{Re} w$ large, $x^w AB$, $x^w BA$ and $Ax^w B$ are trace class (Q0), and their traces are integrals of their kernels over the diagonal. Both $\operatorname{Tr}(x^w BA)$ and $\operatorname{Tr}(Ax^w B)$ equal the absolutely convergent double integral $\iint x(p)^w K_B(p,q) K_A(q,p)\, \mu_b \mu_b$. Hence

$$ \operatorname{Tr}(x^w[A,B]) = \operatorname{Tr}\big((x^w A - Ax^w)B\big) = w\, \operatorname{Tr}\big(x^w C(w) B\big), \qquad C(w) = \frac{A - x^{-w}Ax^w}{w}. $$

By Lemma 14.1, $C(w)$ has kernel $w^{-1}(1 - s^{-w})K_A$. This is an entire family in $x^{-\ell}\Psi_b^{-\infty}$ with $C(0) = D(A)$. By Lemma 19.1, $\operatorname{Tr}(x^w C(w) B)$ has at most a simple pole at 0, with residue $\sum_H \operatorname{Tr}_{\partial,H}(D(A)B)$. Hence $\operatorname{Tr}(x^w[A,B])$ is regular at 0, and its value there is that residue. The explicit form follows from (24): the constant term of $(x^k i\partial_\lambda a_k)(x^{-k} b_{-k})$ is $\alpha^{-k}(i\partial_\lambda a_k) b_{-k}$. □

Theorem 19.3. *Let $\mathcal{M} = x^{-\mathbb{Z}}\Psi_b^{-\infty}(X)$. In the long exact sequence of $0 \to \mathcal{I}^b \to \mathcal{M} \to \mathcal{I}_\partial^b \to 0$, the composite $\operatorname{Tr}\circ\partial : \mathrm{HH}_1(\mathcal{I}_\partial^b) \to \mathbb{C}$ equals $-\sum_H \varphi_H$. Consequently $\mathrm{HH}_0(\mathcal{M}) \cong \mathbb{C}^N$, spanned by the residues $\operatorname{Res}_{w=0}\operatorname{Tr}(x^w \cdot)$ at the components, and $\mathrm{HH}_1(\mathcal{M}) \cong \mathbb{C}^{N-1}$, while $\mathrm{HH}_k(\mathcal{M}) = 0$ for $k \geq 2$.*

Proof. The ideal $\mathcal{I}^b$ is H-unital (Lemma 16.5), and $\mathcal{I}^b$ is a closed ideal of the filtered nuclear algebra $x^{-\mathbb{Z}}\Psi_b^{-\infty}$, whose quotient is identified with $\mathcal{I}_\partial^b$ by Proposition 15.3. So Theorem 16.3 applies. Let $c = \sum a_i \otimes b_i$ be a 1-cycle, with lifts $\tilde{a}_i, \tilde{b}_i$. Then $\partial[c] = [\sum[\tilde{a}_i, \tilde{b}_i]]$. Since ${}^b\operatorname{Tr} = \operatorname{Tr}$ on $\mathcal{I}^b$, Theorem 19.2 gives $\operatorname{Tr}\partial[c] = \sum_{H,i} \operatorname{Tr}_{\partial,H}(D(a_i)b_i) = -\sum_{H,i}\varphi_H(a_i, b_i)$, using the cyclicity in Proposition 18.4. This map $\mathbb{C}^N \to \mathbb{C}$ is surjective. Now insert this into the long exact sequence, together with Proposition 17.1 and Theorem 18.5. □

Remark 19.4 (Dependence on choices). The algebras and extensions depend only on X, because $x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}$ and the ideals are unchanged under $x \mapsto ax$ with $0 < a \in C^\infty(X)$. The Taylor maps T_H depend on x and on the collar. The residue $\operatorname{Tr}_{\partial,H} = \operatorname{Res}_{w=0}\operatorname{Tr}(x^w \cdot)$ does not: since $a^w = 1 + w\log a + O(w^2)$, the extra term $w\operatorname{Tr}(x^w \log a\, A)$ is regular at 0. The b-trace changes by $\operatorname{Tr}_\partial(\log a \cdot)$. The derivation changes to $D' = D + [\log a, \cdot]$, so

$$ \varphi'_H(A,B) = \varphi_H(A,B) + \operatorname{Tr}_{\partial,H}(A[\log a, B]) = \varphi_H(A,B) - (b\psi)(A,B), \qquad \psi(A) = \operatorname{Tr}_{\partial,H}(A\log a), $$

using the trace property. Hence the class of φ_H is independent of the choices.

Remark 19.5. Theorem 19.3 is a homological form of the statement that the b-trace is not a trace. The trace on $\mathcal{I}^b$ admits no extension to a trace on $\mathcal{M}$, and the obstruction is exactly $\sum_H \varphi_H$. The traces of $\mathcal{M}$ are the N boundary residues.

Part III
Assembly and the index functional

20 The interior ideal and the boundary symbol algebra

20.1 Convergence and degeneration

The algebras $\mathcal{I}^b_\sigma$ and $\mathcal{A}^b_{\partial,\sigma}$ are computed by the same order spectral sequence as $\mathcal{A}^b_\sigma$ in Part I; only the coefficient functions change. In this subsection we separate the purely algebraic part of the convergence question from the analytic one.

Let $(C_\bullet, b)$ be a complex such that each C_k is a *subspace* of the space of series $c = \sum_{p\le P} c_p$ with $c_p \in C_{p,k}$, containing every component $C_{p,k}$, where $b = \sum_{l\ge 0} b_l$ and $b_l : C_{p,k} \to C_{p-l,k-1}$. Put $F_pC_k = \{c : c_q = 0 \text{ for } q > p\}$, let $E^1_{p,k} = H_k(C_{p,\bullet}, b_0)$, and let d^1 be induced by b_1. Since $\sum_{i+j=l} b_ib_j = 0$, d^1 is well defined and $(d^1)^2 = 0$; let E^2 be its homology. Because C_k is only a subspace, completeness is a real condition. We call the filtration *complete* if every series $\sum_j e_j$ with $e_j \in F_{p_j}C_k$, $\sup p_j < \infty$ and $p_j \to -\infty$, defines an element of C_k, and b commutes with such sums.

Proposition 20.1 (Algebraic convergence)**.** *Assume the filtration is complete and that, for some integer n, $E^2_{p,k} = 0$ whenever $p \ne k - n$. Then every class in $H_k(C, b)$ has a representative in $F_{k-n}C_k$, and*

$$\Theta : H_k(C, b) \to E^2_{k-n,k}, \qquad [c] \mapsto [c_{k-n}],$$

is an isomorphism. All differentials d^r, $r \ge 2$, vanish.

Proof. This is the argument of [19, Ch. "Hochschild homology", convergence proposition], written out in our notation.

Step (P). Let $c \in F_pC_k$ with $b_0c_p = 0$ and $b_1c_p + b_0c_{p-1} = 0$, and suppose $[c_p] = 0$ in $E^2_{p,k}$. Then $[c_p] = d^1[x]$ for a b_0-cycle $x \in C_{p+1,k+1}$, so $c_p - b_1x = b_0y$ with $y \in C_{p,k+1}$. The components of $b(x + y)$ of degrees $p + 1$ and p are $b_0x = 0$ and $b_1x + b_0y = c_p$, so $c - b(x + y) \in F_{p-1}$.

Representatives. If $c \in F_PC_k$ is a cycle and $P > k - n$, then $E^2_{P,k} = 0$, so (P) lowers P. Finitely many steps reach F_{k-n}.

Well defined. Let $c = be \in F_{k-n}$ with $e \in F_q$. If $q > k + 1 - n$, then (P) applies to e, since $b_0e_q = (be)_q = 0$, $b_1e_q + b_0e_{q-1} = (be)_{q-1} = 0$ and $E^2_{q,k+1} = 0$. It replaces e by an element of F_{q-1} with the same boundary. Once $q \le k + 1 - n$, we have $c_{k-n} = b_1e_{k+1-n} + b_0e_{k-n}$, so $[c_{k-n}] = d^1[e_{k+1-n}] = 0$ in E^2.

Injective. If $bc = 0$, $c \in F_{k-n}$, and $[c_{k-n}] = 0$, apply (P) repeatedly; this is possible because $E^2_{p,k} = 0$ for $p < k - n$. It yields $e_p \in F_{p+1}C_{k+1}$, $p = k - n, k - n - 1, \ldots$, with $c - b(\sum_{p\ge P} e_p) \in F_{P-1}$. By completeness, $c = b\sum_p e_p$.

Surjective. Let $x \in C_{k-n,k}$ be a b_0-cycle with $d^1[x] = 0$. Then $b_1x = -b_0y$ for some y, and $c^{(1)} = x + y$ has $bc^{(1)} \in F_{k-n-2}$. Inductively, if $z = bc^{(r)} \in F_p$ with $p = k - n - r - 1$, then z is a cycle and $p \ne k - 1 - n$, so (P) gives x', y' of degree $\le p + 1 < k - n$ with $z - b(x' + y') \in F_{p-1}$. Set $c^{(r+1)} = c^{(r)} - x' - y'$. By completeness the limit c exists, $bc = 0$, and $\Theta[c] = [x]$.

Degeneration. The differential d^r maps $E^r_{p,k}$ to $E^r_{p-r,k-1}$. Source and target can both be nonzero only if $p = k - n$ and $p - r = (k - 1) - n$, that is, only if $r = 1$. Hence $d^r = 0$ for $r \ge 2$. □

The full chain complex fails the completeness condition of Proposition 20.1. For example, $r^j \otimes r^{-2j}$ has total order $-j$, and the sum over $j \ge 1$ has unbounded first-slot order. The per-factor

stages are also unstable under the boundary. Take an element a with scalar principal symbol of order $N > 0$ whose square is nonzero. At the associated graded level the boundary of $a^{\otimes 3}$ has components $2a^2 \otimes a$ and $-a \otimes a^2$ of respective multidegrees $(2N, N)$ and $(N, 2N)$. Thus no enlargement of the per-factor stage makes it a subcomplex. For flat symbols one can take the principal symbol supported in the interior. We instead use a chain complex with a separate bound on the coefficient slot.

Lemma 20.2 (A mixed complex for Laurent complete symbols)**.** *Let A be either $\mathcal{A}^b_\sigma$ or $\mathcal{A}^b_{\partial,\sigma}$, and let $B \subset A$ be its subalgebra of symbols of order at most zero and without boundary poles. For $L, M \geq 0$ write $A_{L,M}$ for the subspace of pole order at most L and symbol order at most M. The natural inclusion*

$$C_k(B, A) = \varinjlim_{L,M} A_{L,M} \widehat{\otimes} B^{\widehat{\otimes} k} \longrightarrow C_k(A)$$

is a quasi-isomorphism. Here $C(B, A)$ uses the Hochschild differential with coefficients in the B-bimodule A. Every $A_{L,M} \widehat{\otimes} B^{\widehat{\otimes} k}$ is a subcomplex in all degrees and is complete for sums of decreasing total order.

Proof. Use full-symbol coordinates obtained by left quantization in b-coordinate charts. A partition of unity in the left variable and cutoffs equal to one at the diagonal give a quantization of each homogeneous symbol that commutes with left multiplication by scalar base functions. Use product charts and cutoffs near the boundary. Successively subtracting the quantized principal symbols extracts the sequence of homogeneous terms. Asymptotic summation realizes any such sequence modulo smoothing, with continuous coordinate maps at each finite order quotient. The complete-symbol topology at fixed pole bound is the product topology of these terms. The same construction in the collar commutes with Taylor expansion at the boundary. We use (Q0) and (C3) for this calculus. Put $t = r^{-1}$. Take an elliptic complete symbol T of order -1, without poles, with principal symbol t. The symbolic parametrix is an inverse in the complete-symbol algebra and has order 1 and no poles. For $\mathcal{A}^b_{\partial,\sigma}$ take the Taylor image of this construction. Conjugation by x^L preserves order and smooth coefficients for every integer L, by the b-composition formula. Thus with $u_{L,M} = x^{-L} T^{-M}$ one has

$$A_{L,M} = u_{L,M} B = B u_{L,M}.$$

Indeed multiplication by $u_{L,M}$ has the asserted bounds, and its inverse on each side takes a symbol with those bounds into B. The right inverse calculation uses conjugation by x^L. All these maps are continuous on fixed stages by (C3). In particular each $A_{L,M}$ is a B-bimodule. The mixed Hochschild boundary preserves the fixed coefficient stage. On this stage the full-symbol splitting writes the chain space as a product of homogeneous tensor components, using commutation of completed nuclear tensor products with countable products. At each total order there are finitely many multidegrees, since the coefficient has order at most M and every other slot has order at most zero. Thus sums tending to minus infinity in total order converge there, and the boundary commutes with them since it does not increase total order.

For the comparison, use the augmented bar bimodule complexes

$$P_k = A \widehat{\otimes} B^{\widehat{\otimes} k} \widehat{\otimes} A, \qquad Q_k = A \widehat{\otimes} A^{\widehat{\otimes} k} \widehat{\otimes} A,$$

where every appearance of A is completed on bounded stages before taking their algebraic union. Their differentials multiply consecutive slots, without the cyclic face, and the augmentation in degree zero is multiplication. On the subcomplex of P whose rightmost slot lies in Bu for a fixed $u = u_{L,M}$, leave the leftmost slot unrestricted. There is a contraction

$$h_k(a \otimes \beta \otimes bu) = (-1)^{k+1} a \otimes \beta \otimes b \otimes u, \qquad h_{-1}(a) = a u^{-1} \otimes u.$$

Here β denotes the middle slots. The face that multiplies the last two slots of h_k gives the identity. All earlier faces cancel against $h_{k-1}d$, including the augmentation when $k = 0$. These identities extend from elementary tensors by continuity. The rightmost-slot bound is preserved and each fixed bound on the leftmost slot is sent into a bounded stage. In Q the same contraction uses $u = 1$ with middle slots in A.

For clarity, the augmented comparison is valid on the completions as follows. The diagonal stages $A_{N,N}$ split continuously inside $A_{N+1,N+1}$. In full-symbol coordinates first remove the top order and then remove the lowest boundary coefficient of every remaining symbol by extending that coefficient in a collar with a cutoff. This gives a continuous projection on the smaller stage. For formal boundary symbols remove that coefficient in the formal series. The procedure uses only one boundary jet per homogeneous coefficient, continuously in the product topology. It does not require a continuous section for the entire Taylor map. Tensor powers of these projections give splittings of the corresponding nested generator stages of Q_k, and the generator space is their algebraic direct sum of successive increments.

The inclusion $P \to Q$ is the identity on the augmentation. Construct an inverse chain map recursively on the free generators $1 \otimes \beta \otimes 1$ of Q and extend A-bilinearly. In degree zero use the identity. In degree k, the already constructed image of the differential of a generator is a cycle in P_{k-1}. On each generator increment that image has a common rightmost bound, so apply the displayed contraction for that bound to lift the cycle. Extend the resulting map by linearity on the sum of increments. Each original stage involves finitely many increments, hence has continuous image in a bounded stage. The bilinear extension is continuous into a bounded stage by the continuity of multiplication and the projective tensor property. For either composite, construct a homotopy to the identity by the same recursion: on generators subtract the homotopy already applied to the differential from the difference of the two maps, and lift the resulting cycle. For generators of P the entire middle tensor power is a single Fréchet stage. This gives stagewise continuous bimodule comparison maps and homotopies.

Finally send $a \otimes \beta \otimes a'$ to $\tau(a \otimes \beta \otimes a') = a'a \otimes \beta$, and write $s(a \otimes \beta) = a \otimes \beta \otimes 1$. For every bimodule map D just constructed, $\tau D = \tau D s \tau$: move a' from the right to the left under τ, first on elementary tensors and then by density. The bar boundary descends under τ to the Hochschild boundary, its last face becoming the cyclic face. Thus the comparison and both homotopies descend, and the induced map of the inclusion is the claimed mixed inclusion. □

Lemma 20.3 (Lifts in a fixed coefficient stage)**.** *Let A and B be as in Lemma 20.2. Fix $L \geq 1$ and $M \geq 0$. In the mixed stage $A_{L,M}\widehat{\otimes}B^{\widehat{\otimes}k}$, step (P) of Proposition 20.1 stays in that stage whenever $p \leq M - 2$ and $p \neq k - n$.*

Proof. The graded algebra of B is $C^\infty(Y)[t]$ in the smooth case and $C^\infty(Z)[[x]][t]$ in the formal case, with $\deg t = -1$. The graded coefficient module of the mixed stage is the free symmetric module generated by $x^{-L}t^{-M}$. Apply Lemma 5.2 to the graded stage, with that generator shifting the weight by M. In weights $q \leq M - 1$ the resulting forms are exactly homogeneous forms of weight q in

$$L^k_L = x^{-L}\Omega^k \quad \subset \quad W^k_L = x^{-L}\Omega^k_b.$$

In the formal case ordinary forms and b-forms are interpreted by their Taylor series. To see the radial bound, write a form as $r^q(\alpha + d\log t \wedge \beta)$ with α, β on the sphere bundle. The term involving $d\log t$ uses $t^{-q-1}dt$, whose coefficient order is $q + 1$, while the first term uses coefficient order q. There is no additional bound for either term if $q \leq M - 1$. The ordinary boundary differential is dx, so in the b-frame the coefficient of dx/x must have one extra power of x.

The first differential is $i\delta_b$ by the same antisymmetrization as in Section 6, with the first entry allowed in the symmetric graded coefficient module. Both L_L and W_L are preserved by δ_b. For the former this follows directly from (5): the b-Poisson bracket is smooth on ordinary smooth functions and sends $x^{-L}C^\infty \times C^\infty$ into $x^{-L}C^\infty$. For the latter use the smooth b-frame and (6).

The quotient W_L/L_L consists of the leading x^{-L} coefficient of the dx/x component at the boundary. It is acyclic in the homogeneous Poisson complex. In a collar let ξ be the homogeneous coefficient of dx/x in the tautological form. Exterior multiplication by $d\xi$ preserves both spaces, and with the convention of (5) one has

$$\delta_b(d\xi \wedge \cdot) + (d\xi \wedge \cdot)\delta_b = \mathcal{L}_{x\partial_x}.$$

Indeed $[\iota_{\pi_b}, d\xi \wedge \cdot] = -\iota_{x\partial_x}$; use $d(d\xi) = 0$ and Cartan's identity. In the smooth case a collar cutoff equal to one near the boundary makes the exterior multiplication global and does not affect the identity on the quotient. The Lie derivative is multiplication by $-L$ there. The resulting contraction raises fiber weight by one.

Suppose now that $\theta \in L_L^k$ has fiber weight p and $\delta_b\theta = 0$, with $p \neq k - n$. In W_L apply $*_b$, contract with the fiber Euler field, and divide by the nonzero weight $p + n - k$. By (6), with its degree sign, this gives a primitive η of weight $p + 1$ in W_L^{k+1}. Its quotient class is closed and thus has a quotient primitive of weight $p + 2$, by the preceding contraction. Lift this primitive in W_L at weight $p + 2$ by extending its boundary coefficient in the collar (or inserting its formal coefficient), and subtract its δ_b from η. The resulting primitive lies in L_L^{k+1} and has weight $p + 1$. The temporary lift is in the form complex W_L and is not a chain-stage lift.

In step (P), the HKR class of c_p is δ_b-closed by the equation at order $p - 1$. The primitive just constructed, divided by i, is the class of a b_0-cycle x in the same stage at order $p + 1$, since $p + 1 \leq M - 1$. The difference $c_p - b_1 x$ is a b_0-boundary in the same stage at order p, again by the fixed-stage HKR calculation. This gives y at that order. The fixed stage is a subcomplex, so the remainder stays there. □

These two lemmas imply the convergence conclusion of Proposition 20.1 for the full mixed complexes. In the direct limit, the E^1-page is the full Laurent or formal Laurent form complex, since each homogeneous form belongs to some L_L with arbitrarily large radial bound. Its E^2 is concentrated at $p = k - n$ by Propositions 9.1 and 20.8. The finite order reductions in the proof of Proposition 20.1 require no completeness and can be done in the direct limit. For injectivity, perform the initial step (P) at $p = k - n$ there and choose a stage containing the remainder, enlarging M so that all subsequent orders are $\leq M - 2$ and taking $L \geq 1$. For surjectivity, choose the initial two terms in the direct limit, then choose the coefficient stage containing them and enlarge M in the same way. All subsequent steps are off the critical row and stay in the one stage by Lemma 20.3. The infinite sums converge there by Lemma 20.2. The mixed inclusions preserve total order and give the same HKR map on leading symbols. The finite well-definedness argument of Proposition 20.1 applies in the ordinary chain model as well, by its E^1- and E^2-calculations. Consequently the isomorphism with E^2 in the ordinary model is also the leading-symbol map. This construction is natural under the Taylor map from the smooth to the formal symbols.

Lemma 20.4 (The flat symbol ideal)**.** *The same leading-symbol isomorphism holds for $J = \mathcal{I}_\sigma^b$.*

Proof. First J is topologically H-unital in the chain model. Choose a smooth locally finite partition of unity (ψ_j) on the interior of X, using a dyadic decomposition of a boundary collar and one compactly supported interior term. For collar terms $x \asymp 2^{-j}$ on the support. Choose ϕ_j supported

in a slightly larger comparable region and equal to one on $\operatorname{supp}\psi_j$. On the cofinal stages of nonnegative upper symbol order define

$$E(a_0 \otimes c) = \sum_j \phi_j \otimes (\psi_j a_0) \otimes c.$$

The smooth multiplication operators ϕ_j, ψ_j have order zero and are supported in the interior. In the full-symbol topology, each fixed seminorm of ϕ_j grows at most as a power of 2^j. Each fixed flat-symbol seminorm of $\psi_j a_0$ is bounded by $O(2^{-jK})$ times a stronger flat-symbol seminorm of a_0, for arbitrary K. Indeed the cutoff derivatives cost fixed powers of 2^j, and flatness gives an arbitrarily high power of x on the support. These estimates give convergence and continuity in each projective tensor stage, without increasing order. Associativity and $\sum_j \phi_j \psi_j = 1$ on the interior give $d_0 E = 1$ and $d_{i+1} E = E d_i$, hence $b'E + Eb' = 1$. By excision (Q1), $C(J)$ maps quasi-isomorphically to the kernel of the ordinary Taylor map on chains.

Take also the kernel of the Taylor map between the two mixed complexes. These Taylor maps are surjective on chains on every fixed pair of coefficient bounds, since Taylor is surjective on the fixed symbol stages and completed nuclear tensor products preserve the quotients. They are surjective also on associated graded pieces, by the homogeneous coefficient description. The mixed inclusions give a quasi-isomorphism between the mixed kernel and the ordinary kernel by their homology exact sequences. At each fixed mixed stage the b_0-homology of the kernel is the kernel of Taylor on the fixed-stage HKR forms: the Taylor map on those forms is surjective, by coefficientwise Borel extension. This kernel consists of flat forms, and for weights $q \leq M - 1$ it is the full space of flat forms of weight q. The Poisson primitive obtained from $*_b$ and the fiber Euler contraction is flat and needs no boundary correction. Thus the same fixed-stage descent applies to the mixed kernel. The E^2-page is as in Proposition 20.7. On associated graded complexes the inclusion of $C(J)$ into the ordinary kernel is the flat HKR identification, by the exact sequence on graded chains and the surjectivity on forms. These filtered maps therefore identify the leading-symbol maps as well. Equivalently, the finite reduction and well-definedness arguments of Proposition 20.1 apply directly to $C(J)$, and its homology is identified via the two kernel comparisons with the homology computed by fixed stages. This gives the asserted isomorphism in the original flat chain model. □

20.2 Two de Rham lemmas

Lemma 20.5 (Flat forms)**.** *Let V be a compact manifold with boundary, and let $\dot\Omega^*(V)$ be the complex of forms vanishing to infinite order at ∂V. Then $H^*(\dot\Omega^*(V)) \cong H^*(V, \partial V)$.*

Proof. Let J^* be the complex of formal Taylor series at ∂V of forms on V. In a collar these are $\sum_{k\geq 0} x^k(a_k + dx \wedge c_k)$ with a_k, c_k forms on ∂V. By Borel's lemma,

$$0 \to \dot\Omega^*(V) \to \Omega^*(V) \to J^* \to 0$$

is exact. On J^* put $K(\sum x^k(a_k + dx \wedge c_k)) = \sum x^{k+1} c_k/(k+1)$, which is the formal integral from 0. The fundamental theorem of calculus gives $dK + Kd = 1 - \pi^*\rho$, where ρ restricts to $x = 0$ and π is the collar projection. Hence $\rho : J^* \to \Omega^*(\partial V)$ is a quasi-isomorphism. The relative complex $\Omega^*(V, \partial V) = \Omega^*(V) \oplus \Omega^{*-1}(\partial V)$, with $D(\alpha, \beta) = (d\alpha, \alpha|_{\partial V} - d\beta)$, receives the chain map $\alpha \mapsto (\alpha, 0)$ from $\dot\Omega^*(V)$. This map is compatible with the long exact sequences of the displayed short exact sequence and of the pair $(V, \partial V)$. The five lemma gives the claim. □

Lemma 20.6 (Formal Laurent forms)**.** *Let W be a manifold, and let $\Omega^*(W)((x))$ be formal Laurent series in x with coefficients in forms on W, together with the generator dx/x. The complex $\Omega^*(W)((x)) \oplus \frac{dx}{x} \wedge \Omega^{*-1}(W)((x))$ has cohomology $H^*(W) \oplus \frac{dx}{x} H^{*-1}(W) \cong H^*(W \times S_\partial)$.*

Proof. Let $\mathcal{E}_x = x\partial_x$. It preserves the x-degree k, and on degree k we have $d\iota_{\mathcal{E}_x} + \iota_{\mathcal{E}_x} d = \mathcal{L}_{\mathcal{E}_x} = k$. For $k \neq 0$ the map $k^{-1}\iota_{\mathcal{E}_x}$ is a contracting homotopy. These homotopies can be summed over k, because they do not lower the x-degree. In degree 0 the complex is $\Omega^*(W) \oplus \frac{dx}{x} \wedge \Omega^{*-1}(W)$ with $d(\alpha_0 + \frac{dx}{x} \wedge \alpha_1) = d\alpha_0 - \frac{dx}{x} \wedge d\alpha_1$. □

20.3 The E^2-pages

Proposition 20.7 (Interior ideal)**.** *$E^2_{p,q}(\mathcal{I}^b_\sigma) = H^{2n-p-q}(Y \times S_\sigma, Z \times S_\sigma)$ for $q = n$, and $E^2_{p,q}(\mathcal{I}^b_\sigma) = 0$ otherwise. Moreover, $\mathrm{HH}_k(\mathcal{I}^b_\sigma) \cong H^{2n-k}(Y \times S_\sigma, Z \times S_\sigma)$. The ideal $\mathcal{I}^b_\sigma$ is the same algebra for the b- and cusp calculi.*

Proof. The arguments of Sections 5–7 apply verbatim to symbols that vanish to infinite order at $x = 0$, because π_b, $*_b$ and the Euler field preserve flatness. The weight-zero complex is $\dot\Omega^*(Y \times S_\sigma)$, and Lemma 20.5 computes its cohomology. The rows $q \neq n$ vanish as in Proposition 9.1, so the spectral sequence degenerates. Kernels conormal to the diagonal that vanish to infinite order at $\partial(X^2)$ do not depend on the blow-up, so $x^\infty \Psi^{\mathbb{Z}}_b = x^\infty \Psi^{\mathbb{Z}}_c$. Lemma 20.4 identifies this page with homology. □

Proposition 20.8 (Boundary symbol algebra)**.** *$E^2_{p,q}(\mathcal{A}^b_{\partial,\sigma}) = H^{2n-p-q}(Z \times S_\sigma \times S_\partial)$ for $q = n$, and $E^2_{p,q}(\mathcal{A}^b_{\partial,\sigma}) = 0$ otherwise. Moreover, $\mathrm{HH}_k(\mathcal{A}^b_{\partial,\sigma}) \cong H^{2n-k}(Z \times S_\sigma \times S_\partial)$.*

Proof. Elements of $\mathcal{A}^b_{\partial,\sigma}$ are formal Laurent series in x of complete b-symbols at ∂X. On the E^1-page the coefficients are formal Laurent series in x with values in homogeneous forms on ${}^bT^*_{\partial X}X \setminus 0$. The b-Poisson tensor $\pi_b = x\partial_x \wedge \partial_\xi + \sum \partial_{y_j} \wedge \partial_{\eta_j}$ and the duality $*_b$ preserve the x-degree, so d^1 and the symplectic duality of Section 6 apply degree by degree. The fibre Euler field kills the nonzero fibre weights (Section 7); the weight-zero part produces S_σ. Lemma 20.6, with $W = Z \times S_\sigma$, kills the nonzero x-degrees and produces S_∂. The row argument of Proposition 9.1 gives degeneration. The fixed coefficient stages of Lemmas 20.2 and 20.3 give the leading-symbol isomorphism. □

Remark 20.9 ($n = 1$). Let $n = 1$. Then Y is two copies of X ($\xi = \pm 1$), and each Z_H is two points with the orientation of ∂Y. Parts (iv)–(vii) of Theorem 1.1 hold as stated, with $\int_{Z_H}$ the signed sum over the two points. In (viii) the kernel of the signed-sum map from $H^0(Z) \cong \mathbb{C}^{2N}$ gives an additional summand in degree one. Concretely:

- $\mathrm{HH}_1(\mathcal{A}^b) \cong \mathbb{C} \oplus H^1(Y \times S_\sigma) \oplus \ker\big(H^0(Z) \to \mathbb{C}^N\big)$;
- $\mathrm{HH}_2(\mathcal{A}^b) \cong \ker\big(H^0(Y) \to \mathbb{C}^N\big)$, where the map is $\beta \mapsto (\int_{Z_H} \beta)_H$;
- for $k = 1, 2$, $\mathrm{HH}_k(\mathcal{A}^b_\partial)$ is the kernel of the surjection $\mathrm{HH}_k(\mathcal{A}^b_{\partial,\sigma}) \to \mathbb{C}^N$ given by the same signed sums.

For $n = 1$ the surjectivity of the connecting maps of $\mathcal{A}^b_\partial$ and of $\partial^{\mathcal{A}}_1$ is proved by explicit test cycles (Propositions 21.4 and 22.2(d)). For $X = [0, 1]$ with global coordinate s, label the two copies $Y_\pm$ by the sign of the covector relative to ds. They have orientations $\pm ds$. The map from $H^0(Y) = \mathbb{C}^2$ to $\mathbb{C}^2$ of oriented periods is $(c_+, c_-) \mapsto (c_- - c_+,\ c_+ - c_-)$ in endpoint order. By Proposition 21.8, the trace-dual connecting map is $(2\pi)^{-1}$ times this map. It has rank one and kernel the constants, so $\mathrm{HH}_2(\mathcal{A}^b) \cong \mathbb{C}$.

21 The indicial algebra $\mathcal{A}_\partial^b$

21.1 The regularized suspended trace

Let $a \in B'_H$ have order m and put $d = \dim H = n-1$. For $p > m+d$ the operator $\partial_\lambda^p a(\lambda)$ has order $m-p < -d$, so it is trace class. On $\mathbb{R}$ the function $f_p(\lambda) = \operatorname{Tr}_H \partial_\lambda^p a(\lambda)$ has a classical expansion $f_p \sim \sum_j c_j^\pm |\lambda|^{m-p+d-j}$ as $\lambda \to \pm\infty$. To see this, apply the trace formula $\operatorname{Tr} P = (2\pi)^{-d} \iint \sigma_P \, dy \, d\eta$ to the suspended symbol and substitute $\eta = |\lambda|\eta'$ in each homogeneous term. Let g_p be the p-fold iterated integral of f_p from 0. Then $\int_{-T}^{T} g_p \, d\lambda$ has an asymptotic expansion in powers T^j and terms $T^j \log T$ as $T \to \infty$. Define $\overline{\operatorname{Tr}}(a)$ as $1/2\pi$ times its constant term.

Lemma 21.1. *(1) $\overline{\operatorname{Tr}}(a)$ does not depend on p. If a has order $< -n$, then $\overline{\operatorname{Tr}}(a) = \frac{1}{2\pi}\int \operatorname{Tr}_H a \, d\lambda$.*

(2) $\overline{\operatorname{Tr}}([a,b]) = 0$ for $a, b \in B'_H$.

(3) If $a \in B_H$, then $\overline{\operatorname{Tr}}(\partial_\lambda^p a) = 0$ for $p \geq 1$.

Proof. (1) Since $g_{p+1}^{(p+1)} = f_{p+1} = f_p' = g_p^{(p+1)}$, the difference $g_{p+1} - g_p$ is a polynomial. The constant term of $\int_{-T}^{T} \lambda^j \, d\lambda$ is zero for every j. For order $< -n$, g_p differs from $\operatorname{Tr}_H a$ by its Taylor polynomial at 0, and $\operatorname{Tr}_H a$ is integrable. (2) Here $f_p([a,b]) = \sum_j \binom{p}{j} \operatorname{Tr}_H[\partial^j a, \partial^{p-j} b]$. Each term vanishes, because $\operatorname{Tr}(PQ) = \operatorname{Tr}(QP)$ for pseudodifferential operators on a closed manifold whose orders add up to less than $-d$. (3) follows from (1), since $\int \partial_\lambda(\operatorname{Tr} \partial^{p-1} a) \, d\lambda = 0$. □

Lemma 21.2 (Shift defect)**.** *Let $a \in B'_H$ have order m, let $c \in \mathbb{C}$, and let $P > m+n$. Then $a_c(\lambda) = a(\lambda + c)$ lies in B'_H, and*

$$\overline{\operatorname{Tr}}(a_c) - \overline{\operatorname{Tr}}(a) = \sum_{p=1}^{P-1} \frac{c^p}{p!} \overline{\operatorname{Tr}}(\partial_\lambda^p a).$$

Proof. By Taylor's formula, $a_c = \sum_{p<P} \frac{c^p}{p!} \partial^p a + R$, where $R(\lambda) = \frac{c^P}{(P-1)!} \int_0^1 (1-t)^{P-1} (\partial^P a)(\lambda + tc) \, dt$. The family $\partial^P a$ has order $< -n$, uniformly on strips, so by Lemma 21.1(1) we have $\overline{\operatorname{Tr}}(R) = \frac{1}{2\pi}\int \operatorname{Tr} R \, d\lambda$. A contour shift on the strip turns $\int \operatorname{Tr}(\partial^P a)(\lambda + tc) \, d\lambda$ into $\int \partial_\lambda \operatorname{Tr}(\partial^{P-1} a) \, d\lambda$. This vanishes, because $\partial^{P-1} a$ has order $< -d$ and its trace tends to 0 at $\pm\infty$. □

For $A \in \mathcal{A}_\partial^b$ with $T_H(A) = \sum x^k a_k$, put $\widehat{\operatorname{Tr}}_{\partial,H}(A) = \overline{\operatorname{Tr}}(a_0)$. By Lemma 21.1(1), this extends $\operatorname{Tr}_{\partial,H}$ from $\mathcal{I}_\partial^b$.

Proposition 21.3 (The connecting cocycle)**.** *For $A, B \in \mathcal{A}_\partial^b$ with $T_H(A) = \sum x^k a_k$ and $T_H(B) = \sum x^k b_k$,*

$$\omega_H(A,B) := \widehat{\operatorname{Tr}}_{\partial,H}([A,B]) = \sum_{k\neq 0} \sum_{p\geq 1} \frac{(ki)^p}{p!} \overline{\operatorname{Tr}}\big(\partial_\lambda^p (a_k \, \alpha^k(b_{-k}))\big), \tag{27}$$

a finite sum. It vanishes if A or B lies in $\mathcal{I}_\partial^b$, so it is a Hochschild 1-cocycle on $\mathcal{A}_{\partial,\sigma}^b$. In the long exact sequence of $0 \to \mathcal{I}_\partial^b \to \mathcal{A}_\partial^b \to \mathcal{A}_{\partial,\sigma}^b \to 0$ we have $\operatorname{Tr}_{\partial,H} \circ \partial = \langle \omega_H, \cdot \rangle$ on $\mathrm{HH}_1(\mathcal{A}_{\partial,\sigma}^b)$. The cocycle

$$\Omega_H(A,B,C) = -\omega_H(A \, D(B), C)$$

on $\mathcal{A}_{\partial,\sigma}^b$ satisfies $\varphi_H \circ \partial = \langle \Omega_H, \cdot \rangle$ on $\mathrm{HH}_2(\mathcal{A}_{\partial,\sigma}^b)$.

Proof. By (24), the constant term of $[A,B]$ is $\sum_k(\alpha^{-k}(a_k)b_{-k} - \alpha^k(b_{-k})a_k)$. Put $F_k = a_k\alpha^k(b_{-k})$. Then $\alpha^{-k}(F_k) = \alpha^{-k}(a_k)b_{-k}$, and $\overline{\mathrm{Tr}}(\alpha^k(b_{-k})a_k) = \overline{\mathrm{Tr}}(F_k)$ by Lemma 21.1(2). Since $\alpha^{-k}F(\lambda) = F(\lambda + ki)$, Lemma 21.2 gives (27). If $a_k \in B_H$ or $b_{-k} \in B_H$, then $F_k \in B_H$, and each term vanishes by Lemma 21.1(3). The ideal $\mathcal{I}_\partial^b$ is H-unital, so Lemma 16.4 applies with $\tau = \mathrm{Tr}_{\partial,H}$ and $\hat\tau = \widehat{\mathrm{Tr}}_{\partial,H}$.

For φ_H, take $\hat\varphi(A,B) = \widehat{\mathrm{Tr}}_{\partial,H}(AD(B))$. It extends φ_H, because D preserves $\mathcal{I}_\partial^b$. A direct expansion gives $b\hat\varphi(A,B,C) = -\widehat{\mathrm{Tr}}([AD(B),C])$, which vanishes when an argument lies in $\mathcal{I}_\partial^b$. Lemma 16.4 then gives the last assertion. □

Proposition 21.4 (Nonvanishing for $n = 1$)**.** *Let $n = 1$, so that each H is a point and B'_H is the algebra of entire functions with classical symbol expansions on strips. Let $\pi_+(\lambda) = \frac12(1 + \operatorname{erf}\lambda) \in B'_H$, and work in the summand of H. Then*

$$c_H = x^{-1}\pi_+ \otimes x, \qquad c'_H = x^{-1}\pi_+ \otimes h \otimes x - x^{-1}\pi_+ \otimes x \otimes h$$

are cycles in $\mathcal{A}^b_{\partial,\sigma}$, and $\langle \omega_{H'}, c_H\rangle = -\frac{i}{2\pi}\delta_{HH'}$ and $\langle \Omega_{H'}, c'_H\rangle = -\frac{1}{2\pi}\delta_{HH'}$. Hence both connecting maps $\mathrm{HH}_1(\mathcal{A}^b_{\partial,\sigma}) \to \mathrm{HH}_0(\mathcal{I}^b_\partial)$ and $\mathrm{HH}_2(\mathcal{A}^b_{\partial,\sigma}) \to \mathrm{HH}_1(\mathcal{I}^b_\partial)$ are surjective.

Proof. By (24), $(x^{-1}\pi_+)x = \alpha(\pi_+)$ and $x(x^{-1}\pi_+) = \pi_+$. Hence $bc_H = \pi_+(\lambda - i) - \pi_+(\lambda)$. This lies in $\mathcal{E}$, since erf tends to ±1 on strips with Gaussian error, so c_H is a cycle in $\mathcal{A}^b_{\partial,\sigma}$. In (27) only $k = -1$ occurs, with $F_{-1} = \pi_+$. Here $\overline{\mathrm{Tr}}(\pi'_+) = \frac{1}{2\pi}$, while $\overline{\mathrm{Tr}}(\pi_+^{(p)}) = 0$ for $p \ge 2$. Therefore $\omega_H(c_H) = -i/2\pi$.

For c'_H, use $hx = x(h - i)$ and $hx^{-1} = x^{-1}(h + i)$, which follow from (24). A direct computation gives $bc'_H = (\pi_+ - \alpha(\pi_+)) \otimes h$, which is zero in $\mathcal{A}^b_{\partial,\sigma}$. Since $D(h) = i$ and $D(x) = 0$, $\Omega_H(c'_H) = -\omega_H(i\,x^{-1}\pi_+, x) = -\frac{1}{2\pi}$. □

21.2 The connecting maps for $n \ge 2$

Identify the suspended variable λ with the b-covariable ξ; this is legitimate because xD_x has symbol ξ and indicial family λ. Then principal symbols of elements of B'_H are homogeneous functions on ${}^bT^*_HX \setminus 0 = (T^*H \times \mathbb{R}_\xi) \setminus 0$. Write $\zeta = (\eta, \xi)$ and $d = n - 1 = \dim H$, and put

$$\hat\rho_H(g) = (2\pi)^{-n}\int_H\int_{\mathbb{R}^d}\big(g(y,\eta,1) - g(y,\eta,-1)\big)\,d\eta\,\nu_\partial(y)$$

for g homogeneous of degree $1 - n$, with $d\eta$ the density dual to ν_∂. The integrand is $O(|\eta|^{-n})$, so the integral converges.

Lemma 21.5 (Residue formula)**.** *Let $G \in B'_H$. If G has order $\le -n$, then $\overline{\mathrm{Tr}}(\partial_\lambda^p G) = 0$ for $p \ge 1$. If G has order $1 - n$ with principal symbol g, then $\overline{\mathrm{Tr}}(\partial_\lambda G) = \hat\rho_H(g)$ and $\overline{\mathrm{Tr}}(\partial_\lambda^p G) = 0$ for $p \ge 2$.*

Proof. If the order is $\le -n$, then $\partial^p G$ has order $< -n$, so $\overline{\mathrm{Tr}}(\partial^p G) = \frac{1}{2\pi}\int \partial_\lambda \operatorname{Tr}(\partial^{p-1}G)\,d\lambda = 0$ by Lemma 21.1. Now let G have order $1 - n = -d$. The case $p \ge 2$ is the previous one applied to ∂G. Changing G by a family of order $-n$ does not change $\overline{\mathrm{Tr}}(\partial G)$. Since $\overline{\mathrm{Tr}}$ is defined for suspended families on the real line, and its definition does not use holomorphy, we may take $G = \mathrm{Op}(\chi g)$ for real λ, quantized in charts, with χ a cut-off near $\zeta = 0$. This family is not entire. From the definition of $\overline{\mathrm{Tr}}$ applied to $a = \partial G$ (order $-n$) with $p = 1$: here $g_1(\lambda) = \operatorname{Tr}\partial G(\lambda) - \operatorname{Tr}\partial G(0)$, and the term $2T\operatorname{Tr}\partial G(0)$ has no constant part. Hence $\overline{\mathrm{Tr}}(\partial G)$ is $\frac{1}{2\pi}$ times the constant term of $I(T) = \int_{-T}^{T}\operatorname{Tr}\partial_\lambda G\,d\lambda$. Since $\partial_\lambda G$ has order $-n < -d$, its trace is the integral of its symbol, $\operatorname{Tr}\partial_\lambda G = (2\pi)^{-d}\iint\partial_\lambda(\chi g)\,d\eta\,\nu_\partial$. This is absolutely integrable over $\{|\lambda| \le T\} \times \mathbb{R}^d$. By Fubini,

$$I(T) = (2\pi)^{-d}\int_H\int_{\mathbb{R}^d}\big(g(\eta,T) - g(\eta,-T)\big)\,d\eta\,\nu_\partial$$

for T large. Substituting $\eta = T\eta'$ and using homogeneity of degree $-d$ shows that $I(T)$ is independent of T, and equal to $(2\pi)\hat{\rho}_H(g)$. $\square$

For a chain c of $\mathcal{A}^b_{\partial,\sigma}$, let $\chi(c)$ be its HKR form, computed on principal symbols. Let θ_x and $\theta_{\xi x}$ denote the coefficients of dx/x and of $d\xi \wedge dx/x$ in the x-weight-zero part of a form θ.

Lemma 21.6 (Top-order formulas)**.** *(a) ω_H vanishes on 1-chains of total order $\leq -n$. On 1-chains of total order $\leq 1-n$, $\omega_H(c) = -i\,\hat{\rho}_H\big(\chi(c_{1-n})_x\big)$.*

(b) Ω_H vanishes on 2-chains of total order $\leq 1-n$. For 2-chains c of total order $\leq 2-n$ with $b_0 c_{2-n} = 0$, $\Omega_H(c) = -\hat{\rho}_H\big(\chi(c_{2-n})_{\xi x}\big)$.

Proof. (a) In (27) the term $F_k = a_k \alpha^k(b_{-k})$ has order at most the total order, and α^k does not change principal symbols. Lemma 21.5 gives $\omega_H(A \otimes B) = \sum_k ik\,\hat{\rho}_H(\sigma(a_k)\sigma(b_{-k}))$ in total order $1-n$, and 0 below. On the other hand $A\,dB = \sum x^k a_k\, d(x^{-k} b_{-k})$ has weight-zero dx/x-coefficient $-\sum_k k\, a_k b_{-k}$.

(b) The derivation D lowers order by one, so $\Omega_H(A,B,C) = -\omega_H(AD(B),C)$ is computed by (a). On elementary tensors this gives $\Omega_H(a\otimes b\otimes c) = -\sum m\,\hat{\rho}_H(a_j \partial_\xi b_l c_m)$, summed over $j+l+m=0$. Put $P = \partial_\xi$, $E = x\partial_x$, and $\ell = \hat{\rho}_H \circ [x^0]$ on symbols of order $1-n$. On a top elementary tensor $a\otimes b\otimes c$, the difference between Ω_H and $-\ell(\chi_2(\cdot)_{\xi x})$ is $-\frac{1}{2}\ell(a(Pb\,Ec + Eb\,Pc))$. Set $L(a,b) = \ell(aPEb)$. The derivation rules for P, E and commutativity of the graded product give $b_0 L(a,b,c) = -\ell(a(Pb\,Ec + Eb\,Pc))$. Thus the difference is $\frac{1}{2} b_0 L$, which vanishes on top b_0-cycles. These identities extend to completed chains by continuity on the fixed graded pieces. $\square$

Proposition 21.7 (The boundary connecting maps)**.** *Let $n \geq 1$, and assume the inputs of Theorem 1.1(ii) and (Q1) for the indicial extension. Then both connecting maps $\mathrm{HH}_1(\mathcal{A}^b_{\partial,\sigma}) \to \mathrm{HH}_0(\mathcal{I}^b_\partial)$ and $\mathrm{HH}_2(\mathcal{A}^b_{\partial,\sigma}) \to \mathrm{HH}_1(\mathcal{I}^b_\partial)$ are surjective onto $\mathbb{C}^N$. They are given on E^2 by the functionals $\theta \mapsto -i\hat{\rho}_H(\theta_x)$ and $\theta \mapsto -\hat{\rho}_H(\theta_{\xi x})$.*

Proof. By Proposition 21.3, Lemma 21.6, and the isomorphism Θ of Proposition 20.1, the composites $\mathrm{Tr}_{\partial,H} \circ \partial_1$ and $\varphi_H \circ \partial_2$ are the functionals $\theta \mapsto -i\hat{\rho}_H(\theta_x)$ and $\theta \mapsto -\hat{\rho}_H(\theta_{\xi x})$ on E^2. We exhibit δ_b-closed forms on which these are nonzero. Fix a metric on H, let $|\zeta|^2 = |\eta|^2 + \xi^2$, and let $\alpha_H = \eta \cdot dy$ be the tautological form. Put

$$f = 1_H\,\xi|\zeta|^{-n}, \qquad \theta_1 = f\,\frac{dx}{x} + 1_H|\zeta|^{-n}\alpha_H, \qquad \theta_2 = f\,d\xi \wedge \frac{dx}{x} + 1_H|\zeta|^{-n}\,d\xi \wedge \alpha_H.$$

These have fibre weights $1-n$ and $2-n$ and x-weight 0. With $\{a,c\}_b = \pi_b(da,dc)$, (5) gives $\{f, \log x\}_b = -\partial_\xi f$, $\{h, y_j\}_b = -\partial_{\eta_j} h$, $\{f,\xi\}_b = 0$ and $\{\xi, \log x\}_b = -1$. Hence

$$\delta_b\theta_1 = -\operatorname{div}_\zeta V, \qquad \delta_b\theta_2 = (\operatorname{div}_\zeta V)\,d\xi, \qquad V = \zeta|\zeta|^{-n}.$$

Since ζ ranges over $\mathbb{R}^n \setminus 0$, the field V is divergence free: $\operatorname{div} V = n|\zeta|^{-n} - n|\zeta|^{-n-2}|\zeta|^2 = 0$. This holds in any coordinates, because

$$\sum_j \eta_j \partial_{\eta_j}|\zeta|^2 + \xi\partial_\xi|\zeta|^2 = 2|\zeta|^2.$$

So θ_1 and θ_2 define E^2-classes. Now $(\theta_1)_x = (\theta_2)_{\xi x} = f$, and

$$\hat{\rho}_{H'}(f) = \delta_{HH'}\,2(2\pi)^{-n}\int_H \int_{\mathbb{R}^d} (1+|\eta|^2)^{-n/2}\,d\eta\,\nu_\partial \neq 0.$$

This proves surjectivity. $\square$

Proposition 21.8 (Boundary periods)**.** *Use the trace and cocycle coordinates of Theorem 1.1(ii). Under the hypotheses of Proposition 21.7, the two boundary connecting maps are*

$$\beta \longmapsto -i(2\pi)^{-n}\Big(\int_{Z_H}\beta_\partial\Big)_H, \qquad \beta \longmapsto (2\pi)^{-n}\Big(\int_{Z_H}\beta_0\Big)_H.$$

Here the boundary generator is written on the left, as in Lemma 20.6. The orientations of Y and Z_H are given by $\iota_{\mathcal{E}}\mu_b$ on $r=1$ and the induced boundary orientation, respectively.

Proof. On the component H write $\vartheta = dx/x$ and $v = (d\eta\wedge dy)^d/d!$, where the power is that of the tangential symplectic form. Thus $\mu_b = d\xi\wedge\vartheta\wedge v$. We have $\pi_b^\sharp\vartheta = -\partial_{\tilde\xi}$ and $\pi_b^\sharp d\xi = x\partial_x$. Take the x-weight-zero part of the HKR form θ, as allowed by Lemma 20.6. In degree one put $g=\theta_x$ and let α be the coefficient of ϑ in $*_b\theta$. In degree two put $g=\theta_{\tilde\xi x}$ and let α be the part of $*_b\theta$ free of ϑ. The form α on $(T^*H\times\mathbb{R}_{\tilde\xi})\setminus 0$ has degree $2d$ and is closed and invariant under radial dilation. Indeed, the star takes Poisson cycles to closed forms of weight zero, and the two parts of the formal x-weight-zero complex are closed separately. The star sends $g\vartheta$ to $-g\vartheta\wedge v$ and $g\,d\xi\wedge\vartheta$ to gv. In the chosen part of the star all other terms contain $d\xi$, since their contracting multivector does not contain $\partial_{\tilde\xi}$. Thus

$$\alpha|_{\tilde\xi=\pm1} = s\,g(y,\eta,\pm1)v, qquads = -1 \text{ in degree one}, \quad s=1 \text{ in degree two}.$$

Choose $r^2=\xi^2+|\eta|^2$ in the collar and let $t=\xi/r$ on Z_H. In radial coordinates write $\alpha = \gamma + d\log r\wedge\gamma_\sigma$, with both coefficients pulled back from Z_H and closed there. The class of γ is β_∂ in degree one and β_0 in degree two. The radial map from $\xi=\pm1$ identifies $|\eta|\le(\varepsilon^{-2}-1)^{1/2}$ with the cap $\pm t\ge\varepsilon$. The inverse map pulls α back to $\gamma - d\log|t|\wedge\gamma_\sigma$, since $r=1/|t|$ on that hyperplane. On either cap the integral of the second term is

$$-\int_{\partial\{\pm t\ge\varepsilon\}}\log|t|\,\gamma_\sigma = -\log\varepsilon\int_{\partial\{\pm t\ge\varepsilon\}}\gamma_\sigma = 0$$

by Stokes and $d\gamma_\sigma=0$. This controls the logarithmic term before taking the limit. The paired hyperplane integrand also converges absolutely: homogeneity gives $g(y,\eta,1)-g(y,\eta,-1) = \int_{-1}^{1}\partial_{\tilde\xi}g(y,\eta,\xi)\,d\xi = O(|\eta|^{-n})$.

At the boundary the contraction in the outward normal direction gives $\iota_{-x\partial_x}\iota_{\mathcal{E}}\mu_b = -\iota_{\mathcal{E}}(d\xi\wedge v)$. The radial projection pulls $\iota_{\mathcal{E}}(d\xi\wedge v)$ on $r=1$ back to $r^{-n}\xi v$ on $\xi=\pm1$. The positive cap therefore has orientation $-v$ in its hyperplane coordinates and the negative cap has orientation v. Adding the oriented cap integrals and letting $\varepsilon\downarrow 0$ gives $\int_{Z_H}\gamma = -s(2\pi)^n\hat\rho_H(g)$. Combining this with Lemma 21.6 gives the stated constants. □

Theorem 21.9. *Under the inputs of Theorem 1.1(ii) and (Q1) for the indicial extension, $\mathrm{HH}_k(\mathcal{A}_\partial^b)\cong\mathrm{HH}_k(\mathcal{A}_{\partial,\sigma}^b)$ for $k\neq1,2$. For $k=1,2$, the group $\mathrm{HH}_k(\mathcal{A}_\partial^b)$ is the kernel of the surjection of Proposition 21.4 ($n=1$), resp. Proposition 21.7. Proposition 21.8 identifies the maps with the boundary periods.*

Proof. Use the long exact sequence of Theorem 16.3 together with Theorem 18.5. Surjectivity of $\mathrm{HH}_2(\mathcal{A}_{\partial,\sigma}^b)\to\mathrm{HH}_1(\mathcal{I}_\partial^b)$ makes $\mathrm{HH}_1(\mathcal{I}_\partial^b)\to\mathrm{HH}_1(\mathcal{A}_\partial^b)$ zero. Surjectivity of $\mathrm{HH}_1(\mathcal{A}_{\partial,\sigma}^b)\to\mathrm{HH}_0(\mathcal{I}_\partial^b)$ makes $\mathrm{HH}_0(\mathcal{I}_\partial^b)\to\mathrm{HH}_0(\mathcal{A}_\partial^b)$ zero. For $k\ge3$ the groups of $\mathcal{I}_\partial^b$ vanish. □

22 The algebra $\mathcal{A}^b$ and the full operator algebra

Lemma 22.1. *$\mathcal{I}_\sigma^b\cap\mathcal{I}_\partial^b=0$ in $\mathcal{A}^b$, and $\mathcal{A}^b\to\mathcal{A}_\partial^b\times_{\mathcal{A}_{\partial,\sigma}^b}\mathcal{A}_\sigma^b$ is an isomorphism.*

Proof. $x^{\infty}\Psi_b^{\mathbb{Z}} \cap x^{-\mathbb{Z}}\Psi_b^{-\infty} = x^{\infty}\Psi_b^{-\infty}$. For any ring and ideals I, J, the ring $R/(I \cap J)$ is the fibre product $R/I \times_{R/(I+J)} R/J$. □

Proposition 22.2. *Let $\partial_k^{\mathcal{A}} : \mathrm{HH}_k(\mathcal{A}_\sigma^b) \to \mathrm{HH}_{k-1}(\mathcal{I}_\partial^b)$ be the connecting maps of $0 \to \mathcal{I}_\partial^b \to \mathcal{A}^b \to \mathcal{A}_\sigma^b \to 0$, and let $p : \mathcal{A}_\sigma^b \to \mathcal{A}_{\partial,\sigma}^b$ be the quotient map.*

(a) $\partial_k^{\mathcal{A}} = \partial_k^{\partial} \circ p_*$, where ∂^{∂} denotes the connecting maps of $\mathcal{A}_\partial^b$.

(b) The composite $\mathrm{HH}_1(\mathcal{I}_\partial^b) \to \mathrm{HH}_1(\mathcal{A}^b) \to \mathrm{HH}_0(\mathcal{I}^b)$, where the second map is the connecting map of $0 \to \mathcal{I}^b \to x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}} \to \mathcal{A}^b \to 0$, equals $-\sum_H \varphi_H$.

(c) The image of $\partial_2^{\mathcal{A}}$ is contained in $\ker \sum_H \varphi_H$, so its rank is at most $N-1$.

(d) If $n = 1$, then $\partial_1^{\mathcal{A}}$ is surjective.

Proof. (a) and (b) express the naturality of the excision sequence for the morphisms of extensions $(\mathcal{I}_\partial^b \subset \mathcal{A}^b) \to (\mathcal{I}_\partial^b \subset \mathcal{A}_\partial^b)$ and $(\mathcal{I}^b \subset x^{-\mathbb{Z}}\Psi_b^{-\infty}) \to (\mathcal{I}^b \subset x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}})$. Part (b) then follows from Theorem 19.3.

(c) The composite $\mathrm{HH}_2(\mathcal{A}_\sigma^b) \to \mathrm{HH}_1(\mathcal{I}_\partial^b) \to \mathrm{HH}_1(\mathcal{A}^b)$ is zero by exactness, and the second map followed by the connecting map of (b) is $-\sum \varphi_H$.

(d) Choose $P_H \in \Psi_b^0(X)$, supported near H, with full symbol $\chi_H(x)1_{\xi>0}$ and with no lower-order terms and with $\chi_H = 1$ near H. Then $x^{-1}P_H x - P_H$ is smoothing, because the full symbol of $x^{-1}Px$ is $\sum_k \frac{(-i)^k}{k!}\partial_\xi^k p$, by Lemma 14.1 applied to symbols, and $\partial_\xi p = 0$ for $\xi \neq 0$. Hence $x^{-1}P_H \otimes x$ is a cycle in $\mathcal{A}_\sigma^b$. Its image under p is the cycle c_H of Proposition 21.4 in the quotient: the full Taylor symbol is constant in x near H, and its constant term is the complete symbol of $\pi_+(\lambda)$. Here ξ is the symbol of xD_x, whose indicial family is λ. Thus $\mathrm{Tr}_{\partial,H'}\,\partial_1^{\mathcal{A}}[x^{-1}P_H \otimes x] = -\frac{i}{2\pi}\delta_{HH'}$. □

Remark 22.3 (Several boundary components). Take compatible collars and fibre norms in the symbol algebras. On associated graded chains p takes the boundary Taylor series in each slot and commutes with HKR and symplectic duality. In the face splitting of Proposition 8.1 the smooth summand contributes its tangential restriction: a factor dx has positive x-weight in the b-frame. The residue summand is represented near the face by dx/x wedged on the left with its boundary form. Thus after the contraction of the nonzero x-weights the map in degree $j = 2n - k$ is

$$(a, b) \longmapsto a|_{Z \times S_\sigma} + d\theta_\partial \wedge b \quad \text{for } (a, b) \in H^j(Y \times S_\sigma) \oplus H^{j-1}(Z \times S_\sigma).$$

By Proposition 22.2(a), Proposition 21.8, and the naturality of Θ, the map $\partial_2^{\mathcal{A}}$ is $\beta \mapsto (2\pi)^{-n}(\int_{Z_H} \beta)_H$ on the summand $H^{2n-2}(Y)$ of $\mathrm{HH}_2(\mathcal{A}_\sigma^b)$, and it vanishes on the other summands. By Stokes' theorem, $\sum_H \int_{Z_H} \beta = \int_Y d\beta = 0$, which recovers the bound of Proposition 22.2(c). Exactness of $H^{2n-2}(Y) \to H^{2n-2}(\partial Y) \to H^{2n-1}(Y, \partial Y)$, together with Lefschetz duality $H^{2n-1}(Y, \partial Y) \cong H_0(Y)$ (C3), shows that the image is the kernel of the sum map $H^{2n-2}(\partial Y) \to H_0(Y)$. If X is connected and $n \geq 2$, then Y is connected, $H^{2n-2}(\partial Y) = \mathbb{C}^N$, and the rank is $N - 1$. If X has c components, each with nonempty boundary, the rank is $N - c$, and $\operatorname{coker} \partial_2^{\mathcal{A}} \cong \mathbb{C}^c$. In the cusp case Melrose and Nistor find $\partial[i_{\log x}\,\mathrm{Tr}_\partial] = 0$ [21, Prop. 12], because $d\theta_\sigma$ extends over the whole cosphere bundle. For each H separately, the class $d\theta_\sigma \cdot 1_{Z_H}$ does not extend when $N \geq 2$; only the sum over all H does. We mention this only for comparison.

Proof of Theorem 1.1. (i) is Proposition 17.1, (ii) is Theorem 18.5, and (iii) is Theorem 19.3. (iv) and (vi) are Propositions 20.7 and 20.8, and (v) is Theorem 2.1. (vii) is Theorem 21.9, with the components identified by Proposition 21.8. For (viii), insert $\partial_1^{\mathcal{A}} = \partial_1^{\partial} \circ p_*$ and $\partial_2^{\mathcal{A}}$ into the long exact sequence.

By Proposition 21.8 and the naturality of the top-component map Θ, $\partial_1^{\mathcal{A}}$ is $\beta \mapsto -i(2\pi)^{-n}(\int_{Z_H}\beta)_H$ on the summand $H^{2n-2}(Z) \cong \mathbb{C}^N$ of $\mathrm{HH}_1(\mathcal{A}_\sigma^b)$ and vanishes on the other summands. The rank of $\partial_2^{\mathcal{A}}$ is $N-1$ (Remark 22.3). Use also $\mathrm{HH}_*(\mathcal{I}_\partial^b)$. This gives $\mathrm{HH}_0(\mathcal{A}^b) \cong \mathrm{HH}_0(\mathcal{A}_\sigma^b)$, $\mathrm{HH}_1(\mathcal{A}^b) \cong \operatorname{coker}\partial_2^{\mathcal{A}} \oplus \ker\partial_1^{\mathcal{A}}$, and $\mathrm{HH}_2(\mathcal{A}^b) \cong \ker\partial_2^{\mathcal{A}}$. (ix) follows from Corollary 23.7 and the long exact sequence of $0 \to \mathcal{I}^b \to x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}} \to \mathcal{A}^b \to 0$, using (i). □

23 The index functional

Lemma 23.1 (Inverses of indicial families)**.** *Let $a \in B_H'$ have order m and be elliptic. If $a(\lambda)$ is invertible for every $\lambda \in \mathbb{C}$, then $a^{-1} \in B_H'$. If $a = 1 + k$ with $k \in B_H$ and $a(\lambda)$ is invertible for every $\lambda \in \mathbb{C}$, then $a^{-1} - 1 \in B_H$.*

Proof. Fix a strip $S = \{|\operatorname{Im}\lambda| \le C\}$. The suspended calculus (C3) gives $q \in \Psi_{\mathrm{sus}}^{-m}$ on S with $aq = 1 - r$, where r is smoothing and decreases rapidly, in all Sobolev norms, as $|\operatorname{Re}\lambda| \to \infty$. The family q need not be holomorphic. For $|\operatorname{Re}\lambda| \ge R$ the Neumann series gives $a^{-1} = q(1-r)^{-1} = q + qr(1-r)^{-1}$. For $|\operatorname{Re}\lambda| \le R$, the identity $a^{-1} = q + a^{-1}r$ holds with a^{-1} holomorphic, hence bounded on that compact set. So $a^{-1} - q$ is smoothing with the decay of r, and a^{-1} is entire because a is. Hence $a^{-1} \in B_H'$. In the second case, a^{-1} is uniformly bounded on each H^s over S: use the Neumann series where $\|k\| < \frac{1}{2}$, and compactness elsewhere. Then $a^{-1} - 1 = -k + ka^{-1}k$ is smoothing and decreases rapidly on S. □

Lemma 23.2. *Let $A \in \Psi_b^m(X)$ be b-elliptic. If for every H and every $\lambda \in \mathbb{C}$ the operator $I_H(A,\lambda)$ is invertible, then $[A]$ is invertible in $\mathcal{A}^b$. Conversely, if $[A]$ is invertible in $\mathcal{A}^b$, then $I_H(A,\lambda)$ is invertible for every $\lambda \in \mathbb{C}$.*

Proof. By Lemma 22.1 it suffices to invert A in $\mathcal{A}_\sigma^b$, where the usual symbolic parametrix construction works, and in $\mathcal{A}_\partial^b$. Write $T_H(A) = \sum_{k\ge 0} x^k a_k$ with $a_0 = I_H(A)$. By (24), the coefficient of x^k in $T(A)\sum x^l b_l$ is $\alpha^k(a_0)b_k + \sum_{j\ge 1}\alpha^{k-j}(a_j)b_{k-j}$. Solve recursively, using $\alpha^k(a_0)^{-1} = \alpha^k(a_0^{-1}) \in B_H'$ (Lemma 23.1). A left inverse is constructed in the same way, and the two inverses agree. Conversely, let the inverse be $\sum_{l\ge -\ell} x^l b_l$. Its lowest coefficient satisfies $b_{-\ell}a_0 = 0$. Since $a_0(\lambda)$ is invertible for large $|\operatorname{Re}\lambda|$, analytic continuation gives $b_{-\ell} = 0$ if $\ell > 0$. Hence $b_0 a_0 = a_0 b_0 = 1$ on all of $\mathbb{C}$. □

Remark 23.3 (Dirac operators). A b-Dirac operator of product type, $c(dx/x)(x\partial_x + A_\partial)$, has $I(\lambda) = c(dx/x)(i\lambda + A_\partial)$, which fails to be invertible at $\lambda = i\mu$ for every $\mu \in \operatorname{spec} A_\partial$. It is therefore never invertible in $\mathcal{A}^b$. Its index on $x^\delta L_b^2$ depends on δ, and the algebra $\mathcal{A}^b$ cannot see this dependence. This is the precise form of the "problems with completeness" that led Melrose–Nistor to the cusp calculus [21, p. 3], where indicial families need only be invertible on $\mathbb{R}$.

Proposition 23.4 (Fedosov)**.** *Let $A \in x^{-\ell}\Psi_b^m(X)$ and $B \in x^{-\mathbb{Z}}\Psi_b^{\mathbb{Z}}(X)$ with $AB - 1, BA - 1 \in \mathcal{I}^b$. Then $A : x^\delta H_b^s \to x^{\delta-\ell}H_b^{s-m}$ is Fredholm for all δ, s, with kernel and cokernel in $\dot{\mathcal{C}}^\infty(X)$ and index independent of δ and s. Moreover*

$$\operatorname{Ind}(A) = \operatorname{Tr}(AB - BA) = -\operatorname{Tr}\big(\partial[B \otimes A]\big),$$

where $B \otimes A$ is a 1-cycle of $\mathcal{A}^b$.

Proof. Operators in $\mathcal{I}^b$ have kernels in $\dot{\mathcal{C}}^\infty(X^2)$, so they are trace class on every weighted Sobolev space and map into $\dot{\mathcal{C}}^\infty(X)$. Hence $\ker A \subset \ker BA \subset \dot{\mathcal{C}}^\infty$, and similarly for A^*. The mapping properties of b-operators on weighted b-Sobolev spaces, and the Calderón–Fedosov formula, are inputs (C3). The latter gives $\operatorname{Ind} A = \operatorname{Tr}(1 - BA) - \operatorname{Tr}(1 - AB)$. Finally, $b(B \otimes A) = BA - AB$ lifts the boundary. □

Theorem 23.5 (Residual perturbations of the identity)**.** *Let $K \in \Psi_b^{-\infty}(X)$, and suppose that for every H the family $a_H(\lambda) = 1 + I_H(K,\lambda)$ is invertible for every $\lambda \in \mathbb{C}$. Then $A = 1 + K$ is Fredholm on every $x^\delta H_b^s$, and*

$$\operatorname{Ind}(A) = \frac{i}{2\pi} \sum_H \int_{\mathbb{R}} \operatorname{Tr}_H\big(a_H^{-1}\partial_\lambda a_H\big)\, d\lambda = -\frac{1}{2}\sum_H \eta_b(a_H) = -\sum_H \operatorname{wind}\big(\det a_H\big),$$

where $\eta_b(a) = \frac{1}{\pi i}\int_{\mathbb{R}} \operatorname{Tr}(a^{-1}\partial_\lambda a)\, d\lambda$, and wind *is the winding number of the Fredholm determinant along* $\mathbb{R}$.

Proof. The symbol of A is 1, and $a_H^{-1} - 1 \in B_H$ by Lemma 23.1. By Lemma 23.2, the inverse in $\mathcal{A}_\partial^b$ is $1 + \sum_{k\ge 0} x^k l_k$ with $l_0 = a_H^{-1} - 1$ and all $l_k \in B_H$. By Proposition 15.3, there is $L \in \Psi_b^{-\infty}$ with this Taylor series, and then $B = 1 + L$ satisfies $AB - 1, BA - 1 \in \mathcal{I}^b$. By Proposition 23.4 and Theorem 19.2,

$$\operatorname{Ind} A = \operatorname{Tr}([K,L]) = {}^b\operatorname{Tr}([K,L]) = \sum_H \operatorname{Tr}_{\partial,H}(D(K)L).$$

Since $T(D(K))$ and $T(L)$ contain only nonnegative powers of x, the constant term of their product is $i\partial_\lambda a_H \cdot (a_H^{-1} - 1)$. The integral of Tr $\partial_\lambda a_H$ vanishes. Moreover $\operatorname{Tr}(a^{-1}\partial a) = \partial_\lambda \log\det a$, and $\det a_H \to 1$ at $\pm\infty$, so the integral equals $2\pi i$ wind. □

Example 23.6 (Index -1)**.** Let e_H be a rank-one projection on H, and put $a_H(\lambda) = 1 + (e^{2\pi i \pi_+(\lambda)} - 1)e_H$, with π_+ as in Proposition 21.4. Then $e^{\pm 2\pi i\pi_+} - 1 \in \mathcal{E}$, $a_H^{-1} = 1 + (e^{-2\pi i\pi_+} - 1)e_H$, and $\det a_H = e^{2\pi i\pi_+}$ winds once. Taking $a_{H'} = 1$ for $H' \neq H$, the operator $1 + K$ has index -1. It is not a compact perturbation of the identity on L_b^2, because its normal operator is translation invariant in $u = \log x$. Near H, in the variable u, it is a Wiener–Hopf operator on a half-line with symbol a_H. The theorem is the b-calculus form of the Gohberg–Krein index formula [5].

Corollary 23.7. *The index functional* $\operatorname{Tr}\circ\partial : \mathrm{HH}_1(\mathcal{A}^b) \to \mathbb{C}$ *is nonzero.*

Proof. For the operator of Example 23.6, Proposition 23.4 gives $\operatorname{Tr}\partial[B\otimes A] = -\operatorname{Ind}(A) = 1$. □

For operators of positive order, the residual perturbations of Theorem 23.5 are replaced by general fully elliptic operators. Their index is computed in Section 24 (Theorem 24.16): it is the difference of the b-zeta values of A^*A and AA^* at 0, minus half the sum of the b-eta invariants (31) of the indicial families. In order zero the interior term is absent, and the formula reduces to Theorem 23.5 (Remark 24.19).

24 Complex powers, b-zeta functions and the index

Status. This section is pending independent referee review. Its analytic inputs are listed under (C3) in Section 1.4.

This section proves the index formula for fully elliptic operators of positive order. The arguments are those of Loya's thesis [6, Chs. 2–8], which treats manifolds with corners. We specialize them to a manifold with boundary, where they become much simpler, and write them in the conventions of Section 14. We also replace the thesis's tempered-symbol calculus (its Chapters 3–5) by a shorter route. The resolvent is decomposed into an explicit symbolic part, an explicit model part at the front faces, and a remainder that is controlled only through weighted mapping properties (Definition 24.2). This is all that the b-trace needs. The index identity itself is proved at the level of operators (Proposition 24.14), and the heat form of the formula is then obtained by Mellin inversion. Throughout, $n = \dim X$, $m \ge 1$ is an integer, and ${}^b\operatorname{Tr}$ is the finite part at $w = 0$ of $\operatorname{Tr}(x^w\,\cdot\,)$, as in Section 19.

24.1 The reduction lemma

We call $A \in \Psi_b^m(X)$ *fully elliptic* if it is b-elliptic and $I_H(A,\lambda)$ is invertible for every $\lambda \in \mathbb{C}$ and every H. By Lemma 23.2, this means that $[A]$ is invertible in $\mathcal{A}^b$. We write A^* for the formal adjoint with respect to μ_b. The kernel of A^* is $\overline{K_A(p',p)}$. At ff_H this replaces s by $1/s$ and exchanges y and y', so (19) gives

$$I_H(A^*,\lambda) = I_H(A,\bar\lambda)^*. \tag{28}$$

Lemma 24.1 (Reduction). *Let $A \in \Psi_b^m(X)$, $m \geq 1$, be fully elliptic, and write $a_H = I_H(A,\cdot)$.*

(i) There is $B \in \Psi_b^{-m}(X)$ with $AB-1,\ BA-1 \in \mathcal{I}^b$. The adjoint A^* is fully elliptic, and $B^*A^*-1,\ A^*B^*-1 \in \mathcal{I}^b$.

(ii) For all $\delta, s \in \mathbb{R}$, the operator $A : x^\delta H_b^s \to x^\delta H_b^{s-m}$ is Fredholm. Its kernel is $N_A = \ker A \cap \dot{\mathcal{C}}^\infty(X)$, and its range is the annihilator of $N_{A^*} = \ker A^* \cap \dot{\mathcal{C}}^\infty(X)$ under the pairing $\int u\bar v\,\mu_b$. Hence its index is $\operatorname{Ind} A = \dim N_A - \dim N_{A^*}$, independent of δ and s. This is also the index of A on $\dot{\mathcal{C}}^\infty(X)$, and it equals $\operatorname{Tr}(AB-BA)$ (Proposition 23.4).

(iii) As an unbounded operator on L_b^2 with domain H_b^m, A is closed with adjoint A^* (domain H_b^m), and its Fredholm index is $\operatorname{Ind} A$.

(iv) $P = A^*A$ and $P' = AA^*$ lie in $\Psi_b^{2m}(X)$, are b-elliptic with principal symbol $|\sigma_m(A)|^2$, and are self-adjoint on L_b^2 with domain H_b^{2m}. Moreover $\ker P = N_A$ and $\ker P' = N_{A^*}$. For real λ,

$$I_H(P,\lambda) = a_H(\lambda)^* a_H(\lambda), \qquad I_H(P',\lambda) = a_H(\lambda) a_H(\lambda)^*,$$

and both are $\geq c\,(\Delta_H + 1 + \lambda^2)^m$ for some $c > 0$. The orthogonal projections π, π' onto N_A, N_{A^*} lie in $\mathcal{I}^b$, and $P_0 = P + \pi$ and $P_0' = P' + \pi'$ are $\geq c_0$ for some $c_0 > 0$.

Proof. (i) Let $B_0 \in \Psi_b^{-m}$ be a small-calculus parametrix, so $AB_0 = 1 - R_0$ with $R_0 \in \Psi_b^{-\infty}$. The proof of Lemma 23.2 inverts $T(A)$ in $\bigoplus_H B'_H[[x;\alpha]]$, with nonnegative powers only and constant term $a_H^{-1} \in B'_H$ (Lemma 23.1). Since B_H is an ideal in B'_H, the series $T(A)^{-1}T(R_0)$ lies in $\bigoplus_H B_H[[x;\alpha]]$ and has nonnegative powers only. By Proposition 15.3 it is $T(B_1)$ for some $B_1 \in \Psi_b^{-\infty}$. Put $B = B_0 + B_1$. Then $AB - 1 \in \Psi_b^{-\infty}$, and $T(AB-1) = -T(R_0) + T(A)T(B_1) = 0$ by Proposition 15.4. Hence $AB - 1 \in \mathcal{I}^b$. A left inverse modulo $\mathcal{I}^b$ is built in the same way, and it differs from B by an element of $\mathcal{I}^b$. Full ellipticity of A^* follows from (28). The last claims follow by taking adjoints, since $(\mathcal{I}^b)^* = \mathcal{I}^b$.

(ii) The operator B is bounded $x^\delta H_b^{s-m} \to x^\delta H_b^s$ (C3). The operators $AB-1$ and $BA-1$ have kernels in $\dot{\mathcal{C}}^\infty(X^2)$. They therefore map $x^\delta H_b^s$ continuously into $x^{\delta+1}H_b^{s+1}$, whose inclusion into $x^\delta H_b^s$ is compact. By Atkinson's theorem, A is Fredholm. If $Au = 0$, then $u = (1-BA)u \in \dot{\mathcal{C}}^\infty$. The annihilator of the range in the dual space $x^{-\delta}H_b^{m-s}$ consists of the solutions of $A^*v = 0$. For these, $v = (1-B^*A^*)v \in \dot{\mathcal{C}}^\infty$ by (i). Both spaces are thus independent of (δ, s). If $f \in \dot{\mathcal{C}}^\infty$ annihilates N_{A^*}, then $f = Au$ with $u \in L_b^2$, and $u = Bf + (1-BA)u \in \dot{\mathcal{C}}^\infty$. So the cokernel of $A : \dot{\mathcal{C}}^\infty \to \dot{\mathcal{C}}^\infty$ is also dual to N_{A^*}. The trace formula is the Calderón–Fedosov formula, as in Proposition 23.4.

(iii) If $u_j \to u$ in L_b^2 and $Au_j \to f$, then $u = Bf + (1-BA)u \in H_b^m$ and $Au = f$. If $v \in L_b^2$ and $A^*v \in L_b^2$ weakly, then $v = B^*A^*v + (1-B^*A^*)v \in H_b^m$. For a closed Fredholm operator, the index is $\dim\ker A - \dim\ker A^*$, which equals $\operatorname{Ind} A$ by (ii).

(iv) The composite of the Fredholm maps $A : H_b^{2m} \to H_b^m$ and $A^* : H_b^m \to L_b^2$ is Fredholm. The operator P is symmetric, and $(BB^*)P - 1 = B(B^*A^* - 1)A + (BA - 1) \in \mathcal{I}^b$. If $v, Pv \in L_b^2$, then $v = BB^*Pv - (BB^*P - 1)v \in H_b^{2m}$, so P is self-adjoint. Since $\langle Pu, u\rangle = \|Au\|^2$, we get $\ker P = N_A$. The indicial identities follow from (28) and the multiplicativity of I_H. By Lemma 23.1, a_H^{-1} is a

suspended family of order $-m$, so $\|(\Delta_H + 1 + \lambda^2)^{m/2} a_H(\lambda)^{-1}\| \le C$ on $\mathbb{R}$ (C3). This gives the lower bound for $a_H^* a_H$, and the same argument with $(a_H^*)^{-1} = (a_H^{-1})^*$ gives it for $a_H a_H^*$. An orthonormal basis $\{\varphi_i\}$ of N_A lies in $\dot{C}^\infty$, so the kernel $\sum \varphi_i(p)\overline{\varphi_i(p')}$ of π lies in $\dot{C}^\infty(X^2)$. Finally, P is Fredholm and self-adjoint, so 0 is isolated in its spectrum. Hence $P_0 \ge c_0 > 0$. The same arguments apply to P'. □

24.2 Weighted model operators and the b-trace

The resolvent of P is not in the small calculus: $I_H(P,\lambda) - \mu$ is invertible only on a strip. The following classes contain everything we need.

Definition 24.2. Fix $\varepsilon_0 > 0$ and $0 < \delta \le \varepsilon_0/4$. Let $\chi_H \in C^\infty(X)$ be supported in the collar of H and equal to 1 near H.

(a) $\mathcal{R}_\delta^{-K}$, for $K \ge 0$, is the space of operators that are bounded $x^a H_b^s \to x^{a'} H_b^{s+K}$ whenever $|a| \le 2\delta$, $a' = \min(a+\delta, 2\delta)$, and $|s| \le K$.

(b) $\mathcal{M}_H^{-k}$, for $k \in \mathbb{R}$, is the space of dilation-invariant operators $M = \mathfrak{m}(xD_x)$ on $M_H = [0,\infty) \times H$. Their indicial families $\mathfrak{m}$ are holomorphic on $|\operatorname{Im}\lambda| < \varepsilon_0$ and suspended of order $-k$ uniformly on $|\operatorname{Im}\lambda| \le 3\delta$.

(c) $\mathcal{K}_\delta^{-k}$ is the space of operators F such that, for every $K \ge 0$,

$$F = S_K + \sum_H \chi_H M_{K,H} \chi_H + R_K, \qquad S_K \in \Psi_b^{-k}(X),\ M_{K,H} \in \mathcal{M}_H^{-k},\ R_K \in \mathcal{R}_\delta^{-K}.$$

Here Ψ_b^s, $s \in \mathbb{C}$, denotes classical b-operators of complex order, as in [12], and $-k = \operatorname{Re} s$. Holomorphic families are defined by asking that the three pieces be holomorphic. We set $N_H(F) = N_H(S_K) + M_{K,H}$ and $I_H(F,\lambda) = I_H(S_K,\lambda) + \mathfrak{m}_{K,H}(\lambda)$, for $|\operatorname{Im}\lambda| < \varepsilon_0$.

Lemma 24.3. *Let $F \in \mathcal{K}_\delta^{-k}$ and $A \in \Psi_b^{m'}(X)$.*

(i) F is bounded $x^a H_b^s \to x^a H_b^{s+k}$ for $|a| \le 2\delta$. Moreover $N_H(F)$ is determined by F, and so is independent of K.

(ii) AF and FA lie in $\mathcal{K}_\delta^{-k+m'}$, with $N_H(AF) = N_H(A)N_H(F)$ and $N_H(FA) = N_H(F)N_H(A)$. The same holds for holomorphic families.

(iii) If $R \in \mathcal{R}_\delta^{-K}$ and $K \ge |k|$, then FR and RF lie in $\mathcal{R}_\delta^{-(K-|k|)}$.

Proof. (i) For S_K this is (C3). For $M \in \mathcal{M}_H^{-k}$ it follows from Mellin–Plancherel on the line $\operatorname{Im}\lambda = -a$, which corresponds to the weight x^a (see (19)). For R_K it holds by definition. For $u \in C_c^\infty((0,\infty)\times H)$, let $\tau_T u(x,y) = u(e^T x, y)$. This is unitary on L_b^2, and $\tau_T^{-1} F \tau_T u \to N_H(F)u$ as $T \to \infty$. For S_K this follows from the Taylor expansion at ff_H. For $\chi_H M \chi_H$ the identity is exact once T is large. For R_K, $\|R_K \tau_T u\| \le C\|x^\delta \tau_T u\| \le Ce^{-\delta T}$.

(ii) We have $A S_K \in \Psi_b^{-k+m'}$ by (Q0). Next,

$$A\chi_H M \chi_H = \chi_H\big(N_H(A)M\big)\chi_H + [A,\chi_H]M\chi_H + \chi_H\big(A - \tilde{\chi}_H N_H(A)\tilde{\chi}_H\big)M\chi_H,$$

where $\tilde{\chi}_H$ equals 1 on the support of χ_H. The first term is a cut-off model operator with indicial family $I_H(A,\lambda)\mathfrak{m}(\lambda)$. The kernel of $[A,\chi_H]$, $(\chi_H(x') - \chi_H(x))K_A$, vanishes near ff and to infinite order at $\mathrm{lb} \cup \mathrm{rb}$. It therefore maps $x^a H_b^s$ to $x^b H_b^{s-m'}$ for all a, b. By the Taylor expansion at ff_H (C3),

$A - \tilde\chi_H N_H(A)\tilde\chi_H$ equals $xT_1 + T_2$ near H, with $T_1 \in \Psi_b^{m'}$ and T_2 of the type of $[A, \chi_H]$. Both terms therefore lie in $\mathcal{R}_\delta^{-K}$ for $K \le k - m'$. Finally, $AR_K \in \mathcal{R}_\delta^{-(K-m')}$, and K is arbitrary. The case of FA is symmetric. (iii) follows from (i). □

Lemma 24.4 (Trace and residue)**.** *Let $F \in \mathcal{K}_\delta^{-k}$ with $k > n$.*

(i) For $0 < \operatorname{Re} w \le 2\delta$, the operator $x^w F$ is trace class on L^2_b, and $\operatorname{Tr}(x^w F) = \int_X x^w K_F(p,p)\,\mu_b$. The restriction of K_F to the diagonal is continuous on X. Near H, $K_F(p,p) = f_{0,H}(y) + O(x^{\delta/2})$, where

$$f_{0,H}(y) = \frac{1}{2\pi}\int_{\mathbb{R}} I_H(F,\lambda)(y,y)\,d\lambda.$$

(ii) $\operatorname{Tr}(x^w F)$ extends meromorphically to $\operatorname{Re} w > -\delta/2$. It has at most a simple pole at $w = 0$, with residue $\sum_H \frac{1}{2\pi}\int_{\mathbb{R}} \operatorname{Tr}_H I_H(F,\lambda)\,d\lambda$. The same holds for holomorphic families $F(w)$, with residue computed from $F(0)$.

(iii) Define ${}^b\operatorname{Tr}(F) = \operatorname{FP}_{w=0}\operatorname{Tr}(x^w F)$. It is linear. It equals $\operatorname{Tr} F$ if every $N_H(F)$ vanishes, and in particular on $\mathcal{I}^b$. On $x^{-\mathbb{Z}}\Psi_b^{-\infty} \cap \mathcal{K}_\delta^{-k}$ it is the b-trace of Section 19.

Proof. For $b > b'$ and $N - N' > n/2$, the inclusion $x^b H_b^N \hookrightarrow x^{b'} H_b^{N'}$ is Hilbert–Schmidt. This is (C3). In the collar it is the statement that $e^{-(b-b')|u|}\langle D\rangle^{-(N-N')}$ is Hilbert–Schmidt on $\mathbb{R}_u \times H$. By Lemma 24.3(i), $x^w F$ maps $x^{-\operatorname{Re} w/2} H_b^{-k/2}$ to $x^{\operatorname{Re} w/2} H_b^{k/2}$. So on L^2_b it is a product of a Hilbert–Schmidt map, a bounded map, and another Hilbert–Schmidt map. It is therefore trace class, and its trace is the integral of its continuous kernel over the diagonal.

We compute the diagonal piece by piece, taking $K > 2n$. The diagonal value of S_K is smooth up to ∂X. At H it equals the value of its normal operator at $s = 1$, up to $O(x)$, by the Taylor expansion at ff_H. The diagonal value of $\chi_H M \chi_H$ is exactly $\chi_H(x)^2 \kappa_M(1, y, y)$. For R_K, compose with the Hilbert–Schmidt inclusion $x^{-\delta/4} H_b^{-K/4} \hookrightarrow x^{-\delta/2} H_b^{-K/2}$. The composite is Hilbert–Schmidt into $x^{\delta/2} H_b^{K/2}$. Its kernel therefore lies in the Hilbert tensor product of $x^{\delta/2} H_b^{K/2}$ and $x'^{\delta/4} H_b^{K/4}$, and so it is continuous with $|K_{R_K}(p,p')| \le C x(p)^{\delta/2} x(p')^{\delta/4}$. Hence $N_H(F)(1,y,y) = f_{0,H}(y)$ by (21), which proves (i).

For (ii), proceed as in Lemma 19.1. The collar contributes $\int_0^\varepsilon x^{w-1}dx \int_H f_{0,H}\nu_\partial$, plus a term that is holomorphic for $\operatorname{Re} w > -\delta/2$.

For (iii): if every $N_H(F)$ vanishes, then $f_{0,H} = 0$, and $\operatorname{Tr}(x^w F)$ is holomorphic near 0 with value $\operatorname{Tr} F$. □

The defect formula of Theorem 19.2 extends to this class. The following statement and Theorem 19.2 together cover all the cases used in these notes.

Theorem 24.5 (b-trace defect)**.** *Let $A \in \Psi_b^{m'}(X)$ and $F \in \mathcal{K}_\delta^{-k}$ with $k > 2(n + m' + 1)$. Then $\operatorname{Tr}(x^w[A,F])$ is holomorphic near $w = 0$, and*

$${}^b\operatorname{Tr}([A,F]) = \sum_H \frac{i}{2\pi}\int_{\mathbb{R}} \operatorname{Tr}_H\big(\partial_\lambda I_H(A,\lambda)\, I_H(F,\lambda)\big)\,d\lambda.$$

For $A, F \in \Psi_b^{-\infty}$ this is the formula of Theorem 19.2.

Proof. By Lemmas 24.3 and 24.4, for $0 < \operatorname{Re} w \le 2\delta$ the operators $x^w AF$, $x^w FA$ and $Ax^w F$ are trace class. Moreover $\operatorname{Tr}(x^w FA) = \operatorname{Tr}(Ax^w F)$. Indeed, composing $x^w F$ with a Hilbert–Schmidt inclusion as in the proof of Lemma 24.4 shows that its kernel lies in the Hilbert tensor product of $x^r H_b^{k/2}$ and

$x'^{r/2}H_b^{k/2-n/2-1}$, where $r = \operatorname{Re} w/2$. Since $k/2 - n/2 - 1 > n/2 + m'$, both traces are continuous functions of the kernel for this topology. They agree on the dense set of finite sums $\varphi \otimes \psi$ with $\varphi, \psi \in \dot{C}^\infty$, by the definition of the transpose. As in Theorem 19.2, $\operatorname{Tr}(x^w[A,F]) = w \operatorname{Tr}(x^w C(w) F)$, where $C(w) = (A - x^{-w}Ax^w)/w$ is an entire family in $\Psi_b^{m'}$ with $C(0) = D(A)$ (Lemma 14.1). By Lemma 24.3, $C(w)F$ is a holomorphic family in $\mathcal{K}_\delta^{-k+m'}$ with $I_H(C(0)F, \lambda) = i\partial_\lambda I_H(A,\lambda) I_H(F,\lambda)$. Lemma 24.4(ii) now gives the claim. □

Remark 24.6 (Comparison with the thesis). Loya writes $N_M(A)(\tau) = (x^{-i\tau}Ax^{i\tau})|_M$, the Mellin transform $\int s^{-i\tau} \cdot ds/s$ of the kernel at the front face [6, Sect. 2.4]. This is $I_H(A,\tau)$ in our notation, with $\tau = \lambda$. His Theorem 6.6 reads ${}^b\operatorname{Tr}[A,B] = -\frac{1}{2\pi}\sum_H \int {}^b\operatorname{Tr}(D_\tau N_H(A) N_H(B))\, d\tau$. The symbol D_τ is not defined explicitly there, but his proof requires $N(-\log s \cdot A) = D_\tau N(A)$, and so $D_\tau = -i\partial_\tau$. With this, his formula is Theorem 24.5. He works with b-half-densities. The map $u \mapsto u\,\mu_b^{1/2}$ is unitary from L_b^2 onto $L_b^2(X; \Omega_b^{1/2})$, so his traces, b-traces and indices are ours.

24.3 The resolvent

From now on, A is fully elliptic of order $m \geq 1$, and P is one of A^*A, AA^*, with the notation of Lemma 24.1(iv). Write $p_H(\lambda) = I_H(P,\lambda)$. Fix $\theta \in (0, \pi/4]$, and fix $\rho > 0$ smaller than half of c_0 and half of the constant c of Lemma 24.1(iv). Put

$$\Lambda = \{\mu \neq 0 : |\arg \mu| \geq \theta\} \cup \{|\mu| \leq \rho\}.$$

Let Γ be the boundary of $\mathbb{C} \setminus \Lambda$, oriented counterclockwise around $[c_0, \infty)$. It consists of the rays $re^{\pm i\theta}$, $r \geq \rho$, and the arc $\rho e^{i\varphi}$, $|\varphi| \leq \theta$.

Lemma 24.7 (Indicial resolvent)**.** *There are $\varepsilon_0, C > 0$ such that, for $|\operatorname{Im}\lambda| \leq \varepsilon_0$ and $\mu \in \Lambda$, the operator $p_H(\lambda) - \mu$ is invertible. Moreover $\|(p_H(\lambda) - \mu)^{-1}\| \leq C(1 + |\mu|)^{-1}$, and $(p_H(\lambda) - \mu)^{-1}$ is holomorphic in (λ, μ) and suspended of order $-2m$, uniformly in μ.*

Proof. Let $L(\lambda) = (\Delta_H + 1 + (\operatorname{Re}\lambda)^2)^m$. By Lemma 24.1(iv), for real λ the operator $p_H(\lambda)$ is self-adjoint and $\geq cL$. For $t \geq c$ and $\mu \in \Lambda$ we have $|t - \mu| \geq c'(t + |\mu|)$. By the spectral theorem, $\|p_H^s(p_H - \mu)^{-1}\| \leq C(1 + |\mu|)^{s-1}$ for $0 \leq s \leq 1$. Since p_H^{-1} is suspended of order $-2m$ (Lemma 23.1), $\|Lp_H^{-1}\| \leq C$, and by interpolation $\|L^s p_H^{-s}\| \leq C$. For $\lambda \in \mathbb{R}$ and $|\sigma| \leq 1$, the difference $E = p_H(\lambda + i\sigma) - p_H(\lambda)$ is σ times a suspended family of order $2m - 1$ (C3). Hence

$$\|E(p_H(\lambda) - \mu)^{-1}\| \leq C|\sigma|\, \|L^{(2m-1)/2m}(p_H - \mu)^{-1}\| \leq C'|\sigma|.$$

For $|\sigma| \leq \varepsilon_0 = 1/(2C')$, the factorization $p_H(\lambda + i\sigma) - \mu = (1 + E(p_H(\lambda) - \mu)^{-1})(p_H(\lambda) - \mu)$ and a Neumann series give the claims. The suspended estimates follow from the parameter-dependent suspended parametrix (C3). □

Fix ε_0 as in Lemma 24.7 and $\delta \leq \varepsilon_0/4$, and use Definition 24.2 with these constants.

Lemma 24.8 (Weighted resolvent)**.** *Shrinking δ if necessary, for $|a| \leq 3\delta$ and $\mu \in \Lambda$ the operator $P_0 - \mu : x^a H_b^{s+2m} \to x^a H_b^s$ is invertible. Moreover $\|(P_0 - \mu)^{-1}\|_{x^a L_b^2} \leq C(1 + |\mu|)^{-1}$, and the norm from $x^a H_b^s$ to $x^a H_b^{s+2m}$ is bounded uniformly in μ.*

Proof. For $a = 0$ this is the spectral theorem, since $P_0 \geq c_0$, together with elliptic regularity. In general, $x^{-a}(P_0 - \mu)x^a = P_0 - \mu + E_a$. By Lemma 14.1, E_a is the sum of a b-operator with kernel $(s^{-a} - 1)K_P$, which vanishes on Δ_b, and an element of $\mathcal{I}^b$. So E_a is a times a family that is bounded

in $\Psi_b^{2m-1} + \mathcal{I}^b$ for $|a| \le 1$. By interpolation, $\|E_a(P_0-\mu)^{-1}\| \le C|a|\,\|P_0^{(2m-1)/2m}(P_0-\mu)^{-1}\| \le C'|a|$. For $3\delta C' \le 1/2$, a Neumann series inverts $x^{-a}(P_0-\mu)x^a$ on L^2_b for all $\mu \in \Lambda$ at once, with twice the norm. The Sobolev statement follows from elliptic regularity. □

Next comes the symbolic part, which is Seeley's construction [12] carried out in b-coordinates. Near H use coordinates (x, y), and b-covariables $\zeta = (\xi, \eta)$, with ξ dual to $\log x$. Write the b-symbol of P as $p \sim \sum_{k\ge 0} p_{2m-k}$, where p_{2m-k} is homogeneous of degree $2m-k$ for $|\zeta| \ge 1$ and smooth up to $x = 0$. Choose a smooth $\tilde p_{2m}$ that equals p_{2m} for $|\zeta| \ge 1$ and satisfies $\tilde p_{2m} \ge c_2 > 2\rho$ everywhere. This is possible after shrinking ρ. Put $D^b = (xD_x, D_y)$, and define, for $j \ge 1$,

$$q_{-2m} = (\tilde p_{2m} - \mu)^{-1}, \qquad q_{-2m-j} = -q_{-2m} \sum_{\substack{k+l+|\alpha|=j \\ l<j}} \frac{1}{\alpha!}\, \partial_\zeta^\alpha p'_{2m-k}\, (D^b)^\alpha q_{-2m-l}, \tag{29}$$

where $p'_{2m} = \tilde p_{2m}$ and $p'_{2m-k} = p_{2m-k}$ for $k \ge 1$.

Lemma 24.9 (Seeley symbols)**.** *For $j \ge 1$, $q_{-2m-j} = \sum_{l=2}^{2j+1} c_{j,l}(x,y,\zeta)(\tilde p_{2m}-\mu)^{-l}$. Here the $c_{j,l}$ are independent of μ, smooth up to $x = 0$, and homogeneous of degree $2m(l-1) - j$ for $|\zeta| \ge 1$. For $\mu \in \Lambda$,*

$$|\partial_{x,y}^\beta \partial_\zeta^\alpha q_{-2m-j}| \le C_{\alpha\beta j}\langle\zeta\rangle^{-j-|\alpha|}(\langle\zeta\rangle^{2m} + |\mu|)^{-1},$$

uniformly up to $x = 0$. Let $Q_J(\mu) = \mathrm{Op}_b(q^{(J)}(\mu))$, with $q^{(J)} = \sum_{j<J} q_{-2m-j}$, be a b-quantization (small calculus, kernel supported near Δ_b). Then $r_J(\mu) = 1 - (P_0-\mu)Q_J(\mu)$ is a holomorphic family in $\Psi_b^{-J}(X)$. Its Ψ_b^{-J}-seminorms are bounded, and its Ψ_b^{-J+2m}-seminorms are $O((1+|\mu|)^{-1})$. The same holds for the left parametrix $\tilde Q_J(\mu)$, with $\tilde r_J(\mu) = 1 - \tilde Q_J(\mu)(P_0-\mu)$.

Proof. The structure of q_{-2m-j} follows by induction from (29). Each derivative of $(\tilde p_{2m}-\mu)^{-l}$ raises l by one and contributes a homogeneous factor of the matching degree. For $\mu \in \Lambda$ we have $|\tilde p_{2m} - \mu| \ge c(\langle\zeta\rangle^{2m} + |\mu|)$, which gives the estimates. The recursion is the b-composition formula, which is the usual formula in the coordinates $(\log x, y)$. So the symbol of $(P-\mu)Q_J$ is 1 up to a finite sum of terms bounded by $\langle\zeta\rangle^{-J}\langle\zeta\rangle^{2m}(\langle\zeta\rangle^{2m}+|\mu|)^{-1}$, together with the term $(p_{2m} - \tilde p_{2m})q$, which has compact ζ-support. The remainder of the composition formula satisfies the same parameter-dependent bounds. This is the parameter-dependent symbol calculus (C3). Finally, $\pi Q_J \in \mathcal{I}^b$ is $O((1+|\mu|)^{-1})$. □

Proposition 24.10 (Resolvent)**.** *For every $J > 2m$ and $\mu \in \Lambda$,*

$$(P_0-\mu)^{-1} = Q_J(\mu) + \sum_H \chi_H M_{J,H}(\mu)\chi_H + R_J(\mu),$$

where:

- *(a) $M_{J,H}(\mu) \in \mathcal{M}_H^{-(J-2m)}$ has indicial family $\mathfrak{m}_{J,H}(\mu,\lambda) = (p_H(\lambda)-\mu)^{-1} I_H(r_J(\mu),\lambda)$, with seminorms $O((1+|\mu|)^{-2})$;*
- *(b) $R_J(\mu) \in \mathcal{R}_\delta^{-(J-2m)}$, with norms $O((1+|\mu|)^{-2})$;*
- *(c) everything is holomorphic in μ in the interior of Λ and continuous up to Γ.*

In particular $(P_0-\mu)^{-1} \in \mathcal{K}_\delta^{-2m}$, and $I_H((P_0-\mu)^{-1},\lambda) = (p_H(\lambda)-\mu)^{-1}$.

Proof. Put $Q = Q_J + \sum_H \chi_H M_{J,H} \chi_H$. Since $N_H(\pi) = 0$ and $(p_H - \mu) I_H(Q_J) = 1 - I_H(r_J)$, Lemma 24.7 gives

$$I_H(Q_J, \lambda) + \mathfrak{m}_{J,H}(\lambda) = (p_H(\lambda) - \mu)^{-1}.$$

Let $r = 1 - (P_0 - \mu)Q$. Then

$$r = \Big(r_J - \sum_H \chi_H N_H(r_J) \chi_H \Big) - \sum_H \Big([P, \chi_H] + \chi_H \big(P - N_H(P) \big) + \pi \chi_H \Big) M_{J,H} \chi_H,$$

where χ_H is inserted on both sides of $N_H(P)$ as in Lemma 24.3. Each term gains a factor x at ff or is supported away from it, as in the proof of Lemma 24.3(ii). Hence $r \in \mathcal{R}_\delta^{-(J-2m)}$, with norms $O((1+|\mu|)^{-1})$ by Lemma 24.9 and (a). The left parametrix $\tilde{Q}$ and its error $\tilde{r}$ are built in the same way. Write $G = (P_0 - \mu)^{-1}$. Then $G = \tilde{Q} + \tilde{r} G$ and $G = Q + G r$, so

$$G = Q + \tilde{Q}\, r + \tilde{r}\, G\, r.$$

Put $R_J = \tilde{Q} r + \tilde{r} G r$. By Lemma 24.3(iii) and Lemma 24.8, $R_J \in \mathcal{R}_\delta^{-(J-2m)}$. The operator norms of $\tilde{Q}$ on weighted Sobolev spaces are $O((1+|\mu|)^{-1})$, by the symbol estimates of Lemma 24.9 and by (a). This gives the bound in (b). Holomorphy is clear from the construction. □

24.4 Complex powers and heat operators

For $\operatorname{Re} z > 0$, set $P_0^{-z} = \frac{i}{2\pi} \int_\Gamma \mu^{-z} (P_0 - \mu)^{-1}\, d\mu$, with the principal branch. By the spectral theorem this agrees with the functional calculus, and $e^{-tP_0} = \frac{i}{2\pi} \int_\Gamma e^{-t\mu} (P_0 - \mu)^{-1}\, d\mu$ for $t > 0$. From (29) and Lemma 24.9,

$$\frac{i}{2\pi} \int_\Gamma \mu^{-z} q_{-2m-j}\, d\mu = \sigma_{-2mz-j}(z), \qquad \sigma_{-2mz}(z) = \tilde{p}_{2m}^{-z}, \quad \sigma_{-2mz-j}(z) = \sum_{l=2}^{2j+1} c_{j,l} \binom{-z}{l-1} \tilde{p}_{2m}^{-z-l+1}.$$

Indeed, $(\tilde{p} - \mu)^{-l} = \frac{1}{(l-1)!} \partial_{\tilde{p}}^{l-1} (\tilde{p} - \mu)^{-1}$. These symbols are entire in z and homogeneous of degree $-2mz - j$ for $|\zeta| \geq 1$. Moreover $\sigma_{-j}(0) = 0$ for $j \geq 1$, and $\sigma_0(0) = 1$.

Proposition 24.11 (Complex powers)**.** *For every $J > 2m+1$, the formula*

$$P_0^{-z} = \operatorname{Op}_b \Big(\sum_{j<J} \sigma_{-2mz-j}(z) \Big) + \sum_H \chi_H M_{J,H}^{(z)} \chi_H + R_J^{(z)},$$

$$M_{J,H}^{(z)} = \frac{i}{2\pi} \int_\Gamma \mu^{-z} M_{J,H}(\mu)\, d\mu, \qquad R_J^{(z)} = \frac{i}{2\pi} \int_\Gamma \mu^{-z} R_J(\mu)\, d\mu,$$

holds for $\operatorname{Re} z > 0$. The right-hand side is a holomorphic family in $\mathcal{K}_\delta^{-2m \operatorname{Re} z}$ on $\operatorname{Re} z > -1$, which defines P_0^{-z} there, with $P_0^0 = 1$. Its indicial family is $I_H(P_0^{-z}, \lambda) = p_H(\lambda)^{-z}$ for real λ, and every seminorm of the pieces is $O((1 + |\operatorname{Im} z|)^N e^{\theta |\operatorname{Im} z|})$ on vertical strips. The same holds for e^{-tP_0} and $P_0^{-1} e^{-tP_0}$, $t > 0$, with μ^{-z} replaced by $e^{-t\mu}$ and $\mu^{-1} e^{-t\mu}$. These lie in $\bigcap_k \mathcal{K}_\delta^{-k}$, with indicial families $e^{-t p_H}$ and $p_H^{-1} e^{-t p_H}$.

Proof. Integrate Proposition 24.10 against μ^{-z}. The terms with $j \geq 1$, and the pieces $M_{J,H}$ and R_J, are $O(|\mu|^{-2})$, so their integrals converge for $\operatorname{Re} z > -1$ and are holomorphic there. The term $j = 0$ gives $\tilde{p}_{2m}^{-z}$, which is entire. On Γ, $|\mu^{-z}| \leq |\mu|^{-\operatorname{Re} z} e^{\theta |\operatorname{Im} z|}$, and $\binom{-z}{l-1}$ is a polynomial, while $\tilde{p}_{2m} > 0$. This gives the growth bound. At $z = 0$, the integrands of the M, R and $j \geq 1$ terms are $O(|\mu|^{-2})$ and holomorphic in the interior of Λ. Pushing Γ to infinity inside Λ shows that these terms vanish, so $P_0^0 = \operatorname{Op}_b(1) = 1$. The indicial family is $\frac{i}{2\pi} \int_\Gamma \mu^{-z} (p_H(\lambda) - \mu)^{-1}\, d\mu = p_H(\lambda)^{-z}$,

by Proposition 24.10 and the spectral theorem on $L^2(H)$. For $e^{-t\mu}$ the same argument applies, and the symbols $\frac{i}{2\pi}\int e^{-t\mu} q_{-2m-j}\, d\mu = \sum_l c_{j,l} \frac{t^{l-1}}{(l-1)!} e^{-t\tilde{p}_{2m}}$ (with $l = 1$ for $j = 0$) are smoothing for $t > 0$. Since J is arbitrary, the operators lie in every $\mathcal{K}_\delta^{-k}$. □

The continuation to all of $\mathbb{C}$, with $P_0^{-z} = P_0^k P_0^{-z-k}$, is [6, Cor. 5.7]. We need only $\operatorname{Re} z > -1$.

24.5 The b-zeta function

Lemma 24.12 (Mellin transforms [6, Lemma 5.10])**.** *Let u be continuous on $(0,\infty)$ and $O(e^{-ct})$ as $t \to \infty$. Suppose that, as $t \downarrow 0$,*

$$u(t) = \sum_{i \le I} \sum_{q=0,1} u_{i,q} t^{e_i} (\log t)^q + u_0 + O(t^\gamma),$$

with $\gamma > 0$ and $e_i \neq 0$. Then $v(z) = \frac{1}{\Gamma(z)} \int_0^\infty t^{z-1} u(t)\, dt$ extends from $\operatorname{Re} z \gg 0$ to a meromorphic function on $\operatorname{Re} z > -\gamma$. It is holomorphic at $z = 0$, with $v(0) = u_0$.

Proof. The integral over $[1,\infty)$ is entire. Over $[0,1]$, the term $t^e(\log t)^q$ gives $(-1)^q q!\,(z+e)^{-q-1}$, the constant gives $1/z$, and the remainder is holomorphic for $\operatorname{Re} z > -\gamma$. Since $1/\Gamma(z) = z + O(z^2)$, only the constant survives at $z = 0$. □

Theorem 24.13 (b-zeta function [6, Def. 8.1, Thm. 8.2])**.** *The b-zeta function $\zeta_P(z) = {}^b\operatorname{Tr}(P_0^{-z})$, defined for $\operatorname{Re} z > n/2m$, extends meromorphically to $\operatorname{Re} z > -1$. Its only poles there are simple poles at the points $(n-j)/2m \neq 0$, $j \in \mathbb{N}_0$, and it is holomorphic at $z = 0$. Moreover, as $t \downarrow 0$,*

$${}^b\operatorname{Tr}(e^{-tP}) = \sum_{0<(n-j)/2m} \Gamma\tfrac{n-j}{2m} \operatorname{Res}_{z=\frac{n-j}{2m}} \zeta_P \cdot t^{-\frac{n-j}{2m}} + \zeta_P(0) + O(t^\gamma)$$

for some $\gamma > 0$. In particular there is no $\log t$ term at t^0, and $\zeta_P(0)$ is the constant term of ${}^b\operatorname{Tr}(e^{-tP})$.

Proof. By Proposition 24.11 and Lemma 24.4, ${}^b\operatorname{Tr}$ of $\chi_H M_{J,H}^{(z)} \chi_H + R_J^{(z)}$ is holomorphic on $\operatorname{Re} z > -1$, once $J > 2m + n + 1$. For the symbolic terms, the diagonal value of $\operatorname{Op}_b(\sigma_{-2mz-j}(z))$ at (x,y) is $(2\pi)^{-n} \int \sigma_{-2mz-j}(z;x,y,\zeta)\, d\zeta$. The integral over $|\zeta| \le 1$ is entire. By homogeneity, the integral over $|\zeta| \ge 1$ is

$$\frac{1}{2mz + j - n} \int_{S^{n-1}} \sigma_{-2mz-j}(z;x,y,\omega)\, d\omega.$$

Both are smooth up to $x = 0$, and the b-integral of a function that is smooth up to ∂X is holomorphic in parameters. This gives meromorphy with simple poles at $2mz = n - j$. At $z = 0$, the only candidate is $j = n$, with residue proportional to $\int_{S^{n-1}} \sigma_{-n}(0) = 0$.

For the heat expansion, note first that $\Gamma(z)\zeta_P(z) = \int_0^\infty t^{z-1} u(t)\, dt$ for $\operatorname{Re} z > n/2m$, where $u(t) = {}^b\operatorname{Tr}(e^{-tP_0})$. This follows by Fubini applied piecewise to the decompositions of Proposition 24.11, because the Mellin transform of $e^{-t\mu}$ is $\Gamma(z)\mu^{-z}$. Next, $|\zeta_P(z)| \le C(1 + |\operatorname{Im} z|)^N e^{\theta|\operatorname{Im} z|}$ on vertical strips away from the poles, by Proposition 24.11 and the explicit symbols. Since $\theta < \pi/2$, Stirling's formula makes $\Gamma\zeta_P$ integrable on vertical lines. Mellin inversion gives $u(t) = \frac{1}{2\pi i} \int_{\operatorname{Re} z = c} \Gamma(z)\zeta_P(z) t^{-z}\, dz$. Move the line to $\operatorname{Re} z = -\gamma$, with $0 < \gamma < 1/2m$. The residues are those listed, plus $\zeta_P(0)$ from the pole of Γ at 0, and the new integral is $O(t^\gamma)$. Finally, $e^{-tP} = e^{-tP_0} + (1 - e^{-t})\pi$, and ${}^b\operatorname{Tr}\pi = \operatorname{Tr}\pi$ by Lemma 24.4(iii). The extra term $(1 - e^{-t})\dim\ker P$ has no constant term. □

24.6 The b-eta function and the index theorem

For each H, write $a = a_H$, and define for $\operatorname{Re} z > (n-1)/2m$ and $t > 0$

$$\eta_H(z) = \frac{1}{\pi i}\int_{\mathbb{R}} \operatorname{Tr}_H\big(\partial_\lambda a\, a^{-1}(aa^*)^{-z}\big)\, d\lambda, \qquad \eta_H(t) = \frac{1}{\pi i}\int_{\mathbb{R}} \operatorname{Tr}_H\big(\partial_\lambda a\, a^{-1} e^{-taa^*}\big)\, d\lambda. \tag{30}$$

The integrals converge absolutely: $\partial_\lambda a\, a^{-1}$ is suspended of order -1, and $(aa^*)^{-z}$ of order $-2m\operatorname{Re} z$. Put $\eta(z) = \sum_H \eta_H(z)$ and $\eta(t) = \sum_H \eta_H(t)$. Let $h(t) = {}^b\operatorname{Tr}(e^{-tA^*A}) - {}^b\operatorname{Tr}(e^{-tAA^*})$ and $h_0(t) = {}^b\operatorname{Tr}(e^{-tP_0}) - {}^b\operatorname{Tr}(e^{-tP_0'})$, with $P = A^*A$ and $P' = AA^*$.

Proposition 24.14 (McKean–Singer–Melrose identity)**.** *For $t > 0$ and $\operatorname{Re} z > n/2m$,*

$$e^{-t}\operatorname{Ind} A = h_0(t) - \tfrac{1}{2}\eta(t), \qquad \operatorname{Ind} A = h(t) - \tfrac{1}{2}\eta(t),$$

$$\operatorname{Ind} A = \zeta_{A^*A}(z) - \zeta_{AA^*}(z) - \tfrac{1}{2}\eta(z).$$

These are [6, Lemma 7.3, Lemma 8.3 and (89)], with the sign of the boundary term corrected as explained in Remark 24.17.

Proof. Let f be one of μ^{-z-1}, $\mu^{-1}e^{-t\mu}$. Polar decomposition gives $g(AA^*)A = Ag(A^*A)$ on $(\ker A)^\perp$ for bounded Borel g. Using $\pi' A = 0$, this yields

$$P_0 f(P_0) = \pi\, f(1) + A^* f(P_0')A, \qquad P_0' f(P_0') = \pi' f(1) + AA^* f(P_0').$$

Put $G = A^* f(P_0')$. By Proposition 24.11 and Lemma 24.3, $G \in \mathcal{K}_\delta^{-k}$ with $k = 2m\operatorname{Re} z + m$ for $f = \mu^{-z-1}$, and with every k for $f = \mu^{-1}e^{-t\mu}$. For real λ

$$I_H(G,\lambda) = a^* f(aa^*) = a^{-1}(aa^*) f(aa^*).$$

Since $f(1) = 1$ for $f = \mu^{-z-1}$ and $f(1) = e^{-t}$ for $f = \mu^{-1}e^{-t\mu}$, Lemma 24.4(iii) and Theorem 24.5 give

$$\begin{aligned} {}^b\operatorname{Tr}(P_0 f(P_0)) - {}^b\operatorname{Tr}(P_0' f(P_0')) &= f(1)\operatorname{Ind} A - {}^b\operatorname{Tr}[A, G] \\ &= f(1)\operatorname{Ind} A - \frac{i}{2\pi}\sum_H \int_{\mathbb{R}} \operatorname{Tr}_H\big(\partial_\lambda a\, a^{-1}(aa^*) f(aa^*)\big)\, d\lambda. \end{aligned}$$

Since $-\frac{i}{2\pi} = \frac{1}{2}\cdot\frac{1}{\pi i}$, this is the first identity, and also the third for $\operatorname{Re} z$ so large that $k > 2(n+m+1)$. Both sides of the third identity are holomorphic for $\operatorname{Re} z > n/2m$ (Theorem 24.13 and (30)), so it holds there by analytic continuation. The second follows from $h_0(t) = h(t) + (e^{-t}-1)\operatorname{Ind} A$, because $e^{-tP_0} = e^{-tP} + (e^{-t}-1)\pi$. □

Lemma 24.15 (b-eta function [6, Lemma 8.4])**.** *Each $\eta_H(z)$ extends meromorphically to $\operatorname{Re} z > -1$, with at most a simple pole at $z = 0$. The sum $\eta(z)$ is holomorphic at $z = 0$, so the residues at 0 add up to zero. Moreover $\eta(0)$ is the constant term of $\eta(t)$ as $t \downarrow 0$, and $\eta(t)$ has no $\log t$ term at t^0.*

Proof. Let $F(z) = D(A)A^*(P_0')^{-z-1}$. By Proposition 24.11 and Lemma 24.3, this is a holomorphic family in $\mathcal{K}_\delta^{-1-2m\operatorname{Re} z}$. For real λ, its indicial family is $i\partial_\lambda a\, a^{-1}(aa^*)^{-z}$. Hence, for $\operatorname{Re} z > (n-1)/2m$, Lemma 24.4(i) gives $\eta_H(z) = -2\int_H f_{0,H}(z,y)\,\nu_\partial$, where $f_{0,H}(z,\cdot)$ is the value at H of the diagonal of $F(z)$. Write $F(z)$ as in Definition 24.2, using Proposition 24.11 and the composition formula. The symbolic part has symbol $\sum_j \tau_{-1-2mz-j}(z)$, homogeneous of degree $-1-2mz-j$ for $|\zeta| \geq 1$. As in Theorem 24.13, its diagonal at $x = 0$ is meromorphic with simple poles at $2mz = n-1-j$.

This includes a possible pole at $z = 0$ from $j = n - 1$. The remaining pieces are holomorphic. The sum $\eta(z)$ equals $2(\zeta_{A^*A}(z) - \zeta_{AA^*}(z) - \operatorname{Ind} A)$ by Proposition 24.14, and this is holomorphic at 0 by Theorem 24.13. Finally, $\eta(t) = 2(h_0(t) - e^{-t} \operatorname{Ind} A)$, and Theorem 24.13 gives its expansion. Its constant term is $2(\zeta_{A^*A}(0) - \zeta_{AA^*}(0) - \operatorname{Ind} A) = \eta(0)$. The same conclusion follows from Lemma 24.12, since $\frac{1}{\Gamma(z)} \int_0^\infty t^{z-1} \eta(t)\, dt = \eta(z)$ for $\operatorname{Re} z \gg 0$ by Fubini. □

For a family $a = I_H(A, \cdot)$ as above, define

$$\eta_b(a) = \operatorname{FP}_{z=0} \frac{1}{\pi i} \int_{\mathbb{R}} \operatorname{Tr}_H\big(\partial_\lambda a\, a^{-1} (aa^*)^{-z}\big)\, d\lambda. \tag{31}$$

If $a = 1 + k$ with $k \in B_H$, as in Theorem 23.5, the integrand is trace class and integrable for every z, and is entire in z. So (31) agrees with the definition used in Theorem 23.5.

Theorem 24.16 (General index formula)**.** *Let $A \in \Psi_b^m(X)$, $m \geq 1$, be fully elliptic, and let* Ind A *be its index (Lemma 24.1): on any $x^\delta H_b^s$, on L_b^2, or on $\dot{C}^\infty(X)$, equal to* $\operatorname{Tr}(AB - BA)$*. Then*

$$\operatorname{Ind} A = \zeta_{A^*A}(0) - \zeta_{AA^*}(0) - \frac{1}{2} \sum_H \eta_b\big(I_H(A)\big).$$

Here $\zeta_P(0)$ is the value at 0 of the b-zeta function of Theorem 24.13, which is the constant term of ${}^b\operatorname{Tr}(e^{-tP})$ as $t \downarrow 0$. The b-eta invariants are those of (31)*, and their sum is the constant term of $\eta(t)$.*

Proof. Take the value at $z = 0$ in Proposition 24.14, using Theorem 24.13 and Lemma 24.15. The finite parts of the η_H add up to the value of η at 0, because the residues cancel. □

This is the codimension-one case of [6, Thm. 8.6], which extends Piazza's formula [11] to manifolds with corners. Here are the hypotheses there. X is a compact manifold with corners; $A \in \Psi_{b,os}^m(X, \Omega_b^{1/2})$, where "os" means one-step polyhomogeneous, that is, classical; $m > 0$ is real; and $A : H_b^m \to L_b^2$ is Fredholm. By [6, Cor. 2.6], the last condition holds exactly when A is elliptic and every $N_H(A)(\tau)$, $\tau \in \mathbb{R}$, is invertible. A fully elliptic A satisfies all of these, by Lemma 24.1. Full ellipticity is used above only in Lemma 24.1: to put π, π' in $\mathcal{I}^b$, to get the lower bound in (iv), and to get the parametrix B and the identification of the index with $\operatorname{Tr}(AB - BA)$.

Remark 24.17 (Conventions and a sign in the thesis). In the notation of Remark 24.6, $N_H(A)(\tau) = a(\tau)$, $D_\tau = -i\partial_\tau$, and $N_H(AA^*)(\tau) = a(\tau)a(\tau)^*$ for real τ. The thesis defines the b-eta function of a face M by [6, (82)]:

$${}^b\eta_M(t) = \frac{2}{2\pi} \int_t^\infty \int_{\mathbb{R}} {}^b\operatorname{Tr}\big(D_\tau N(A) N(A^*) N(e^{-sAA^*})\big)\, d\tau\, ds.$$

Since $N(A^*)e^{-saa^*} = -a^{-1}\partial_s e^{-saa^*}$ and $e^{-saa^*} \to 0$ as $s \to \infty$, the inner s-integral is $+D_\tau a\, a^{-1} e^{-taa^*}$. The last line of the computation preceding [6, Lemma 7.3] has exactly this sign. So (82) equals our $\eta_H(t)$. The second expression [6, (83)], $-\frac{2}{2\pi} \int {}^b\operatorname{Tr}(D_\tau N(A) N(A)^{-1} N(e^{-tAA^*}))\, d\tau$, has the opposite sign, and so equals $-\eta_H(t)$. The thesis carries (83) into Lemma 7.4, Lemma 8.3, (89) and the displayed formula for ${}^b\eta_M(z)$ in Theorem 8.6. In our notation that formula is $-\eta_H(z)$, which would give Ind $A = \zeta - \zeta + \frac{1}{2}\eta$. This is wrong. The correct statement is Theorem 24.16, as the explicit check in Example 24.18 confirms. The results that use (82) directly, namely [6, Lemma 7.9, Thms. 7.10, 8.7, Cor. 8.8], have the correct sign: they reduce to the Atiyah–Patodi–Singer formula. Two smaller slips occur in [6, Cor. 5.12, Thm. 8.2]. The residue at a pole $z_k = (n - k)/m$ coming from t^{-z_k} is $\int \gamma_k / \Gamma(z_k)$, not $\int \gamma_k / \Gamma(-z_k)$. The logarithmic terms give poles at $z_k \in -\mathbb{N}$, not at $z_k \in \mathbb{N}$. Both follow from [6, Lemma 5.10], which is stated correctly. Finally, [6, Lemma 8.4] asserts holomorphy at 0 only for the sum over faces, and the individual ${}^b\eta_M(z)$ may have poles there. This is why (31) uses a finite part.

Example 24.18 (Sign check)**.** Let $X = [0,1]$, with the b-structure at both ends, and let $u = \log(x/(1-x))$, so that $\mu_b = du$. Put $A = \partial_u + \tanh u$, a b-differential operator. It is Fredholm on L^2_b, but not fully elliptic: $a_0(-i) = 0$ below. So this example checks the sign of the identity of Proposition 24.14 in the generality of the thesis. For this A, the conclusions of Lemma 24.1(iv) still hold, except that π, π' lie only in $\mathcal{R}_\delta^{-K}$ for every K (the kernel of A is $O(x)$). With this change, the proofs of Propositions 24.10, 24.11 and 24.14 go through unchanged. The kernel of A is spanned by $1/\cosh u \in L^2$, and $\ker A^*$, spanned by $\cosh u$, is trivial in L^2. Hence $\operatorname{Ind} A = 1$.

Near $x = 0$ we have $\partial_u = x(1-x)\partial_x$, and $\tanh u \to -1$, so $a_0(\lambda) = i\lambda - 1$ by (20). Near $x = 1$, with $x' = 1 - x$, we have $\partial_u \to -x'\partial_{x'}$, and so $a_1(\lambda) = -i\lambda + 1$. For $a(\lambda) = \pm i\lambda + c$, c real,

$$\frac{1}{\pi i}\int_{\mathbb{R}} \frac{\pm i}{\pm i\lambda + c}(\lambda^2 + c^2)^{-z}\, d\lambda = \pm \frac{c}{|c|^{1+2z}}\,\frac{\Gamma(z+\frac{1}{2})}{\sqrt{\pi}\,\Gamma(z+1)},$$

since the odd part integrates to zero. So $\eta_b(i\lambda - 1) = -1$ and $\eta_b(-i\lambda + 1) = -1$, and the boundary term is $-\frac{1}{2}(-1-1) = +1$. The interior term vanishes. Indeed, $A^*A = -\partial_u^2 + 1 - 2\operatorname{sech}^2 u$ and $AA^* = -\partial_u^2 + 1$ have the same normal operators. So $h(t) = \operatorname{Tr}(e^{-tA^*A} - e^{-tAA^*})$, which has the small-time expansion $(4\pi t)^{-1/2} e^{-t}\sum_{k\geq 1} c_k t^k$, with $c_1 = -\int(-2\operatorname{sech}^2 u)\, du = 4$. This expansion has no constant term. Hence $\operatorname{Ind} A = 0 + 1 = 1$, as it should be. The printed form of [6, Thm. 8.6] would give -1. The order-zero case gives the same sign: see Remark 24.19.

Remark 24.19 (Order zero and Theorem 23.5). This remark is a consistency check, and we only indicate the argument. For $A = 1 + K$ as in Theorem 23.5, $P_0 - 1 \in \Psi_b^{-\infty} + \mathcal{I}^b$. The operator P_0 is bounded with spectrum in a compact subset of $(0,\infty)$, so a compact contour may be used, and

$$P_0^{-z} - 1 = -\frac{i}{2\pi}\oint \mu^{-z}(1-\mu)^{-1}(P_0 - 1)(P_0 - \mu)^{-1}\, d\mu$$

is an entire family of residual operators. Since K is residual, the trace-class conditions in the defect formula hold automatically. The proof of Proposition 24.14 then gives

$${}^b\operatorname{Tr}(P_0^{-z} - 1) - {}^b\operatorname{Tr}(P_0'^{-z} - 1) = \operatorname{Ind} A - \frac{i}{2\pi}\sum_H \int_{\mathbb{R}} \operatorname{Tr}_H\big(\partial_\lambda a\, a^{-1}(aa^*)^{-z}\big)\, d\lambda.$$

Here ${}^b\operatorname{Tr}[A, G] = {}^b\operatorname{Tr}[K, G]$, and K is residual. At $z = 0$ the left side vanishes, which is Theorem 23.5 with the same sign. In order zero there is no interior term.

Remark 24.20 (Residue form; not proved here). On closed manifolds, $\zeta_{A^*A}(0) - \zeta_{AA^*}(0)$ can be rewritten as a Wodzicki residue built from the symbols of A and a parametrix, the "Atiyah–Singer term" $\operatorname{AS}(A)$. The earlier sketch of this section stated the b-analogue. It requires the meromorphy of $\operatorname{Tr}(x^w[A,B]Q^{-\tau})$ in two variables, and we have not proved it. Theorem 24.16 is stated, and proved, with the zeta values.

Remark 24.21 (Relation to the companion notes). The companion notes [10] quote the b-heat kernel theorem only for product-type generalized b-Laplacians, through the parabolic heat calculus. The operators e^{-tA^*A} used here have A pseudodifferential and are not of product type, so they lie outside that theorem. We do not use a heat calculus. The heat operators are contour integrals of the resolvent (Proposition 24.11), and their small-time behaviour on the diagonal is read off from the b-zeta function by Mellin inversion (Theorem 24.13). The b-trace defect formula of [10] is the case of Theorem 24.5 with both operators in the full calculus.

Remark 24.22. Moroianu and Nistor relate index and homology for manifolds with boundary [24]. Liu and Ma refine eta invariants through differential K-theory [18]. Mazzeo and Melrose treat the fibred-boundary analogue of the calculus [23].

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